

**GEETHANJALI INSTITUTE OF SCIENCE AND TECHNOLOGY-NELLORE
(AUTONOMOUS)**



DEPARTMENT OF CSE (AI&ML)

REGULATION: RG23

With effective from

2023-2027 Batch

COURSE STRUCTURE AND SYLLABUS



GEETHANJALI INSTITUTE OF SCIENCE & TECHNOLOGY

(Autonomous)

Unit of USHODAYA EDUCATIONAL SOCIETY

An ISO 9001:2015 certified Institution: Recognized under Sec. 2(f)& 12(B) of UGC Act, 1956
3rd Mile, Bombay Highway, Gangavaram (V), Kovur(M), SPSR Nellore (Dt), Andhra Pradesh, India- 524137,
E-Mail: geethanjali@gist.edu.in, Website: www.gist.edu.in

DEPARTMENT OF CSE (AI&ML)

Vision

To become a learning-resource center producing Computer Engineers suitable for career roles in Artificial Intelligence and Machine Learning domains.

Mission

DM1: Imparting quality education through industry relevant curriculum and effective teaching-learning methodologies.

DM2: Organizing Professional skill development programmes to cater to the societal needs through Institute-Industry interface.

DM3: Incorporating life skill development programmes aimed at promoting social responsibility on ethical foundations.

DM4: Fostering employability skills by incorporating cutting-edge technologies in the curriculum.

Program Specific Outcomes

PSO 1: Apply the expertise in adaptive algorithms to develop quality software applications.

PSO 2: Demonstrate the capabilities in basic and advanced technologies towards getting employed or to become an entrepreneur

Program Educational Objectives (PEOs)

PEO1: Carving successful careers in industry, academia with expertise in Artificial Intelligence & Machine Learning (AI & ML)

PEO2: Excelling as socially committed Artificial Intelligence & Machine Learning (AI & ML) Engineers.

PEO3: Engaged in the successful execution of multi-disciplinary projects for the welfare and progress of humankind.

PEO4: Pursuing higher studies, research activities and initiatives of entrepreneurship.

Program Outcomes (POs)

- PO1: Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2: Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3: Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4: Conduct investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5: Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- PO6: The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7: Environment and Sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8: Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9: Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10: Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11: Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



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B.Tech-I-Year-I-Semester

S.No.	Category	Course Code	Title	L	T	P	Credits
1	BS	23A0003T	Engineering Physics	3	0	0	3
2	BS	23A0001T	Linear Algebra & Calculus	3	0	0	3
3	ES	23A0201T	Basic Electrical & Electronics Engineering	3	0	0	3
4	ES	23A0301T	Engineering Graphics	1	0	4	3
5	ES	23A0501T	Introduction to Programming	3	0	0	3
6	ES	23A0503P	IT Workshop	0	0	2	1
7	BS	23A0006P	Engineering Physics Lab	0	0	2	1
8	ES	23A0202P	Electrical & Electronics Engineering Workshop	0	0	3	1.5
9	ES	23A0502P	Computer Programming Lab	0	0	3	1.5
10	BS&H	23ANS01P	NSS/NCC/Scouts & Guides/Community Service	-	--	1	0.5
TOTAL				13	00	15	20.5

Category	CREDITS
Basic Science Course(BSC)	7.5
Engineering Science Course(ESC)	13
Humanities and Social Science(HSMC)	-
TOTAL CREDITS	20.5


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B.Tech-I-Year-II-Semester

S.No.	Category	Course Code	Title	L	T	P	Credits
1	BS	230009T	Communicative English	2	0	0	2
2	BS	23A0004T	Chemistry	3	0	0	3
3	ES	23A0002T	Differential Equations & Vector Calculus	3	0	0	3
4	ES	23A0101T	Basic Civil & Mechanical Engineering	3	0	0	3
5	PC	23A0504T	Data structures	3	0	0	3
6	BS	2300010P	Communicative English Lab	0	0	2	1
7	BS	23A0007P	Chemistry Lab	0	0	2	1
8	ES	23A0302P	Engineering Workshop	0	0	3	1.5
9	PC	23A0505P	Data structures Lab	0	0	3	1.5
10	BS&HM	23AYG01P	Health and Wellness, Yoga and Sports	-	--	1	0.5
TOTAL				14	00	11	19.5

Category	CREDITS
Basic Science Course(BS)	7.5
Engineering Science Course(ES)	7.5
Professional Core Course(PC)	4.5
TOTAL CREDITS	19.5


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B.Tech-II-Year-I- Semester

Semester-3(Theory-5,Lab-2,SEC-1,MC-1)

S. No.	Category	Course Code	Course Title	Hours per Week			Credits
				L	T	P	
1	BS	23A0015T	Discrete Mathematics & Graph Theory	3	0	0	3
2	HS	23A0021T	Universal Human Values 2 Understanding Harmony and Ethical Human Conduct	2	1	0	3
3	ES	23A3301T	Artificial Intelligence	3	0	0	3
4	PC	23A0506T	Advanced Data Structures & Algorithm Analysis	3	0	0	3
5	PC	23A0507T	Object-Oriented Programming Through JAVA	3	0	0	3
6	PC	23A0508P	Advanced Data structures And Algorithms Analysis Lab	0	0	3	1.5
7	PC	23A0509P	Object-Oriented Programming through JAVA Lab	0	0	3	1.5
8	SEC	23A0510P	Python Programming	0	1	2	2
9	MC	23A0109T	Environmental Science	2	0	0	-
TOTAL				16	2	8	20

Category	Credits
Basic Science Course(BSC)	3
Professional Core Course(PCC)	9
Engineering Science Course(ESC)	3
Humanities and Social Science Course(HSC)	3
Skill Enhancement Course(SEC)	2
TOTAL CREDITS	20

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B.Tech-II-Year-II- Semester

Semester-4(Theory-5,Lab-2,SEC-1,MC-1)							
S. No.	Category	Course Code	Course Title	Hours per week			Credits
				L	T	P	
1	HS	23A0020T	Optimization Techniques	2	0	0	2
2	BS	23A0017T	Probability & Statistics	3	0	0	3
3	PC	23A3302T	Machine Learning	3	0	0	3
4	PC	23A0512T	Database Management Systems	3	0	0	3
5	PC	23A0406T	Digital Logic and Computer Organization	3	0	0	3
6	PC	23A3303P	Machine Learning Lab	0	0	3	1.5
7	PC	23A0515P	Database Management Systems Lab	0	0	3	1.5
8	SEC	23A0516P	FullStackDevelopment-1	0	1	2	2
9	ES	23A0413T	Design Thinking & Innovation	1	0	2	2
Total				15	1	10	21
Mandatory Community Service Project Internship of 08 weeks duration during summer vacation							

Category	Credits
Basic Science Course(BSC)	3
Professional Core Course(PCC)	12
Engineering Science Course(ESC)	2
Skill Enhancement Course(SEC)	2
Management Elective Course(HS)	2
TOTAL CREDITS	21

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B.Tech-III-Year-I Semester

Semester-5 (Theory-5, Lab-2, SEC-1, AC-1)							
S. No.	Category	Course Code	Course Title	Hours per week			Credits
				L	T	P	
1	PC	23A3304T	Natural Language Processing	3	0	0	3
2	PC	23A3305T	System Software Programming	3	0	0	3
3	PC	23A3306T	Computer Vision & Image Processing	3	0	0	3
4	PC	23A0522T	Introduction to Quantum Technologies and Applications	3	0	0	3
5	PE	23A3307Ta 23A3307Tb 23A3307Tc 23A3307Td	Professional Elective-I 1. Data Visualization 2. Soft computing 3. Exploratory Data Analysis with Python 4. Computational Intelligence	3	0	0	3
6	OE		Open Elective-I	3	0	0	3
7	PC	23A3308P	Computer Vision & Machine Learning Lab	0	0	3	1.5
8	PC	23A3309P	AI & System Programming Lab	0	0	3	1.5
9	SEC	23A0527P	Skill Enhancement Course Full Stack Development-II	0	1	2	2
10	MC	23A0420	Tinkering Lab	0	0	2	1
11	PC	23A3310	Evaluation of Community Service Internship	-	-	-	2
Total				18	1	10	26

Category	Credits
Professional Core Course(PCC)	17
Professional Elective Course(PEC)	3
Open Elective Course(OEC)	3
Skill Enhancement Course(SEC)	2
Audit Course(MC)	1
TOTAL CREDITS	26

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Open Elective – I

S.No.	Course Code	Course Title	Offered by the Dept
1	23A0148T	Green Buildings	CIVIL
2	23A0149T	Construction Technology and Management	CIVIL
3	23A0222T	Electrical Safety Practices and Standards	EEE
4	23A0319T	Sustainable Energy Technologies	ME
5	23A0442T	Electronic Circuits	ECE
6	23A0027T	Mathematics for Machine Learning and AI	Mathematics
7	23A0034T	Materials Characterization Techniques	Physics
8	23A0040T	Chemistry of Energy Systems	Chemistry
9	23A0044T	English for Competitive Examinations	Humanities
10	23A0051T	Entrepreneurship and New Venture Creation	Humanities
11	23A0547T	Quantum Technologies and Applications	CSE


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B.Tech-III-Year-II- Semester

Semester-6 (Theory-6, Lab-2,SEC-1, AC-1)							
Sl. No.	Category	Course Code	Course Title	Hours per Week			Credits
				L	T	P	
1	PC	23A3311T	Cloud Computing for AI	3	0	0	3
2	PC	23A3312T	Big Data Analytics & AI Applications	3	0	0	3
3	PC	23A3313T	Full Stack AI Development	3	0	0	3
4	PE	23A3314Ta 23A3314Tb 23A3314Tc 23A3314Td	Professional Elective-II 1. Graph Neural Networks 2. Recommender Systems 3. Predictive Analytics 4. Block chain for AI	3	0	0	3
5	PE	23A3315Ta 23A3315Tb 23A3315Tc 23A3315Td	Professional Elective-III 1. Introduction to Quantum Computing 2. AI for Finance 3. Social Network Analysis 4. Cyber security &AI-driven Threat Detection	3	0	0	3
6	OE		Open Elective-II	3	0	0	3
7	PC	23A3316P	Big Data & Cloud Computing Lab	0	0	3	1.5
8	PC	23A3317P	Full Stack AI Lab	0	0	3	1.5
9	SEC	23A0026T	Skill Enhancement Course Soft skills	0	1	2	2
10	MC	23A0053T	Audit Course Technical Paper Writing &IPR	2	0	0	-
11	PC	23A3318	Workshop				
Total				20	1	08	23
Mandatory Industry Internship of 08 weeks duration during summer vacation							



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B.Tech-III-Year-II- Semester

Category	Credits
Professional Core Course(PCC)	12
Professional Elective Course(PEC)	6
Open Elective Courses(OEC)	3
Skill Enhancement Course(SEC)	2
Audit Course(AC)	-
Total	23


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Open Elective – II

S.No.	Course Code	Course Title	Offered by the Dept
1	23A0150T	Disaster Management	CIVIL
2	23A0151T	Sustainability in Engineering Practices	CIVIL
3	23A0232T	Renewable Energy Sources	EEE
4	23A0334Tb	Automation and Robotics	ME
5	23A0443T	Digital Electronics	ECE
6	23A0030T	Optimization Techniques	Mathematics
7	23A0035T	Physics of Electronic Materials and Devices	Physics
8	23A0041T	Chemistry Of Polymers And Applications	Chemistry
9	23A0045T	Academic Writing and Public Speaking	Humanities
10	23A0029T	Mathematical Foundations of quantum technologies	Mathematics


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B.Tech-IV-Year-I- Semester

Semester-7 (Theory-6, SEC-1, AC-1)							
Sl. No.	Category	Course Code	Course Title	Hours per week			Credits
				L	T	P	
1	PC	23A3319T	Generative AI	3	0	0	3
2	HS	23A0049T 23A0050T 23A0048T	Management Course-II 1. Business Ethics and Corporate Governance 2. E-Business 3. Management Science	2	0	0	2
3	PE	23A3320Ta 23A3320Tb 23A3320Tc 23A3320Td	Professional Elective-IV 1. Explainable AI & Model Interpretability 2. AI in Cyber Security 3. AI-driven Software Engineering & DevOps 4. AI for Robotics	3	0	0	3
4	PE	23A3321Ta 23A3321Tb 23A3321Tc 23A3321Td	Professional Elective-V 1. ML Ops & AI Model Deployment 2. Data Wrangling 3. Healthcare AI 4. AI for Smart Cities & IoT Systems	3	0	0	3
5	OE		Open Elective-III	3	0	0	3
6	OE		Open Elective-IV	3	0	0	3
7	SEC	23A0540P	Skill Enhancement Course Prompt Engineering	0	1	2	2
8	MC	23A0054T	Audit Course Gender Sensitization	2	0	0	-
9	PC	23A3322	Evaluation of Industry Internship	-	-	-	2
Total				19	1	02	21



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B.Tech-IV-Year-I- Semester

Category	Credits
Professional Core Course(PCC)	5
Professional Elective Course(PEC)	6
Management Elective Course(HC)	2
Open Elective Courses(OEC)	6
Skill Enhancement Course(SEC)	2
Audit Course(MC)	-
TOTAL CREDITS	21


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Open Elective – III

Sl.No.	Course Code	Course Title	Offered by the Dept
1	23A0152T	Building Materials and Services	CIVIL
2	23A0121T	Environmental Impact Assessment	CIVIL
3	23A0241T	Smart Grid Technologies	EEE
4	23A0335T	3D Printing Technologies	ME
5	23A0416T	Microprocessors and Microcontrollers	ECE
6	23A0027T	Wavelet transforms and its Applications	Mathematics
7	23A0036T	Smart Materials And Devices	Physics
8	23A0042T	Green Chemistry And Catalysis For Sustainable Environment	Chemistry
9	23A0046T	Employability Skills	Humanities
10	23A0037T	Introduction to Quantum Mechanics	Physics

Open Elective – IV

S.No.	Course Code	Course Title	Offered by the Dept
1	23A0153T	Geo-Spatial Technologies	CIVIL
2	23A0154T	Solid Waste Management	CIVIL
3	23A0242T	Electric Vehicles	EEE
4	23A0334Td	Total Quality Management	ME
5	23A0444T	Transducers and Sensors	ECE
6	23A0032T	Financial Mathematics	Mathematics
7	23A0038T	Sensors And Actuators For Engineering Applications	Physics
8	23A0043T	Chemistry Of Nanomaterials and Applications	Chemistry
9	23A0047T	Literary Vibes	Humanities
10	23A3315Ta	Introduction to Quantum Computing	CSE(AI&ML)


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B.Tech-IV-Year-II- Semester

Semester-8							
Sl. No.	Category	Course Code	Course Title	Hours per week			Credits
				L	T	P	C
1	PC	23A3323	Internship	-	-	24	4
			Project	-	-	-	8
TOTAL CREDITS							12


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DEPARTMENT OF CSE(AI&ML)

REGULATION: R23

FOR

2023-2027 BATCH

COURSE STRUCTURE AND SYLLABUS

GEETHANJALI INSTITUTE OF SCIENCE AND TECHNOLOGY – NELLORE**DEPARTMENT OF CSE(AI&ML)****REGULATION- R23****COURSE STRUCTURE AND SYLLABUS****B. Tech-I-Year-I-Semester (GROUP-B)**

S.No.	Category	Course Code	Title	L/D	T	P	Credits
1	BS&H	23A0003T	Engineering Physics	3	0	0	3
2	BS&H	23A0001T	Linear Algebra & Calculus	3	0	0	3
3	Engineering Science	23A0201T	Basic Electrical & Electronics Engineering	3	0	0	3
4	Engineering Science	23A0301T	Engineering Graphics	1	0	4	3
5	Engineering Science	23A0501T	Introduction to Programming	3	0	0	3
6	Engineering Science	23A0503P	IT Workshop	0	0	2	1
7	BS&H	23A0006P	Engineering Physics Lab	0	0	2	1
8	Engineering Science	23A0202P	Electrical & Electronics Engineering Workshop	0	0	3	1.5
9	Engineering Science	23A0502P	Computer Programming Lab	0	0	3	1.5
10	BS&H	23ANS01P	NSS/NCC/Scouts & Guides/Community Service	-	--	1	0.5
TOTAL				13	00	15	20.5

B. Tech-I-Year-II-Semester (GROUP-B)

S.No.	Category	Course Code	Title	L/D	T	P	Credits
1	BS&H	230009T	Communicative English	2	0	0	2
2	BS&H	23A0004T	Chemistry	3	0	0	3
3	Engineering Science	23A0002T	Differential Equations & Vector Calculus	3	0	0	3
4	Engineering Science	23A0101T	Basic Civil & Mechanical Engineering	3	0	0	3
5	Professional Core	23A0504T	Data structures	3	0	0	3
6	BS&H	2300010P	Communicative English Lab	0	0	2	1
7	BS&H	23A0007P	Chemistry Lab	0	0	2	1
8	Engineering Science	23A0302P	Engineering Workshop	0	0	3	1.5
9	Professional Core	23A0505P	Data structures Lab	0	0	3	1.5
10	BS&H	23AYG01P	Health and wellness, Yoga and Sports	-	--	1	0.5
TOTAL				14	00	11	19.5



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Department of Computer science & Engineering CSE (AIML)

B.Tech – I Year I Semester

ENGINEERING PHYSICS

(Common to all branches)

Course Code	L: T: P	Credits	Exam Marks	Exam Duration	Course Type
23A0003T	3: 0: 0	3	CIE: 30 EE:70	3 Hours	BS

Prerequisite: Student should know about fundamental and basic principles in physics

Course Objectives:

To bridge the gap between the Physics in school at 10+2 level and UG level engineering courses by identifying the importance of the optical phenomenon like interference, diffraction etc., enlightening the periodic arrangement of atoms in crystalline solids and concepts of quantum mechanics, introduce novel concepts of dielectric and magnetic materials, physics of semiconductors.

Course Outcomes:

On completion of this course, the students are able to

CO1: Analyze the intensity variation of light due to polarization, interference and diffraction.

CO2: Familiarize with the basics of crystals and their structures.

CO3: Summarize various types of polarization of dielectrics and classify the magnetic materials.

CO4: Apply fundamentals of quantum mechanics to band theory of solids.

CO5: Identify the type of semiconductor using Hall effect.

SYLLABUS		Total Hours:48
Unit - I	WAVE OPTICS	10
Interference: Introduction - Principle of superposition –Interference of light - Interference in thin films (Reflection Geometry) & applications - Colors in thin films- Newton's Rings- Determination of wavelength and refractive index. Diffraction: Introduction - Fresnel and Fraunhofer diffractions - Fraunhofer diffraction due to single slit, double slit & N-slits (Qualitative) – Diffraction Grating - Dispersive power and resolving power of Grating (Qualitative). Polarization: Introduction -Types of polarization - Polarization by reflection, refraction and Double refraction - Nicol's Prism -Half wave and Quarter wave plates		
Unit- II	CRYSTALLOGRAPHY AND X-RAY DIFFRACTION	8
Crystallography: Space lattice, Basis, Unit Cell and lattice parameters – Bravais Lattices – crystal systems (3D) – coordination number - packing fraction of SC, BCC & FCC - Miller indices – separation between successive(hkl) planes. X-ray diffraction: Bragg's law - X-ray Diffractometer – crystal structure determination by Laue's and powder methods		

Unit- III	DIELECTRIC AND MAGNETIC MATERIALS	10
Dielectric Materials: Introduction - Dielectric polarization - Dielectric polarizability, Susceptibility, Dielectric constant and Displacement Vector – Relation between the electric vectors - Types of polarizations- Electronic (Quantitative), Ionic (Quantitative) and Orientation polarizations (Qualitative) - Lorentz internal field - Clausius-Mossotti equation - complex dielectric constant – Frequency dependence of polarization – dielectric loss Magnetic Materials: Introduction - Magnetic dipole moment - Magnetization-Magnetic susceptibility and permeability – Atomic origin of magnetism - Classification of magnetic materials: Dia, para, Ferro, anti-ferro & Ferri magnetic materials - Domain concept for Ferromagnetism & Domain walls (Qualitative) - Hysteresis -soft and hard magnetic materials		
Unit- IV	QUANTUM MECHANICS AND FREE ELECTRON THEORY	10
Quantum Mechanics: Dual nature of matter – Heisenberg's Uncertainty Principle – Significance and properties of wave function – Schrodinger's time independent and dependent wave equations– Particle in a one-dimensional infinite potential well. Free Electron Theory: Classical free electron theory (Qualitative with discussion of merits and demerits) – Quantum free electron theory – electrical conductivity based on quantum free electron theory - Fermi-Dirac distribution Density of states - Fermi energy.		
Unit- V	SEMICONDUCTORS	10
Semiconductors: Formation of energy bands – classification of crystalline solids - Intrinsic semiconductors: Density of charge carriers – Electrical conductivity – Fermi level – Extrinsic semiconductors: density of charge carriers – dependence of Fermi energy on carrier concentration and temperature - Drift and diffusion currents – Einstein's equation - Hall effect and its applications. Superconductors- Introduction – Properties of superconductors – Meissner effect – Type I and Type II superconductors – BCS theory – High T_c superconductors– Applications of superconductors		

Textbooks:

1. A Text book of Engineering Physics - M. N. Avadhanulu, P.G.Kshirsagar & TVS Arun Murthy, S. Chand Publications, 11th Edition 2019.
2. Engineering Physics - D.K.Bhattacharya and Poonam Tandon, Oxford press (2015).
3. Engineering Physics – K. Thyagarajan, McGraw Hill Publishers

Reference Books:

1. Engineering Physics - B.K. Pandey and S. Chaturvedi, Cengage Learning
2. Engineering Physics - Shatendra Sharma, Jyotsna Sharma, Pearson Education, 2018.
3. Engineering Physics” - Sanjay D. Jain, D. Sahasrabudhe and Girish, University Press.
4. Engineering Physics - M.R. Srinivasan, New Age international publishers (2009).

E-resources:

- <https://www.textbooks.com/Catalog/MG5/Applied-Physics.php>
- https://edurev.in/courses/9596_Electromagnetic-Theory-Notes--Videos--MCQs--PPTs
- <https://libguides.ntu.edu.sg/c.php?g=867756&p=6226561>
- <https://bookauthority.org/books/best-applied-physics-books>
- <https://www.electronicsforu.com/resources/16-free-ebooks-on-material-science/2>

LINEAR ALGEBRA & CALCULUS
(Common to All Branches of Engineering)

Course Code	L: T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23A0001T	3:0:0:0	3	CIE:30 SEE:70	3Hours	BS

Course Objectives:

To equip the students with standard concepts and tools at an intermediate to advanced level mathematics to develop the confidence and ability among the students to handle various real-world problems and their applications.

Course Outcomes: At the end of the course, the student will be able to

CO1: Solving systems of linear equations that is needed by engineers for practical applications.

CO2: Find the eigen values and eigen vectors to facilitate the calculation of matrix characteristics.

CO3: Utilize mean value theorems to real life problems.

CO4: Apply the technique of partial differentiation to find the Jacobian and the extreme values of functions of several variables.

CO5: Apply the techniques of multiple integrals to find the areas and volumes.

SYLLABUS		Total Hours: 45
Unit-I	Matrices	9 Hrs
Rank of a matrix by echelon form, normal form. Cauchy–Binet formulae (without proof). Inverse of non-singular matrices by Gauss-Jordan method, System of linear equations: Solving system of Homogeneous and Non-Homogeneous equations - Gauss elimination method, Iteration Methods: Gauss- Jacobi and Gauss Seidel Iteration Methods. Applications: Finding the current in electrical circuits.		
Unit-II	Eigenvalues, Eigenvectors and Orthogonal Transformation	9 Hrs
Eigenvalues, Eigenvectors and their properties, Diagonalization of a matrix, Cayley-Hamilton Theorem (without proof), finding inverse and power of a matrix by Cayley-Hamilton Theorem, Quadratic forms and Nature of the Quadratic Forms, Reduction of Quadratic form to canonical forms by Orthogonal Transformation.		
Unit-III	Calculus	9 Hrs
Mean Value Theorems: Rolle’s Theorem (Without Proof), Lagrange’s mean value theorem (Without Proof)with their geometrical interpretation, Cauchy’s mean value theorem (Without Proof), Taylor’s and Maclaurintheorems with remainders (Without Proof), Problems and applications on the above theorems.		
Unit-IV	Partial differentiation and Applications (Multi variable calculus)	9 Hrs
Functions of several variables: Continuity and Differentiability, Partial derivatives, total derivatives, chain rule, Taylor’s and Maclaurin’s series expansion of functions of two variables. Jacobians, Functionaldependence, maxima and minima of functions of two variables, method of Lagrange multipliers.		

Unit-V	Multiple Integrals (Multi variable Calculus)	9 Hrs
Double integrals, triple integrals, change of order of integration (Cartesian Coordinate only), change of variables to polar, cylindrical and spherical coordinates. Finding areas (by double integrals) and volumes (by double integrals and triple integrals)		

Textbooks:

1. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, 2017, 44th Edition
2. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley & Sons, 2018, 10th Edition.

Reference Books:

1. Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition.
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint).
3. Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4. Advanced Engineering Mathematics, Micheael Greenberg, Pearson publishers, 9th edition
5. Higher Engineering Mathematics, H. K Das, Er. Rajnish Verma, S. Chand Publications, 2014, Third Edition (Reprint 2021)
6. Engineering Mathematics III by N.P. Bali, Dr. K.L. Sai Prasad, University Science Press.
7. Engineering Mathematics I by T.K.V. Iyengar, B. Krishna Gandhi, S. Chand Publications, 2019 Edition.
8. Mathematical Methods by T.K.V. Iyengar, B. Krishna Gandhi, S. Ranganatham and M.V.S.S.N. Prasad, S. Chand Publications.
9. Higher Engineering Mathematics, B. V. Ramana, McGraw Hill Education, 2017.

BASIC ELECTRICAL & ELECTRONICS ENGINEERING

(Common to All branches of Engineering)

Course Code	L: T:P	Credits	Exam marks	Exam Duration	Course Type
23A0201T	3:0:0	3	CIE:30 & SEE:70	3 Hours	PCC

Course Objectives:

The objectives of the course are to make the students learn about:

To expose to the field of electrical & electronics engineering, laws and principles of electrical/electronic engineering and to acquire fundamental knowledge in the relevant field.

Course Outcomes:

At the end of studying the course, the student should be able to:

CO1: Remember the fundamental laws, operating principles of motors, generators, MC and MI instruments.

CO2: Understand the problem-solving concepts associated to AC and DC circuits, construction and operation of AC and DC machines, measuring instruments; different power generation mechanisms, Electricity billing concept and important safety measures related to electrical operations.

CO3: Apply mathematical tools and fundamental concepts to derive various equations related to machines, circuits and measuring instruments; electricity bill calculations and layout representation of electrical power systems.

CO4: Analyze different electrical circuits, performance of machines and measuring instruments.

CO5: Evaluate different circuit configurations, Machine performance and Power systems operation.

PART A: BASIC ELECTRICAL ENGINEERING

SYLLABUS

		48Hrs
Unit-I	DC & AC Circuits	10 Hrs
DC Circuits: Electrical circuit elements (R, L and C), Ohm’s Law and its limitations, KCL & KVL, series,parallel, series-parallel circuits, Super Position theorem, Simple numerical problems.		
AC Circuits: A.C. Fundamentals: Equation of AC Voltage and current, waveform, time period, frequency, amplitude, phase, phase difference, average value, RMS value, form factor, peak factor, Voltage and current relationship with phasor diagrams in R, L, and C circuits, Concept of Impedance, Active power, reactive powerand apparent power, Concept of power factor (Simple Numerical problems).		
Unit-II	Machines and Measuring Instruments	8 Hrs
Machines: Construction, principle and operation of (i) DC Motor, (ii) DC Generator, (iii) Single PhaseTransformer, (iv) Three Phase Induction Motor and (v) Alternator, Applications of electrical machines.		
Measuring Instruments: Construction and working principle of Permanent Magnet Moving Coil (PMMC),Moving Iron (MI) Instruments and Wheat Stone bridge.		
Unit -III	Energy Resources, Electricity Bill & Safety Measures	6 Hrs

Energy Resources: Conventional and non-conventional energy resources; Layout and operation of various Power Generation systems: Hydel, Nuclear, Solar & Wind power generation.

Electricity bill: Power rating of household appliances including air conditioners, PCs, Laptops, Printers, etc.

Definition of “unit” used for consumption of electrical energy, two-part electricity tariff, calculation of electricity bill for domestic consumers.

Equipment Safety Measures: Working principle of Fuse and Miniature circuit breaker (MCB), merits and demerits.

Personal safety measures: Electric Shock, Earthing and its types, Safety Precautions to avoid shock.

Text Books:

1. Basic Electrical Engineering, D. C. Kulshreshtha, Tata McGraw Hill, 2019, First Edition
2. Power System Engineering, P.V. Gupta, M.L. Soni, U.S. Bhatnagar and A. Chakrabarti, Dhanpat Rai & Co, 2013
3. Fundamentals of Electrical Engineering, Rajendra Prasad, PHI publishers, 2014, Third Edition

Reference Books:

1. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, Mc Graw Hill, 2019, Fourth Edition
2. Principles of Power Systems, V.K. Mehtha, S.Chand Technical Publishers, 2020
3. Basic Electrical Engineering, T. K. Nagsarkar and M. S. Sukhija, Oxford University Press, 2017
4. Basic Electrical and Electronics Engineering, S. K. Bhattacharya, Person Publications, 2018, Second Edition.

Web Resources:

1. <https://nptel.ac.in/courses/108105053>
2. <https://nptel.ac.in/courses/108108076>

PART B: BASIC ELECTRONICS ENGINEERING

Course Objectives:

The objectives of the course are to make the students learn about:

This course provides the student with the fundamental skills to understand the principles of digital electronics, basics of semiconductor devices like diodes & transistors, characteristics and its applications.

Course Outcomes (CO):

At the end of studying the course, the student should be able to:

CO1: Apply the concept of science and mathematics to understand the working of diodes, transistors, and their applications.

CO2: Explain the characteristics of diodes and transistors.

CO3: Familiarize with the number systems, codes, Boolean algebra and logic gates.

CO4: Understand the working mechanism of different combinational, sequential circuits and their role in the digital systems

SYLLABUS

Unit-I	SEMICONDUCTOR DEVICES	6 Hrs
Introduction - Evolution of electronics – Vacuum tubes to nano electronics - Characteristics of PN Junction Diode — Zener Effect — Zener Diode and its Characteristics. Bipolar Junction Transistor — CB, CE, CC Configurations and Characteristics — Elementary Treatment of Small Signal CE Amplifier		
Unit-II	BASIC ELECTRONIC CIRCUITS AND INSTRUMENTATION	10 Hrs
Rectifiers and power supplies: Block diagram description of a dc power supply, working of a full wave bridge rectifier, capacitor filter (no analysis), working of simple zener voltage regulator. Amplifiers: Block diagram of Public Address system, Circuit diagram and working of common emitter (RC coupled) amplifier with its frequency response. Electronic Instrumentation: Block diagram of an electronic instrumentation system.		
Unit -III	DIGITAL ELECTRONICS	8 Hrs
Overview of Number Systems, Logic gates including Universal Gates, BCD codes, Excess-3 code, Gray code, Hamming code. Boolean Algebra, Basic Theorems and properties of Boolean Algebra, Truth Tables and Functionality of Logic Gates – NOT, OR, AND, NOR, NAND, XOR and XNOR. Simple combinational circuits–Half and Full Adder, Introduction to sequential circuits, Flip flops, Registers and counters (Elementary Treatment only).		

Textbooks:

1. R.L. Boylestad & Louis Nashlesky, Electronic Devices & Circuit Theory, Pearson Education, 2021.
2. R. P. Jain, Modern Digital Electronics, 4th Edition, Tata McGraw Hill, 2009

Reference Books:

1. R. S. Sedha, A Textbook of Electronic Devices and Circuits, S. Chand & Co, 2010.
2. Santiram Kal, Basic Electronics- Devices, Circuits and IT Fundamentals, Prentice Hall, India, 2002.
3. R. T. Paynter, Introductory Electronic Devices & Circuits – Conventional Flow Version, Pearson Education, 2009

ENGINEERING GRAPHICS (Common to All Branches of Engineering)					
Course Code	L: T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23A0301T	3:0:0:0	3	CIE:30 SEE:70	3Hours	BS

Course Objectives:

The students completing the course are expected to:

- Understand the basic principles and conventions of engineering drawing, use engineering instruments and draw engineering curves.
- Use orthographic projections and make the students draw the projections of lines and planes inclined to both the planes.
- Draw the projections of the solids in different positions with respect to the reference planes.
- Understand the importance of sectioning and concept of development of surfaces.
- Represent and convert isometric views to orthographic views and vice versa.

Course Outcomes:

On completion of the course, the student should be able to:

CO1: Understand the principles of engineering drawing, including engineering curves, scales, orthographic and isometric projections.

CO2: Draw and interpret orthographic projections of points, lines, planes and solids in front, top and side views.

CO3: Understand and apply concepts of sectional views to represent details of solids in simple positions.

CO4: Gain a clear understanding of the principles behind development of surfaces and to understand how to unfold basic geometric shapes into flat patterns.

CO5: Develop the ability to draw isometric views and orthographic views and should be able to convert isometric views to orthographic views and vice versa.

SYLLABUS		Total Hours: 45
Unit-I	Introduction	9 Hrs
Lines, Lettering and Dimensioning, Geometrical Constructions and Constructing regular polygons by general methods. Curves: construction of ellipse, parabola and hyperbola by general, Cycloids, Involute, Normal and tangent to Curves. Scales: Plain scales, diagonal scales and vernier scales.		
Unit-II	Orthographic Projections, Projections of Straight lines, Projections of Planes	9 Hrs

Orthographic Projections: Reference plane, importance of reference lines or Plane, Projections of a point situated in any one of the four quadrants.

Projections of Straight Lines: Projections of straight lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane. Projections of Straight Line Inclined to both the reference planes

Projections of Planes: regular planes Perpendicular to both reference planes, parallel to one reference plane and inclined to the other reference plane; plane inclined to both the reference planes.

Unit-III	Projections of Solids	9 Hrs
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Projections of Solids: Types of solids: Polyhedra and Solids of revolution. Projections of solids in simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical plane and Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to other and axes inclined to both the reference planes.

Unit-IV	Sections of Solids, Development of Surfaces	9 Hrs
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Sections of Solids: Perpendicular and inclined section planes, Sectional views and True shape of section, Sections of solids in simple position only.

Development of Surfaces: Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

Unit-V	Conversion of Views, Computer Graphics	9 Hrs
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Conversion of Views: Conversion of isometric views to orthographic views; Conversion of orthographic views to isometric views.

Computer graphics: Creating 2D & 3D drawings of objects including PCB and Transformations using Auto CAD (*Not for end examination*).

Textbooks:

1. N. D. Bhatt, Engineering Drawing, Charotar Publishing House, 2016.

Reference Books:

4. Engineering Drawing, K.L. Narayana and P. Kannaiah, Tata McGraw Hill, 2013.
5. Engineering Drawing, M.B. Shah and B.C. Rana, Pearson Education Inc, 2009.
6. Engineering Drawing with an Introduction to AutoCAD, Dhananjay Jolhe, Tata McGraw Hill, 2017.

INTRODUCTION TO PROGRAMMING
(Common to All Branches of Engineering)

Course Code	L: T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23A0301T	3:0:0:0	3	CIE:30 SEE:70	3Hours	BS

Course Objectives:

- To introduce students to the fundamentals of computer programming. To provide hands-on experience with coding and debugging.
- To foster logical thinking and problem-solving skills using programming.
- To familiarize students with programming concepts such as data types, control structures, functions, and arrays.
- To encourage collaborative learning and teamwork in coding projects.

Course Outcomes:

CO1: Understand basics of computers, the concept of algorithm and algorithmic thinking.

CO2: Analyze a problem and develop an algorithm to solve it.

CO3: Implement various algorithms using the C programming language.

CO4: Understand more advanced features of C language.

CO5: Develop problem-solving skills and the ability to debug and optimize the code.

SYLLABUS		Total Hours: 45
Unit-I	Introduction to Programming and Problem Solving	9 Hrs
History of Computers, Basic organization of a computer: ALU, Input-Output units, Memory, Program Counter, Introduction to Programming Languages, Basics of a Computer Program, Algorithms, Flowcharts (Using Dia Tool), Pseudo code. Introduction to Compilation and Execution, Primitive DataTypes, Variables, and Constants, Basic Input and Output, Operations, Type Conversion, and Casting.Problem solving techniques: Algorithmic approach, Characteristics of algorithm, Problem Solving Strategies: Top-down approach, Bottom-up approach, Time and space complexities of algorithms.		
Unit-II	Control Structures	9 Hrs
Simple sequential programs Conditional Statements (if, if-else, switch), Loops (for, while, do while) Break and Continue.		
Unit-III	Arrays and Strings	9 Hrs
Arrays indexing, memory model, programs with array of integers, two dimensional arrays, Introduction to Strings.		
Unit-IV	Pointers & User Defined Data types	9 Hrs

Pointers, dereferencing and address operators, pointer and address arithmetic, array manipulation using pointers, User-defined data types-Structures and Unions.		
Unit-V	Functions & File Handling	9 Hrs
Introduction to Functions, Function Declaration and Definition, Function call Return Types and Arguments, modifying parameters inside functions using pointers, arrays as parameters. Scope and Lifetime of Variables, Command Line Arguments , Preprocessor Directives , Basics of File Handling		

Textbooks:

1. "The C Programming Language", Brian W. Kernighan and Dennis M. Ritchie, Prentice Hall, 1988
2. "Programming with C", Byron S Gottfried, McGraw-Hill Education, 1996

Reference Books:

1. Computing fundamentals and C Programming, Balagurusamy, E., McGraw-Hill Education, 2008.
2. Programming in C, Rema Theraja, Oxford, 2016, 2nd edition
3. C Programming, A Problem-Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE, 3rd edition

IT WORKSHOP
(Common to All Branches of Engineering)

Course Code	L: T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23A0503P	3:0:0:0	3	CIE:30 SEE:70	3Hours	ES

Course Objectives:

- To introduce the internal parts of a computer, peripherals, I/O ports, connecting cables
- To demonstrate configuring the system as Dual boot both Windows and other Operating Systems Viz. Linux, BOSS
- To teach basic command line interface commands on Linux.
- To teach the usage of Internet for productivity and self-paced life-long learning
- To introduce Compression, Multimedia and Antivirus tools and Office Tools such as Wordprocessors, Spread sheets and Presentation tools.

Course Outcomes:

CO1: Perform Hardware troubleshooting.

CO2: Understand Hardware components and inter dependencies.

CO3: Safeguard computer systems from viruses/worms.

CO4: Document/ Presentation preparation.

CO5: Perform calculations using spreadsheets. PC Hardware & Software Installation

List of Experiments

Task 1:

Identify the peripherals of a computer, components in a CPU and its functions. Draw the block diagram of the CPU along with the configuration of each peripheral and submit to your instructor.

Task 2:

Every student should disassemble and assemble the PC back to working condition. Lab instructors should verify the work and follow it up with a Viva. Also, students need to go through the video which shows the process of assembling a PC. A video would be given as part of the course content.

Task 3:

Every student should individually install MS windows on the personal computer. Lab instructors should verify the installation and follow it up with a Viva.

Task 4:

Every student should install Linux on the computer. This computer should have windows installed. The system should be configured as dual boot (VMWare) with both Windows and Linux. Lab instructors should verify the installation and follow it up with a Viva

Task 5:

Every student should install BOSS on the computer. The system should be configured as dual boot (VMWare) with both Windows and BOSS. Lab instructors should verify the installation and follow it up with a Viva

Internet & World Wide Web**Task 1:**

Orientation & Connectivity Boot Camp: Students should get connected to their Local Area Network and access the Internet. In the process they configure the TCP/IP setting. Finally, students should demonstrate, to the instructor, how to access the websites and email. If there is no internet connectivity preparations need to be made by the instructors to simulate the WWW on the LAN.

Task 2:

Web Browsers, Surfing the Web: Students customize their web browsers with the LAN proxy settings, bookmarks, search toolbars and pop-up blockers. Also, plug-ins like Macromedia Flash and JRE for applets should be configured.

Task 3:

Search Engines & Netiquette: Students should know what search engines are and how to use the search engines. A few topics would be given to the students for which they need to search on Google. This should be demonstrated to the instructors by the student.

Task 4:

Cyber Hygiene: Students would be exposed to the various threats on the internet and would be asked to configure their computer to be safe on the internet. They need to customize their browsers to block pop ups, block active x downloads to avoid viruses and/or worms.

Task 5:

Install any anti-virus software on your computer

LaTeX and WORD

Task 1 – Word Orientation: The mentor needs to give an overview of LaTeX and Microsoft (MS) office or equivalent (FOSS) tool word: Importance of LaTeX and MS office or equivalent (FOSS) tool Word as word Processors, Details of the four tasks and features that would be covered in each, Using LaTeX and word – Accessing, overview of toolbars, saving files, Using help and resources, rulers, format painter in word.

Task 2:

Using LaTeX and Word to create a project certificate. Features to be covered: - Formatting Fonts in word, Drop Cap in word, Applying Text effects, Using Character Spacing, Borders and Colors, Inserting Header and Footer, Using Date and Time option in both LaTeX and Word.

Task 3:

Creating project abstract Features to be covered: -Formatting Styles, inserting table, Bullets and Numbering, Changing Text Direction, Cell alignment, Footnote, Hyperlink, Symbols, Spell Check, Track Changes.

Task 4:

Creating a Newsletter: Features to be covered: - Table of Content, Newspaper columns, Images from files and clipart, Drawing toolbar and Word Art, Formatting Images, Textboxes, Paragraphs and Mail Merge in word.

EXCEL**Excel Orientation:**

The mentor needs to tell the importance of MS office or equivalent (FOSS) tool Excel as a Spreadsheet tool, give the details of the four tasks and features that would be covered in each. Using Excel – Accessing, overview of toolbars, saving excel files, Using help and resources.

Task 1:

Creating a Scheduler - Features to be covered: Gridlines, Format Cells, Summation, auto fill, Formatting Text

Task 2:

Calculating GPA -. Features to be covered: - Cell Referencing, Formulae in excel – average, std. deviation, Charts, Renaming and Inserting worksheets, Hyper linking, Count function

LOOKUP/VLOOKUP**Task 3:**

Split cells, freeze panes, group and outline, Sorting, Boolean and logical operators, Conditional formatting

POWER POINT**Task 1:**

Students will be working on basic power point utilities and tools which help them create basic power point presentations. PPT Orientation, Slide Layouts, Inserting Text, Word Art, Formatting Text, Bullets and Numbering, Auto Shapes, Lines and Arrows in PowerPoint.

Task 2:

Interactive presentations - Hyperlinks, Inserting –Images, Clip Art, Audio, Video, Objects, Tables and Charts.

AI TOOLS – ChatGPT**Task 1:**

Prompt Engineering: Experiment with different types of prompts to see how the model responds. Try asking questions, starting conversations, or even providing incomplete sentences to see how the model completes them.

Ex: Prompt: "You are a knowledgeable AI. Please answer the following question: What is the capital of France?"

Task 2:

Creative Writing: Use the model as a writing assistant. Provide the beginning of a story or a description of a scene, and let the model generate the rest of the content. This can be a fun way to brainstorm creative ideas

Ex: Prompt: "In a world where gravity suddenly stopped working, people started floating upwards. Write a story about how society adapted to this new reality."

Task 3:

Language Translation: Experiment with translation tasks by providing a sentence in one language and asking the model to translate it into another language. Compare the output to see how accurate and fluent the translations are.

Ex: Prompt: "Translate the following English sentence to French: 'Hello, how are you doing today?'"

Reference Books:

1. Comdex Information Technology course tool kit, Vikas Gupta, WILEY Dream tech, 2003
2. The Complete Computer upgrade and repair book, Cheryl A Schmidt, WILEY Dream tech, 2013, 3rd edition
3. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education, 2012, 2nd edition
4. PC Hardware - A Handbook, Kate J. Chase, PHI (Microsoft)
5. LaTeX Companion, Leslie Lamport, PHI/Pearson.
6. IT Essentials PC Hardware and Software Companion Guide, David Anfinson and Ken Quamme. – CISCO Press, Pearson Education, 3rd edition
7. IT Essentials PC Hardware and Software Labs and Study Guide, Patrick Regan – CISCO Press, Pearson Education, 3rd edition

ENGINEERING PHYSICS LAB**(Common to All Branches of Engineering)**

Course Code	L : T : P	Credits	Exam Marks	Exam Duration	Course Type
23A0006P	0 : 0 : 2	1	CIE: 30 EE:70	3 Hours	BS

Prerequisite: Student should know about fundamental and basic principles in physics**Course Objectives:**

To study the concepts of optical phenomenon like interference, diffraction etc., recognize the importance of energy gap in the study of conductivity and Hall effect in semiconductors and study the parameters and applications of dielectric and magnetic materials by conducting experiments.

Course Outcomes:

On completion of this course, the students are able to: -

CO1: Operate optical instruments like travelling microscope and spectrometer.

CO2: Estimate dielectric constant of capacitor and magnetic induction of current carrying coil

CO3: Identify the type of semiconductor and calculate band gap of it.

CO4: Evaluate different modulus of materials.

CO5: Measure the frequency of tuning fork and verify the laws in Sonometer.

SYLLABUS**Total Hours: 32****List of Experiments**

1. Determination of radius of curvature of a given plano convex lens by Newton's rings.
2. Determination of wavelengths of different spectral lines in mercury spectrum using diffraction grating in normal incidence configuration.
3. Verification of Brewster's law
4. Determination of wavelength of Laser light using diffraction grating.
5. Estimation of Planck's constant using photoelectric effect.
6. Magnetic field along the axis of a current carrying circular coil by Stewart Gee's Method.
7. Determination of dielectric constant using charging and discharging method.
8. Study the variation of B versus H by magnetizing the magnetic material (B-H curve).
9. Determination of magnetic susceptibility by Kundt's tube method.
10. Determination of the resistivity of semiconductors by four probe methods.
11. Determination of energy gap of a semiconductor using p-n junction diode.
12. Determination of Hall voltage and Hall coefficient of a given semiconductor using Hall effect.
13. Determination of temperature coefficients of a thermistor.

14. Determination of rigidity modulus of the material of the given wire using Torsional pendulum.
15. Determination of young's modulus for the given material of wooden scale by non-uniform bending(or double cantilever) method.
16. Determination of Frequency of electrically maintained tuning fork by Melde's experiment.
17. Sonometer: Verification of laws of stretched string.
18. Determination of acceleration due to gravity and radius of Gyration by using a compound pendulum.

Note: Any TEN of the listed experiments are to be conducted. Out of which any TWO experiments may be conducted in virtual mode.

References: A Textbook of Practical Physics - S. Balasubramanian, M.N. Srinivasan, S. Chand Publishers, 2017.

URL:www.vlab.co.in

ELECTRICAL & ELECTRONICS ENGINEERING WORKSHOP

(Common to All branches of Engineering)

Course Code	L: T:P	Credit s	Exam marks	Exam Duration	Course Type
23A0202P	0:0:3	1.5	CIE:30 &SEE:70	3 Hours	PCC

Course Objectives:

This course will enable students to:

1. To impart knowledge on the fundamental laws & theorems of electrical circuits, functions of electrical machines and energy calculations

Course Outcomes (CO):

At the end of the course, the student will be able to:

CO1: Understand the Electrical circuit design concept; measurement of resistance, power, power factor; concept of wiring and operation of Electrical Machines and Transformer.

CO2: Apply the theoretical concepts and operating principles to derive mathematical models for circuits, Electrical machines and measuring instruments; calculations for the measurement of resistance, power and power factor.

CO3: Apply the theoretical concepts to obtain calculations for the measurement of resistance, power and power factor.

CO4: Analyze various characteristics of electrical circuits, electrical machines and measuring instruments.

CO5: Design suitable circuits and methodologies for the measurement of various electrical parameters; Household and commercial wiring.

PART A: ELECTRICAL ENGINEERING LAB

List of experiments:

1. Verification of KCL and KVL
2. Verification of Superposition theorem
3. Measurement of Resistance using Wheat stone bridge
4. Magnetization Characteristics of DC shunt Generator
5. Measurement of Power and Power factor using Single-phase wattmeter
6. Measurement of Earth Resistance using Megger
7. Calculation of Electrical Energy for Domestic Premises.

Reference Books:

1. Basic Electrical Engineering, D. C. Kulshreshtha, Tata McGraw Hill, 2019, First Edition
2. Power System Engineering, P.V. Gupta, M.L. Soni, U.S. Bhatnagar and A. Chakrabarti,

Dhanpat Rai & Co, 2013

3. Fundamentals of Electrical Engineering, Rajendra Prasad, PHI publishers, 2014, Third Edition

Note: Minimum Six Experiments to be performed

PART B: ELECTRONICS ENGINEERING LAB

Course Objectives:

- To impart knowledge on the principles of digital electronics and fundamentals of electron devices & its applications

Course Outcomes (CO):

At the end of the course, the student will be able to

CO1: Identify & testing of various electronic components.

CO2: Understand the usage of electronic measuring instruments.

CO3: Plot and discuss the characteristics of various electron devices. **CO4:** Explain the operation of a digital circuit.

List of Experiments:

1. Plot V-I characteristics of PN Junction diode A) Forward bias B) Reverse bias.
2. Plot V – I characteristics of Zener Diode and its application as voltage Regulator.
3. Implementation of half wave and full wave rectifiers
4. Plot Input & Output characteristics of BJT in CE and CB configurations
5. Frequency response of CE amplifier.
6. Simulation of RC coupled amplifier with the design supplied
7. Verification of Truth Table of AND, OR, NOT, NAND, NOR, Ex-OR, Ex-NOR gates using ICs.
8. Verification of Truth Tables of S-R, J-K & D flip flops using respective ICs.

Tools / Equipment Required: DC Power supplies, Multi meters, DC Ammeters, DC Voltmeters, AC Voltmeters, CROs, all the required active devices.

Reference Books:

1. R. L. Boylestad & Louis Nashlesky, Electronic Devices & Circuit Theory, Pearson Education, 2021.
2. R. P. Jain, Modern Digital Electronics, 4th Edition, Tata Mc Graw Hill, 2009
3. R. T. Paynter, Introductory Electronic Devices & Circuits – Conventional Flow Version, Pearson Education, 2009.

Note: Minimum Six Experiments to be performed. All the experiments shall be implemented using both Hardware and Software.

COMPUTER PROGRAMMING LAB**(Common to All Branches of Engineering)**

Course Code	L: T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23A0502P	3:0:0:0	3	CIE:30 SEE:70	3Hours	ES

Course Objectives:

The course aims to give students hands – on experience and train them on the concepts of the C-programming language.

Course Outcomes:

CO1: Read, understand, and trace the execution of programs written in C language.

CO2: Select the right control structure for solving the problem.

CO3: Develop C programs which utilize memory efficiently using programming constructs like pointers.

CO4: Develop, Debug and Execute programs to demonstrate the applications of arrays, functions, basic concepts of pointers in C.

List of Experiments**WEEK 1**

Objective: Getting familiar with the programming environment on the computer and writing the first program.

Suggested Experiments/Activities:

Tutorial 1: Problem-solving using Computers.

Lab1: Familiarization with programming environment

Basic Linux environment and its editors like Vi, Vim & Emacs etc.

Exposure to Turbo C, gcc

Writing simple programs using printf(), scanf()

WEEK 2

Objective: Getting familiar with how to formally describe a solution to a problem in a series of finite steps both using textual notation and graphic notation.

Suggested Experiments /Activities:

Tutorial 2: Problem-solving using Algorithms and Flow charts.

Lab1: Converting algorithms/flow charts into C Source code. Developing the algorithms/flowcharts for the following sample programs

- i. Sum and average of 3 numbers
- ii. Conversion of Fahrenheit to Celsius and vice versa
- iii. Simple interest calculation

WEEK 3

Objective: Learn how to define variables with the desired data-type, initialize them with appropriate values and how arithmetic operators can be used with variables and constants.

Suggested Experiments/Activities:

Tutorial 3: Variable types and type conversions:

Lab 3: Simple computational problems using arithmetic expressions.

- i. Finding the square root of a given number
- ii. Finding compound interest
- iii. Area of a triangle using heron's formulae
- iv. Distance travelled by an object

WEEK 4

Objective: Explore the full scope of expressions, type-compatibility of variables & constants and operators used in the expression and how operator precedence works.

Suggested Experiments/Activities:

Tutorial 4: Operators and the precedence and as associativity:

Lab 4: Simple computational problems using the operator' precedence and associativity

- i. Evaluate the following expressions. a. $A+B*C+(D*E) + F*G$ b. $A/B*C-B+A*D/3$ c. $A+++B---$
A d. $J = (i++) + (++i)$
- ii. Find the maximum of three numbers using conditional operator
- iii. Take marks of 5 subjects in integers, and find the total, average in float

WEEK 5

Objective: Explore the full scope of different variants of "if construct" namely if-else, null else, if-else if*-else, switch and nested-if including in what scenario each one of them can be used and how to use them. Explore all relational and logical operators while writing conditionals for "if construct".

Suggested Experiments/Activities:

Tutorial 5: Branching and logical expressions

Lab 5: Problems involving if-then-else structures.

- i) Write a C program to find the max and min of four numbers using if-else.
- ii) Write a C program to generate electricity bill.
- iii) Find the roots of the quadratic equation.
- iv) Write a C program to simulate a calculator using switch case.

- iv) Write a C program to find the given year is a leap year or not.

WEEK 6

Objective: Explore the full scope of iterative constructs namely while loop, do-while loop and for loop in addition to structured jump constructs like break and continue including when each of these statements is more appropriate to use.

Suggested Experiments/Activities:

Tutorial 6: Loops, while and for loops

Lab 6: Iterative problems e.g., the sum of series

- i) Find the factorial of given number using any loop.
- ii) Find the given number is a prime or not.
- iii) Compute sine and cos series
- iv) Checking a number palindrome
- v) Construct a pyramid of numbers.

WEEK 7:

Objective: Explore the full scope of Arrays construct namely defining and initializing 1-D and 2-D and more generically n-D arrays and referencing individual array elements from the defined array. Using integer 1-D arrays, explore search solution linear search.

Suggested Experiments/Activities:

Tutorial 7: 1 D Arrays: searching.

Lab 7: 1D Array manipulation, linear search

Find the min and max of a 1-D integer array.

Perform linear search on 1D array.

The reverse of a 1D integer array

Find 2's complement of the given binary number.

Eliminate duplicate elements in an array.

WEEK 8:

Objective: Explore the difference between other arrays and character arrays that can be used as Strings by using null character and get comfortable with string by doing experiments that will reverse a string and concatenate two strings. Explore sorting solution bubble sort using integer arrays.

Suggested Experiments/Activities:

Tutorial 8: 2 D arrays, sorting and Strings.

Lab 8: Matrix problems, String operations, Bubble sort

- i) Addition of two matrices
- ii) Multiplication two matrices
- iii) Sort array elements using bubble sort
- iv) Concatenate two strings without built-in functions
- v) Reverse a string using built-in and without built-in string functions

WEEK 9:

Objective: Explore pointers to manage a dynamic array of integers, including memory allocation & value initialization, resizing changing and reordering the contents of an array

and memory de-allocation using malloc (), calloc (), realloc () and free () functions. Gain experience processing command-line arguments received by C

Suggested Experiments/Activities:

Tutorial 9: Pointers, structures and dynamic memory allocation

Lab 9: Pointers and structures, memory dereference.

- i) Write a C program to find the sum of a 1D array using malloc ()
- ii) Write a C program to find the total, average of n students using structures
- iii) Enter n students data using calloc () and display failed students list
- iv) Read student name and marks from the command line and display the student details alongwith the total. v) Write a C program to implement realloc ()

WEEK 10:

Objective: Experiment with C Structures, Unions, bit fields and self-referential structures (Singly linked lists) and nested structures

Suggested Experiments/Activities:

Tutorial 10: Bitfields, Self-Referential Structures, Linked lists

Lab10: Bitfields, linked lists Read and print a date using dd/mm/yyyy format using bit-fields and differentiate the same without using bit- fields

- i) Create and display a singly linked list using self-referential structure.
- ii) Demonstrate the differences between structures and unions using a C program.
- iii) Write a C program to shift/rotate using bitfields.
- iv) Write a C program to copy one structure variable to another structure of the same type.

WEEK 11:

Objective: Explore the Functions, sub-routines, scope and extent of variables, doing some experiments by parameter passing using call by value. Basic methods of numerical integration

Suggested Experiments/Activities:

Tutorial 11: Functions, call by value, scope and extent,

Lab 11: Simple functions using call by value, solving differential equations using Euler's theorem.

Write a C function to calculate NCR value.

Write a C function to find the length of a string.

Write a C function to transpose of a matrix.

Write a C function to demonstrate numerical integration of differential equations using Euler's method

WEEK 12:

Objective: Explore how recursive solutions can be programmed by writing recursive functions that can be invoked from the main by programming at-least five distinct problems that have naturally recursive solutions.

Suggested Experiments/Activities:

Tutorial 12: Recursion, the structure of recursive calls

Lab 12: Recursive functions

Write a recursive function to generate Fibonacci series.

Write a recursive function to find the lcm of two numbers.

Write a recursive function to find the factorial of a number.

Write a C Program to implement Ackermann function using recursion.

Write a recursive function to find the sum of series.

WEEK 13:

Objective: Explore the basic difference between normal and pointer variables, Arithmetic operations using pointers and passing variables to functions using pointers

Suggested Experiments/Activities:

Tutorial 13: Call by reference, dangling pointers

Lab 13: Simple functions using Call by reference, Dangling pointers.

- i) Write a C program to swap two numbers using call by reference.
- ii) Demonstrate Dangling pointer problem using a C program.
- iii) Write a C program to copy one string into another using pointer.
- iv) Write a C program to find no of lowercase, uppercase, digits and other characters using pointers.

WEEK14:

Objective: To understand data files and file handling with various file I/O functions. Explore the differences between text and binary files.

Suggested Experiments/Activities:**Tutorial 14:** File handling**Lab 14:** File operations

- i) Write a C program to write and read text into a file.
- ii) Write a C program to write and read text into a binary file using fread () and fwrite ()
- iii) Copy the contents of one file to another file.
- iv) Write a C program to merge two files into the third file using command-line arguments.
- v) Find no. of lines, words and characters in a file
- vi) Write a C program to print last n characters of a given file.

Textbooks:

1. Ajay Mittal, Programming in C: A practical approach, Pearson.
2. Byron Gottfried, Schaum' s Outline of Programming with C, McGraw Hill

Reference Books:

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India
2. C Programming, A Problem-Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE

NSS/NCC/SCOUTS GUIDES/COMMUNITY SERVICE**(Common to all branches of Engineering)**

Course Code	L: T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23ANS01P	0:0:1:0	0.5	CIE:30 SEE:70	3Hours	BS

Course Objectives:

The objective of introducing this course is to impart discipline, character, fraternity, teamwork, social consciousness among the students and engaging them in selfless service.

Course Outcomes: At the end of the course, the students will be able to

CO1: Understand the importance of discipline, character and service motto.

CO2: Solve some societal issues by applying acquired knowledge, facts, and techniques.

CO3: Explore human relationships by analyzing social problems.

CO4: Determine to extend their help for the fellow beings and downtrodden people.

CO5: Develop leadership skills and civic responsibilities.

SYLLABUS**TotalHours:45****Unit-I****Orientation****9Hrs**

General Orientation on NSS/NCC/ Scouts & Guides/Community Service activities, career guidance.

Activities:

- i) Conducting –ice breaking sessions-expectations from the course-knowing personal talents and skills
- ii) Conducting orientations programs for the students –future plans-activities-releasing road map etc.
- iii) Displaying success stories-motivational biopics- award winning movies on societal issues etc.
- iv) Conducting talent show in singing patriotic songs-paintings- any other contribution.

Unit-II**Nature & Care****9Hrs****Activities:**

- i) Best out of waste competition.

- ii) Poster and signs making competition to spread environmental awareness.
- iii) Recycling and environmental pollution article writing competition.
- iv) Organising Zero-waste day.
- v) Digital Environmental awareness activity via various social media platforms.
- vi) Virtual demonstration of different eco-friendly approaches for sustainable living.
- vii) Write a summary on any book related to environmental issues.

Unit-III	Community Service	9Hrs
Activities: i) Conducting One Day Special Camp in a village contacting village-area leaders- Survey in the village, identification of problems- helping them to solve via media authorities- experts-etc. ii) Conducting awareness programs on Health-related issues such as General Health, Mental health, Spiritual Health, HIV/AIDS, iii) Conducting consumer Awareness. Explaining various legal provisions etc. iv) Women Empowerment Programmes- Sexual Abuse, Adolescent Health and Population Education. v) Any other programmes in collaboration with local charities, NGOs etc.		
Reference Books: 1. Nirmalya Kumar Sinha & Surajit Majumder, <i>A Text Book of National Service Scheme</i> Vol. I, Vidya Kutir Publication, 2021 (ISBN 978-81-952368-8-6) 2. <i>Red Book - National Cadet Corps</i> – Standing Instructions Vol I & II, Directorate General of NCC, Ministry of Defence, New Delhi 3. Davis M. L. and Cornwell D. A., “Introduction to Environmental Engineering”, McGraw Hill, New York 4/e 2008 4. Masters G. M., Joseph K. and Nagendran R. “Introduction to Environmental Engineering and Science”, Pearson Education, New Delhi. 2/e 2007 5. Ram Ahuja. <i>Social Problems in India</i>, Rawat Publications, New Delhi		

B. Tech-I-Year-II-Semester (GROUP-B)

COMMUNICATIVE ENGLISH (Common to all Branches of Engineering)					
Course Code	L:T: P: S	Credits	Exam marks	Exam Duration	Course Type
230009T	2: 0: 0: 0	2	CIE:30 SEE:70	3 Hours	HS
Course Objectives: - Facilitate effective listening skills for better comprehension of academic lectures and English spoken by native speakers - Help improve speaking skills motivating the learners to participate in activities such as role plays, discussions and structured talks/oral presentations - Focus on appropriate reading skills for comprehension of various academic texts and authentic materials - Impart effective strategies for good writing skills in summarizing, writing well organized essays, drafting formal letters and designing well-structured reports Broaden the knowledge base of grammatical structures and vocabulary and encourage their appropriate use in speech and writing					
Course Outcomes: On completion of this course, student will be able to CO1: The learner will be able to speak and write grammatically accurate sentences through applications of principles of English grammar CO2: The learner will enhance vocabulary skills to build strong language skills. CO3: The learner acquires the ability to understand the academic text from multiple dimensions employing ethical and logical reasoning based on accurate comprehension CO4: The learner gains evaluation potential by employing standard reading & listening strategies to grasp the core essence and spirit of the text CO5: The learner will gain mastery on speaking & writing skills through the application of relevant guidelines, through consistent practice of functional English expression					
SYLLABUS					Total Hours:32
Module - I	HUMAN VALUES: Gift of Magi (Short Story)				8 Hrs

Listening: Identifying the topic, the context and specific pieces of information by listening to short audio texts and answering a series of questions.
Speaking: Asking and answering general questions on familiar topics such as home, family, work, studies and interests; introducing oneself and others.
Reading: Skimming to get the main idea of a text Scanning to look for specific pieces of information.
Writing: Mechanics of Writing-Capitalization, Spellings, Punctuation-Parts of Sentences.
Grammar: Parts of Speech, Basic Sentence Structures-forming questions
Vocabulary: Synonyms, Antonyms, Affixes (Prefixes/Suffixes), Root words.

Module - II	The Brook by Alfred Tennyson (Poem)	7Hrs
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Listening: Answering a series of questions about main idea and supporting ideas after listening to audio texts.
Speaking: Discussion in pairs/small groups on specific topics followed by short structured talks.
Reading: Identifying sequence of ideas; recognizing verbal techniques that help to link the ideas in a paragraph together.
Writing: Structure of a paragraph - Paragraph writing (specific topics)
Grammar: Cohesive devices - linkers, use of articles and zero article; prepositions.
Vocabulary: Homonyms, Homophones, Homographs.

Module - III	BIOGRAPHY: Elon Musk	6 Hrs
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Listening: Listening for global comprehension and summarizing what is listened to.
Speaking: Discussing specific topics in pairs or small groups and reporting what is discussed
Reading: Reading a text in detail by making basic inferences -recognizing and interpreting specific context clues; strategies to use text clues for comprehension.
Writing: Summarizing, Note-making, paraphrasing
Grammar: Verbs - tenses; subject-verb agreement; Compound words,
Vocabulary: Compound words, Collocations

Module - IV	INSPIRATION: The Toys of Peace -Saki	6 Hrs
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Listening: Making predictions while listening to conversations/ transactional dialogues without video; listening with video.
Speaking: Role plays for practice of conversational English in academic contexts (formal and informal) - asking for and giving information/directions.
Reading: Studying the use of graphic elements in texts to convey information, reveal trends/patterns/relationships, communicate processes or display complicated data
Writing: Letter Writing: Official Letters, Resumes
Grammar : Reporting verbs, Direct & Indirect speech, Active & Passive Voice
Vocabulary: Words often confused, Jargons

Module - V	MOTIVATION: The Power of Intrapersonal Communication (An Essay)	5 Hrs
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Listening: Identifying key terms, understanding concepts and answering a series of relevant questions that test comprehension.
Speaking: Formal oral presentations on topics from academic contexts

Reading: Reading for Comprehension

Writing: Writing structured essays on specific topics.

Grammar: Editing short texts –identifying and correcting common errors in grammar and usage (articles, prepositions, tenses, subject verb agreement)

Vocabulary: Technical Jargons

Textbooks:

1. Pathfinder: Communicative English for Undergraduate Students, 1st Edition, Orient Black Swan, 2023 (Units 1,2 & 3)
2. Empowering with Language by Cengage Publications, 2023 (Units 4 & 5)

Reference Books:

1. Dubey, Sham Ji & Co. English for Engineers, Vikas Publishers, 2020
2. Bailey, Stephen. Academic writing: A Handbook for International Students. Routledge, 2014.
3. Murphy, Raymond. English Grammar in Use, Fourth Edition, Cambridge University Press, 2019.
- Lewis, Norman. Word Power Made Easy- The Complete Handbook for Building a Superior Vocabulary. Anchor, 2014.

Web Resources:

GRAMMAR:

1. www.bbc.co.uk/learningenglish
2. <https://dictionary.cambridge.org/grammar/british-grammar/>
3. www.eslpod.com/index.html
4. <https://www.learngrammar.net/>
5. <https://english4today.com/english-grammar-online-with-quizzes/>
6. <https://www.talkenglish.com/grammar/grammar.aspx>

VOCABULARY

1. <https://www.youtube.com/c/DailyVideoVocabulary/videos>
2. https://www.youtube.com/channel/UC4cmBAit8i_NJZE8qK8sfpA

CHEMISTRY (Common to CSE, AI&ML, CS, ECE, EEE, DS)					
Course Code	L: T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23A0004T	3:0:0:0	3	CIE: 30 SEE:70	3 Hours	BSC
Course Objectives: Student will be able to • To familiarize chemistry and its applications. • To train the students on the principles and applications of electrochemistry and polymers. • To introduce instrumental methods.					
Course Outcomes: After completion of the course, students will be able to CO1: Describe Planck's quantum theory, dual nature of matter, Schrodinger equation, molecularorbital Theory and molecular orbital energy level diagram of different molecules CO2: Explain Crystal field theory, splitting in octahedral and tetrahedral geometry and the magneticbehavior, Oxidation state, coordination and color of complexes. CO3: Explain the principle of Band diagrams of conductors, superconductor, semiconductors and insulatorand nonmaterial CO4: Discuss the principles of electrochemistry in potentiometry, conductometry, battery andelectrochemical sensors CO5: Explain polymerization and the preparation, properties, and applications ofthermoplastics &thermosetting, elastomers, & conducting polymers CO6: Discuss the different applications of analytical instruments					
SYLLABUS				Total Hours: 48 Hrs	
Unit- I	Structure and Bonding Models				9 Hrs
Fundamentals of Quantum mechanics, Schrodinger Wave equation, significance of Ψ and Ψ^2 , particle in one dimensional box, molecular orbital theory – bonding in homo- and hetero-nuclear diatomic molecules – energylevel diagrams of O2, CO, and NO. π -molecular orbitals of butadiene and benzene, calculation of bond order.					
Unit-II	Modern Engineering materials				10 Hrs

Semiconductors – Introduction, basic concept, application
 Superconductors: Introduction, Basic concept and Applications. Supercapacitors: Introduction, Basic concept, Classification and Applications.
 Nanomaterials: Introduction, classification, properties and applications of Fullerenes, carbon nanotubes and Graphene nanoparticles.

Unit-III	Electrochemistry and Applications	10 Hrs
Electrochemical cell, Nernst equation, cell potential calculations and numerical problems, potentiometry- potentiometric titrations (redox titrations), concept of conductivity, conductivity cell, conductometric titrations (acid-base titrations). Electrochemical sensors – potentiometric sensors with examples, amperometric sensors with examples. Primary cells – Zinc-air battery, Secondary cells –lithium-ion batteries- working of the batteries including cell reactions; Fuel cells, hydrogen-oxygen fuel cell– working of the cells. Polymer Electrolyte Membrane Fuel cells (PEMFC).		
Unit-IV	Polymer Chemistry	10 Hrs
Introduction to polymers, functionality of monomers, chain growth and step growth polymerization, coordination polymerization, with specific examples and mechanisms of polymer formation. Plastics –Thermo and Thermosetting plastics, Preparation, properties and applications of – PVC, Teflon, Bakelite, Nylon-6,6, carbon fibres. Elastomers–Buna-S, Buna-N–preparation, properties and applications. Conducting polymers – polyacetylene, polyaniline, – mechanism of conduction and applications. Biodegradable polymers - poly dioxanone, Polyglycolic Acid (PGA), Polylactic Acid (PLA).		
Unit-V	Instrumental Methods and applications	9 Hrs
Electromagnetic spectrum. Absorption of radiation: Beer-Lambert's law. UV-Visible Spectroscopy, electronic transition, Instrumentation, IR spectroscopies, fundamental modes and selection rules, Instrumentation. Chromatography-Basic Principle, Classification, Gas chromatography , HPLC: Principle, Instrumentation and applications.		

Text Books:

1. Jain and Jain, Engineering Chemistry, 16/e, Dhanpat Rai, 2013.
2. Peter Atkins, Julio de Paula and James Keeler, Atkins' Physical Chemistry, 10/e, Oxford University Press, 2010.

Reference Books:

1. G.V.Subba Reddy, K.N.Jayaveera and C. Ramachandraiah, Engineering Chemistry, Mc Graw Hill, 2020.
2. Skoog and West, Principles of Instrumental Analysis, 6/e, Thomson, 2007.
 J.M.Lehn, Supra Molecular Chemistry, VCH Publications

DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS
(Common to all branches of Engineering)

Course Code	L:T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23A0002T	3:0:0:0	3	CIE:30 SEE:70	3Hours	BS

Course Objectives:

- To enlighten the learners in the concept of differential equations and multivariable calculus.
- To furnish the learners with basic concepts and techniques at plus two level to lead them into advanced level by handling various real-world applications.

Course Outcomes:

At the end of the course, the student will be able to

CO1: Solve the first order differential equations related to various engineering fields.

CO2: Solve the linear differential equations of higher order with constant coefficients

CO3: Identify solution methods for partial differential equations that model physical processes.

CO4: Interpret the physical meaning of different operators such as gradient, curl and divergence.

CO5: Apply Green's, Stokes and Divergence theorem in work done, circulation, flux and triple integrals.

SYLLABUS**Total Hours: 45**

Unit-I	Differential equations of first order and first degree	9Hrs
Linear differential equations – Bernoulli's equations- Exact equations and equations reducible to exact form. Applications: Newton's Law of cooling – Law of natural growth and decay Electrical circuits.		
Unit-II	Linear differential equations of higher order (Constant Coefficients)	9Hrs
Definitions, homogenous and non-homogenous, complimentary function, general solution, particular integral, Wronskian, Method of variation of parameters. Simultaneous linear equations, Applications to L-C-R Circuit problems and Simple Harmonic motion.		
Unit-III	Partial Differential Equations	9Hrs
Introduction and formation of Partial Differential Equations by elimination of arbitrary constants and arbitrary functions, solutions of first order linear equations using Lagrange's method. Homogeneous Linear Partial differential equations with constant coefficients.		
Unit-IV	Vector differentiation	9Hrs
Scalar and vector point functions, vector operator Del, Del applies to scalar point functions-Gradient, Directional derivative, del applied to vector point functions-Divergence and Curl, vector identities.		
Unit-V	Vector integration	9Hrs
Line integral-circulation-work done, surface integral-flux, Green's theorem in the plane (without proof), Stoke's theorem (without proof), volume integral, Divergence theorem (without proof) and related problems		

Textbooks:

1. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, 2017, 44th Edition
2. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley & Sons, 2018, 10th Edition.

Reference Books:

1. Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition.
2. Advanced Engineering Mathematics, Dennis G. Zill and Warren S. Wright, Jones and Bartlett, 2018.
3. Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint).
5. Higher Engineering Mathematics, B. V. Ramana, , McGraw Hill Education, 2017
6. Engineering Mathematics I by T.K.V. Iyengar, B.Krishna Gandhi, S. Chand Publications, 2015 Edition

BASIC CIVIL & MECHANICAL ENGINEERING							
Year/Sem -I-I(G-A) -I-II(G-B)		(Common for all branches)			Batch:2023-27		
Course Code	L:T:P	Credits	Exam. Marks		Exam Duration	REG	Course Type
23A0101T	3:0:0	3	CIE:30	SEE:70	3 Hours	RG23	ESC
Course Objectives: • Get familiarized with the scope and importance of Civil Engineering sub-divisions • Introduce the preliminary concepts of surveying. • Acquire preliminary knowledge on Transportation and its importance in nation’s economy. • Get familiarized with the importance of quality, conveyance and storage of water • Introduction to basic civil engineering materials and construction techniques							
SYLLABUS							
	PART-A(Civil)						Total Hours: 48
Unit –I							9 Hrs
Basics of Civil Engineering: Role of Civil Engineers in Society- Various Disciplines of Civil Engineering- Structural Engineering- Geo-technical Engineering- Transportation Engineering Hydraulics and Water Resources Engineering - Environmental Engineering-Scope of each discipline - Building Construction and Planning- Construction Materials-Cement - Aggregate - Bricks- Cement concrete- Steel. Introduction to Prefabricated construction Techniques							
Unit –II							10 Hrs
Fluid Mechanics: Properties of fluids and types of fluids. Surveying: Objectives of Surveying- Horizontal Measurements- Angular Measurements- Introduction to Bearings Levelling instruments used for levelling -Simple problems on levelling and bearings- Contour mapping.							
Unit –III							9 Hrs
Transportation Engineering Importance of Transportation in Nation's economic development- Types of Highway Pavements- Flexible Pavements and Rigid Pavements - Simple Differences. Basics of Harbour, Tunnel, Airport, and Railway Engineering. Water Resources and Environmental Engineering: Introduction, Sources of water- Quality of water- Specifications- Introduction to Hydrology–Rainwater Harvesting-Water Storage and Conveyance Structures (Simple introduction to Dams and Reservoirs)							
Unit –IV	PART-B (Mechanical)						9 Hrs
Fluid Mechanics: Properties of fluids and types of fluids. Surveying: Objectives of Surveying- Horizontal Measurements- Angular Measurements- Introduction to Bearings Levelling instruments used for levelling -Simple problems on levellingand bearings-Contour mapping.							
Unit –IV	PART-B (Mechanical)						9 Hrs

Basics of Civil Engineering: Role of Civil Engineers in Society- Various Disciplines of Civil Engineering- Structural Engineering- Geo-technical Engineering- Transportation Engineering Hydraulics and Water Resources Engineering - Environmental Engineering-Scope of each discipline - Building Construction and Planning- Construction Materials-Cement - Aggregate -Bricks- Cement concrete- Steel. Introduction to Prefabricated construction Techniques

Text Books:

1. Basic Civil Engineering, M.S.Palanisamy, , Tata Mcgraw Hill publications (India) Pvt. Ltd. Fourth Edition.
2. Introduction to Civil Engineering, S.S. Bhavikatti, New Age International Publishers. 2022. First Edition.

Basic Civil Engineering, Satheesh Gopi, Pearson Publications, 2009, First Edition

Reference Books:

1. Surveying, Vol- I and Vol-II, S.K. Duggal, Tata McGraw Hill Publishers 2019. Fifth Edition.
2. Hydrology and Water Resources Engineering, Santosh Kumar Garg, Khanna Publishers, Delhi. 2016
3. Irrigation Engineering and Hydraulic Structures - Santosh Kumar Garg, Khanna Publishers, Delhi 2023. 38th Edition.
4. Highway Engineering, S.K.Khanna, C.E.G. Justo and Veeraraghavan, Nemchand and Brothers Publications 2019. 10th Edition.

Indian Standard DRINKING WATER — SPECIFICATION IS10500-2012.

E-Resources:

<https://archive.nptel.ac.in/courses/105/106/105106201/>

DATA STRUCTURES (Common to CSE,IT & Allied Branches)					
Course Code	L: T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23A0002T	3:0:0:0	3	CIE:30 SEE:70	3Hours	BS
Course Objectives: • To provide the knowledge of basic data structures and their implementations. • To understand importance of data structures in context of writing efficient programs. • To develop skills to apply appropriate data structures in problem solving.					
Course Outcomes: CO1: Explain the role of linear data structures in organizing and accessing data efficiently in algorithms. CO2: Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation. CO3: Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems. CO4: Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between dequeues and priority queues, and apply them appropriately to solve data management challenges. CO5: Devise novel solutions to small scale programming challenges involving data structures such as stacks, queues, Trees. CO6: Recognize scenarios where hashing is advantageous, and design hash-based solutions for specific problems.					
SYLLABUS					TotalHours:45
Unit-I					9Hrs
Introduction to Linear Data Structures: Definition and importance of linear data structures, Abstract datatypes (ADTs) and their implementation, Overview of time and space complexity analysis for linear data structures. Searching Techniques: Linear & Binary Search, Sorting Techniques: Bubble sort, Selection sort, Insertion Sort					
Unit-II					9Hrs
Linked Lists: Singly linked lists: representation and operations, doubly linked lists and circular linked lists, Comparing arrays and linked lists, Applications of linked lists.					

Unit-III		9Hrs
Stacks: Introduction to stacks: properties and operations, implementing stacks using arrays and linked lists, Applications of stacks in expression evaluation, backtracking, reversing list etc.		
Unit-IV		9Hrs
Queues: Introduction to queues: properties and operations, implementing queues using arrays and linkedlists, Applications of queues in breadth-first search, scheduling, etc. Deques: Introduction to deques (double-ended queues), Operations on deques and their applications.		
Unit-V		9Hrs
Trees: Introduction to Trees, Binary Tree-Insertion, Deletion & Traversal , Binary Search Tree – Insertion, Deletion & Traversal, Hashing: Brief introduction to hashing and hash functions, Collision resolution techniques: chaining and open addressing, Hash tables: basic implementation and operations, Applications of hashing in unique identifier generation, caching, etc.		

Text Books:

1. Data Structures and algorithm analysis in C, Mark Allen Weiss, Pearson, 2nd Edition.
2. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Susan Anderson Freed, SiliconPress, 2008

Reference Books:

1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders
2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft
3. "Problem Solving with Algorithms and Data Structures" by Brad Miller and David Ranum
4. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein
5. Algorithms in C, Parts 1-5 (Bundle): Fundamentals, Data Structures, Sorting, Searching, and Graph Algorithms" by Robert Sedgewick

COMMUNICATIVE ENGLISH LAB**(Common to all Branches of Engineering)**

Course Code	L:T:P:S	Credits	Exam Marks	Exam Duration	Course Type
2300010P	0:0:2:0	1	CIE:30 SEE:70	3H	HS

Course Objectives:

The main objective of introducing this course, Communicative English Laboratory, is to expose the students to a variety of self-instructional, learner friendly modes of language learning students will get trained in the basic communication skills and also make them ready to face job interviews

List of Experiments**Total Hours:32**

1. Vowels & Consonants
2. Neutralization/ Accent Rules
3. Communication Skills & Jam
4. Role Play or Conversational Practice
5. Email Writing
6. Resume Writing, Cover Letter, Sop
7. Group Discussion-Methods & Practice
8. Debate - Method & Practice
9. PPT Presentation / Poster Presentation
10. Interview Skills

Suggested Software: Walden InfoTech / Young India Films**Online Learning Resources/Virtual Labs:****Spoken English:** 1. www.esl-lab.com2. www.englishmedialab.com3. www.englishinteractive.net4. <https://www.britishcouncil.in/english/online>5. <http://www.letstalkpodcast.com/>6. https://www.youtube.com/c/mmmEnglish_Emma/featured7. <https://www.youtube.com/c/ArnelsEverydayEnglish/featured>8. <https://www.youtube.com/c/engvidAdam/featured>9. <https://www.youtube.com/c/EnglishClass101/featured>10. <https://www.youtube.com/c/SpeakEnglishWithTiffani/playlists>11. https://www.youtube.com/channel/UCV1h_cBE0Drdx19qkTM0WNw**Voice & Accent:**1. <https://www.youtube.com/user/letstalkaccent/videos>2. <https://www.youtube.com/c/EngLanguageClub/featured>3. https://www.youtube.com/channel/UC_OskgZBoS4dAnVUgJVexchttps://www.youtube.com/channel/UCNfm92h83W2i2ijc5Xwp_IA

CHEMISTRY LAB

(Common to CSE, AI&ML, ECE, EEE, DS)

Course Code	L:T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23A0007P	0:0:2:0	1	CIE:30 SEE:70	3 Hours	BSC

Course Objectives:

- Verify the fundamental concepts with experiments

Syllabus**Total Hours: 48****List of Experiments**

1. Measurement of 10Dq by spectrophotometric method
2. Conductometric titration of strong acid vs. strong base
3. Conductometric titration of weak acid vs. strong base
4. Determination of cell constant and conductance of solutions
5. Potentiometry - determination of redox potentials and emfs
6. Determination of Strength of an acid in Pb-Acid battery
7. Preparation of a Bakelite
8. Verify Lambert-Beer's law
9. Simultaneous estimation of Mn and Cr ions by spectrophotometry in water samples.
10. Wavelength measurement of sample through UV-Visible Spectroscopy
11. Identification of functional groups in organic compounds by IR Spectroscopy.
12. Preparation of nanomaterials by precipitation method
13. Estimation of Ferrous Iron by Dichrometry
14. Determination of Hardness of a groundwater sample
15. pH metric titration of strong acid vs strong base

(Any 10 experiments from the above list)**Course Outcomes:**

On completion of this course, the students are able to:

- Determine the cell constant and conductance of solutions and the strength of an acid by conductometry
- Synthesize of advanced polymer materials
- Measure the strength of an acid present in secondary battery and Ferrous ion using volumetric analysis
- Determine the potentials and EMFs of solutions by Potentiometry
- Identify some organic and inorganic compounds by instrumental methods
- Synthesize of nanomaterials by simple methods

Text Book(s):

1. A Textbook of Quantitative Analysis, Arthur J. Vogel.
2. Jain & Jain. Engineering Chemistry: Dhanapath Publications., 2015.
3. S.S.Dara, Experiments and Calculations in Engineering Chemistry: S-Chand Publications, Revised edition, 2008.

Reference Book(s):

- "Vogel's Quantitative Chemical Analysis 6th Edition 6th Edition" Pearson Publications by J. Mendham, R.C.Denney, J.D.Barnes and B. Sivasankar

ENGINEERING WORKSHOP**(Common to All Branches of Engineering)**

Course Code	L: T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23A0302P	3:0:0:0	3	CIE:30 SEE:70	3Hours	ES

Course Objectives:

To familiarize students with wood working, sheet metal operations, fitting and electrical house wiring skills

Course Outcomes:

CO1: Identify workshop tools and their operational capabilities.

CO2: Practice on manufacturing of components using workshop trades including fitting, carpentry, foundry and welding.

CO3: Apply fitting operations in various applications.

CO4: Apply basic electrical engineering knowledge for House Wiring Practice

List of Experiments

Demonstration: Safety practices and precautions to be observed in workshop.

Wood Working: Familiarity with different types of woods and tools used in wood working and make following joints.

Half-Lap joint b) Mortise and Tenon joint c) Corner Dovetail joint or Bridle joint

Sheet Metal Working: Familiarity with different types of tools used in sheet metal working, Developments of following sheet metal job from GI sheets.

Tapered tray b) Conical funnel c) Elbow pipe d) Brazing

Fitting: Familiarity with different types of tools used in fitting and do the following fitting exercises.

V-fit b) Dovetail fit c) Semi-circular fit d) Bicycle tire puncture and change of two-wheel tire

Electrical Wiring: Familiarity with different types of basic electrical circuits and make the following connections.

a) Parallel and series b) Two-way switch c) Godown lighting

d) Tubelight e) Three phase motor f) Soldering of wires

Foundry Trade: Demonstration and practice on Moulding tools and processes, Preparation of Green Sand Moulds for given Patterns.

Welding Shop: Demonstration and practice on Arc Welding and Gas welding. Preparation of Lap joint and Butjoint.

Plumbing: Demonstration and practice of Plumbing tools, Preparation of Pipe joints with coupling for same diameter and with reducer for different diameters.

Textbooks:

1. Basic Workshop Technology: Manufacturing Process, Felix W.; Independently Published, 2019. Workshop Processes, Practices and Materials; Bruce J. Black, Routledge publishers, 5th Edn. 2015.
2. A Course in Workshop Technology Vol I. & II, B.S. Raghuvanshi, Dhanpath Rai & Co., 2015 & 2017.

Reference Books:

1. Elements of Workshop Technology, Vol. I by S.K. Hajra Choudhury & Others, Media Promoters and Publishers, Mumbai. 2007, 14th edition
2. Workshop Practice by H. S. Bawa, Tata-McGraw Hill, 2004.
3. Wiring Estimating, Costing and Contracting; Soni P.M. & Upadhyay P.A.; Atul Prakashan, 2021-22.

DATA STRUCTURES LAB (Common to All Branches of Engineering)					
Course Code	L: T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23A0505P	3:0:0:0	3	CIE:30 SEE:70	3Hours	ES
Course Objectives: The course aims to strengthen the ability of the students to identify and apply the suitable data structure for the given real-world problem. It enables them to gain knowledge in practical applications of data structures.					
Course Outcomes: At the end of the course, Student will be able to CO1: Explain the role of linear data structures in organizing and accessing data efficiently in algorithms. CO2: Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation. CO3: Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems. CO4: Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between dequeues and priority queues and apply them appropriately to solve data management challenges. CO5: Recognize scenarios where hashing is advantageous, and design hash-based solutions for specific problems.					
List of Experiments					
Exercise 1: Array Manipulation i) Write a program to reverse an array. ii) C Programs to implement the Searching Techniques – Linear & Binary Search iii) C Programs to implement Sorting Techniques – Bubble, Selection and Insertion Sort					

Exercise 2: Linked List Implementation

- i) Implement a singly linked list and perform insertion and deletion operations.
- ii) Develop a program to reverse a linked list iteratively and recursively.
- iii) Solve problems involving linked list traversal and manipulation.

Exercise 3: Linked List Applications

- i) Create a program to detect and remove duplicates from a linked list.
- ii) Implement a linked list to represent polynomials and perform addition.
- iii) Implement a double-ended queue (deque) with essential operations.

Exercise 4: Double Linked List Implementation

- i) Implement a doubly linked list and perform various operations to understand its properties and applications.
- ii) Implement a circular linked list and perform insertion, deletion, and traversal.

Exercise 5: Stack Operations

- i) Implement a stack using arrays and linked lists.
- ii) Write a program to evaluate a postfix expression using a stack.
- iii) Implement a program to check for balanced parentheses using a stack.

Exercise 6: Queue Operations

- i) Implement a queue using arrays and linked lists.
- ii) Develop a program to simulate a simple printer queue system.
- iii) Solve problems involving circular queues.

Exercise 7: Stack and Queue Applications

- i) Use a stack to evaluate an infix expression and convert it to postfix.
- ii) Create a program to determine whether a given string is a palindrome or not.
- iii) Implement a stack or queue to perform comparison and check for symmetry.

Exercise 8: Binary Tree

- i) Implementing a Binary tree using Linked List
- ii) Traversing of Binary tree

Exercise 9: Binary Search Tree

- i) Implementing a BST using Linked List.
- ii) Traversing of BST.

Exercise 10: Hashing

- i) Implement a hash table with collision resolution techniques.
- ii) Write a program to implement a simple cache using hashing.

Text Books

- 1. Data Structures and algorithm analysis in C, Mark Allen Weiss, Pearson, 2nd Edition.
- 2. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Susan

Reference Books

- 1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders
- 2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft
- 3. Problem Solving with Algorithms and Data Structures" by Brad Miller and David Ranum
- 4. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein
- 5. Algorithms in C, Basics, Data Structures, Sorting, Searching, Graph Algorithms by Robert.

Textbooks:

- 3. Ajay Mittal, Programming in C: A practical approach, Pearson.
- 4. Byron Gottfried, Schaum' s Outline of Programming with C, McGraw Hill

Reference Books:

- 3. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India
- 4. C Programming, A Problem-Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE

HEALTH AND WELLNESS, YOGA AND SPORTS					
(Common to all branches of Engineering)					
Course Code	L: T:P:S	Credits	Exam Marks	Exam Duration	Course Type
23AYG01P	0:0:1:0	0.5	CIE:30 SEE:70	3Hours	BS
Course Objectives: The main objective of introducing this course is to make the students maintain their mental and physical wellness by balancing emotions in their life. It mainly enhances the essential traits required for the development of the personality.					
CO1: Understand the importance of yoga and sports for Physical fitness and sound health.					
CO2: Demonstrate an understanding of health-related fitness components.					
CO3: Compare and contrast various activities that help enhance their health.					
CO4: Assess current personal fitness levels.					
CO5: Develop Positive Personality					
SYLLABUS					TotalHours:45
Unit-I					9Hrs
Concept of health and fitness, Nutrition and Balanced diet, basic concept of immunity Relationship between diet and fitness, Globalization and its impact on health, Body Mass Index (BMI) of all age groups.					
Activities: i) Organizing health awareness programmes in community ii) Preparation of health profile iii) Preparation of chart for balance diet for all age groups					
Unit-II					9Hrs
Concept of yoga, need for and importance of yoga, origin and history of yoga in Indian context, classification of yoga, Physiological effects of Asanas-Pranayama and meditation, stress management and yoga, Mental health and yoga practice.					
Activities: Yoga practices –Asana, Kriya, Mudra, Bandha, Dhyana, Surya Namaskar					
Unit-III					9Hrs
Concept of Sports and fitness, importance, fitness components, history of sports, Ancient and Modern Olympics, Asian games and Commonwealth games.					
Activities: i) Participation in one major game and one individual sport viz., Athletics, Volleyball,					

Basketball, Handball, Football, Badminton, Kabaddi, Kho-kho, Table tennis, Cricket etc.
Practicing general and specific warm up, aerobics
ii) Practicing cardio respiratory fitness, treadmill, run test, 9 min walk, skipping and running.

Reference Books:

1. Gordon Edlin, Eric Golanty. Health and Wellness, 14th Edn. Jones & Bartlett Learning, 2022
2. T.K.V.Desikachar. The Heart of Yoga: Developing a Personal Practice
3. Archie J.Bahm. Yoga Sutras of Patanjali, Jain Publishing Company, 1993
4. Wiseman, John Lofty, SAS Survival Handbook: The Ultimate Guide to Surviving Anywhere Third Edition, William Morrow Paperbacks, 2014
5. The Sports Rules Book/ Human Kinetics with Thomas Hanlon. -- 3rd ed. Human Kinetics, Inc.2014

**GEETHANJALI INSTITUTE OF SCIENCE AND TECHNOLOGY
NELLORE**

DEPARTMENT OF CSE(AI&ML)

REGULATION: RG23

II – YEAR

COURSE STRUCTURE AND SYLLABUS



GEETHANJALI INSTITUTE OF SCIENCE & TECHNOLOGY

Unit of USHODAYA EDUCATIONAL SOCIETY

An ISO 9001:2015 certified Institution: Recognized under Sec. 2(f)& 12(B) of UGC Act, 1956
3rd Mile, Bombay Highway, Gangavaram (V), Kovur(M), SPSR Nellore (Dt), Andhra Pradesh, India- 524137
Ph. No. 08622-212769, E-Mail: geethanjali@gist.edu.in, Website: www.gist.edu.in

Semester-3 (Theory-5, Lab-2, SEC-1, MC-1)							
Sl. No.	Category	Course Code	Course Title	Hours per week			Credits
				L	T	P	
1	BSC	23A0015T	Discrete Mathematics & Graph Theory	3	0	0	3
2	HSC	23A0021T	Universal Human Values 2 Understanding Harmony and Ethical human conduct	2	1	0	3
3	ESC	23A3301T	Artificial Intelligence	3	0	0	3
4	PCC	23A0506T	Advanced Data Structures & Algorithm Analysis	3	0	0	3
5	PCC	23A0507T	Object-Oriented Programming Through JAVA	3	0	0	3
6	PCC(Lab)	23A0508P	Advanced Data structures and Algorithms Analysis Lab	0	0	3	1.5
7	PCC(Lab)	23A0509P	Object-Oriented Programming Through JAVA Lab	0	0	3	1.5
8	SEC	23A0510P	Python programming	0	1	2	2
9	AC	23A0109T	Environmental Science	2	0	0	-
Total				16	2	8	20

Category	Credits
Basic Science Course (BSC)	3
Professional Core Courses (PCC)	9
Engineering Science Courses (ESC)	3
Humanities and Social Science Course (HSC)	3
Skill Enhancement Course (SEC)	2
Total	20



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Ph. No. 08622-212769, E-Mail: geethanjali@gist.edu.in, Website: www.gist.edu.in

Semester-4 (Theory-5, Lab-2, SEC-1, MC-1)							
Sl. No.	Category	Course Code	Course Title	Hours per week			Credits
				L	T	P	
1	MEC	23A0020T	Optimization Techniques	2	0	0	2
2	BSC	23A0017T	Probability & Statistics	3	0	0	3
3	PCC	23A3302T	Machine Learning	3	0	0	3
4	PCC	23A0512T	Database Management Systems	3	0	0	3
5	PCC	23A0406T	Digital Logic and Computer Organization	3	0	0	3
6	PCC(Lab)	23A3303P	Machine Learning Lab	0	0	3	1.5
7	PCC(Lab)	23A0515P	Database Management Systems Lab	0	0	3	1.5
8	SEC	23A0516P	Full Stack Development-1	0	1	2	2
9	ESC	23A0413T	Design Thinking & Innovation	1	0	2	2
Total				15	1	10	21
Mandatory Community Service Project Internship of 08 weeks duration during summer vacation							

Category	Credits
Basic Science Course (BSC)	3
Professional Core Courses (PCC)	12
Engineering Science Courses (ESC)	2
Skill Enhancement Course (SEC)	2
Management Elective Course (MEC)	2
Total	21



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DISCRETE MATHEMATICS & GRAPH THEORY

(Common to CSE and CSE allied branches)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0015T	3:0:0:3	3	CIE: 30 SEE:70	3 Hours	BSC
Course Outcomes (CO):					
On completion of this course, student will be able to					
• Apply mathematical logic to solve problems. • Understand the concepts and perform the operations related to sets, relations and functions. • Gain the conceptual background needed and identify structures of algebraic nature. • Apply basic counting techniques to solve combinatorial problems. • Formulate problems and solve recurrence relations. • Apply Graph Theory in solving computer science problems					
Syllabus				Total Hours:48	
Module-I	Mathematical Logic			9Hrs	
Mathematical Logic: Introduction, Statements and Notation, Connectives, Well-formed formulas, Tautology, Duality law, Equivalence, Implication, Normal Forms, functionally complete set of connectives, Inference Theory of Statement Calculus, Predicate Calculus, Inference theory of Predicate Calculus					
Module-II	Set theory and algebraic structures			10Hrs	
Set theory: Sets and its operations, The Principle of Inclusion- Exclusion, Pigeon hole principle and its application, Functions composition of functions, Inverse Functions, Recursive Functions, Lattices and its properties. Algebraic structures: Algebraic Systems-Examples and General Properties, Semi groups and Monoids, groups, sub groups, homomorphism, Isomorphism...					
Module-III	Elementary Combinatorics			10Hrs	
Combinations and Permutations, Enumeration of Combinations and Permutations, Enumerating Combinations and Permutations with Repetitions, Enumerating Permutations with Constrained Repetitions, Binomial Coefficients, The Binomial and Multinomial Theorems.					
Module-IV	Recurrence Relations			10Hrs	
Generating Functions of Sequences, Calculating Coefficients of Generating Functions, Recurrence relations, Solving Recurrence Relations by Substitution and Generating functions, The Method of Characteristic roots, Solutions of Inhomogeneous, Recurrence Relations.					
Module-V	Graphs			9Hrs	
Basic Concepts, Isomorphism and Subgraphs, Trees and their Properties, Spanning Trees, Directed Trees, Binary Trees, Planar Graphs, Euler’s Formula, Multigraphs and Euler Circuits, Hamiltonian Graphs					

Text Books:

1. J.P. Tremblay and R. Manohar, Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill, 2002.
2. Kenneth H. Rosen, Discrete Mathematics and its Applications with Combinatorics and Graph Theory, 7th Edition, McGraw Hill Education (India) Private Limited.

Reference Books:

1. Joe L. Mott, Abraham Kandel and Theodore P. Baker, Discrete Mathematics for Computer Scientists & Mathematicians, 2nd Edition, Pearson Education.
2. Narsingh Deo, Graph Theory with Applications to Engineering and Computer Science.

Web References:

1. <http://www.cs.yale.edu/homes/aspnes/classes/202/notes.pdf>



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UNIVERSAL HUMAN VALUES – UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT

(Common to all branches of Engineering)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0021T	2:1:0:3	3	CIE: 30 SEE:70	3 Hours	HSC

Course Objectives:

This course will enable students to:

- To help the students appreciate the essential complementary between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
- To facilitate the development of a Holistic perspective among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of the Human reality and the rest of existence. Such holistic perspective forms the basis of Universal Human Values and movement towards value-based living in a natural way.
- To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behaviour and mutually enriching interaction with Nature.

Course Outcomes (CO):

On completion of this course, student will be able to

- Define the terms like Natural Acceptance, Happiness and Prosperity.
- Identify one's self, and one's surroundings (family, society nature)
- Apply what they have learnt to their own self in different day-to-day settings in real life.
- Relate human values with human relationship and human society.
- Justify the need for universal human values and harmonious existence
- Develop as socially and ecologically responsible engineers

Course Topics

Course Topics

The course has 28 lectures and 14 tutorials in 5 modules. The lectures and tutorials are of 1- hour duration. Tutorial sessions are to be used to explore and practice what has been proposed during the lecture sessions.

The Teacher's Manual provides the outline for lectures as well as practice sessions. The teacher is expected to present the issues to be discussed as propositions and encourage the students to have a dialogue

Syllabus		Total Hours:48
Module-I	Introduction to Value Education (6 lectures and 3 tutorials for practice session)	9Hrs
Lecture 1: Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education)		
Lecture 2: Understanding Value Education		

Tutorial 1: Practice Session PS1 Sharing about Oneself Lecture 3: self-exploration as the Process for Value Education Lecture4: Continuous Happiness and Prosperity – the Basic Human Aspirations Tutorial 2: Practice Session PS2 Exploring Human Consciousness Lecture 5: Happiness and Prosperity – Current Scenario Lecture 6: Method to Fulfill the Basic Human Aspirations Tutorial 3: Practice Session PS3 Exploring Natural Acceptance		
Module-II	Harmony in the Human Being (6 lectures and 3 tutorials for practice session)	10Hrs
Lecture 7: Understanding Human being as the Co-existence of the self and the body. Lecture 8: Distinguishing between the Needs of the self and the body Tutorial 4: Practice Session PS4 Exploring the difference of Needs of self and body. Lecture 9: The body as an Instrument of the self-Lecture 10: Understanding Harmony in the self Tutorial 5: Practice Session PS5 Exploring Sources of Imagination in the self Lecture 11: Harmony of the self with the body Lecture 12: Programme to ensure self-regulation and Health Tutorial 6: Practice Session PS6 Exploring Harmony of self with the body		
Module-III	Harmony in the Family and Society (6 lectures and 3 tutorials for practice session)	10Hrs
Lecture 13: Harmony in the Family – the Basic Unit of Human Interaction Lecture 14: 'Trust' – the Foundational Value in Relationship Tutorial 7: Practice Session PS7 Exploring the Feeling of Trust Lecture 15: 'Respect' – as the Right Evaluation Tutorial 8: Practice Session PS8 Exploring the Feeling of Respect Lecture 16: Other Feelings, Justice in Human-to-Human Relationship Lecture 17: Understanding Harmony in the Society Lecture 18: Vision for the Universal Human Order Tutorial 9: Practice Session PS9 Exploring Systems to fulfill Human Goal.		
Module-IV	Harmony in the Nature/Existence (4 lectures and 2 tutorials for practice session)	10Hrs
Lecture 19: Understanding Harmony in the Nature Lecture 20: Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature Tutorial 10: Practice Session PS10 Exploring the Four Orders of Nature Lecture 21: Realizing Existence as Co-existence at All Levels Lecture 22: The Holistic Perception of Harmony in Existence Tutorial 11: Practice Session PS11 Exploring Co-existence in Existence.		
Module-V	Implications of the Holistic Understanding – a Look at Professional Ethics (6 lectures and 3 tutorials for practice session)	9Hrs
Lecture 23: Natural Acceptance of Human Values Lecture 24: Definitiveness of (Ethical) Human Conduct Tutorial 12: Practice Session PS12 Exploring Ethical Human Conduct Lecture 25: A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order Lecture 26: Competence in Professional Ethics Tutorial 13: Practice Session PS13 Exploring Humanistic Models in Education Lecture 27: Holistic Technologies, Production Systems and Management Models-Typical Case Studies		

Lecture 28: Strategies for Transition towards Value-based Life and Profession

Tutorial 14: Practice Session PS14 Exploring Steps of Transition towards Universal Human Order

Practice Sessions for UNIT I – Introduction to Value Education PS1 Sharing about Oneself

PS2 Exploring Human Consciousness PS3 Exploring Natural Acceptance

Practice Sessions for UNIT II – Harmony in the Human Being PS4 Exploring the difference of Needs of self and body

PS5 Exploring Sources of Imagination in the self PS6 Exploring Harmony of self with the body

Practice Sessions for UNIT III – Harmony in the Family and Society PS7 Exploring the Feeling of Trust

PS8 Exploring the Feeling of Respect

PS9 Exploring Systems to fulfil Human Goal

Practice Sessions for UNIT IV – Harmony in the Nature (Existence) PS10 Exploring the Four Orders of Nature

PS11 Exploring Co-existence in Existence

Practice Sessions for UNIT V – Implications of the Holistic Understanding – a Look at Professional Ethics

PS12 Exploring Ethical Human Conduct

PS13 Exploring Humanistic Models in Education

PS14 Exploring Steps of Transition towards Universal Human Order

Mode of Conduct:

Lecture hours are to be used for interactive discussion, placing the proposals about the topics at hand and motivating students to reflect, explore and verify them.

Tutorial hours are to be used for practice sessions.

While analyzing and discussing the topic, the faculty mentor's role is in pointing to essential elements to help in sorting them out from the surface elements. In other words, help the students explore the important or critical elements.

In the discussions, particularly during practice sessions (tutorials), the mentor encourages the student to connect with one's own self and do self-observation, self-reflection and self- exploration.

Scenarios may be used to initiate discussion. The student is encouraged to take up "ordinary" situations rather than "extra-ordinary" situations. Such observations and their analyses are shared and discussed with other students and faculty mentor, in a group sitting.

Tutorials (experiments or practical) are important for the course. The difference is that the laboratory is everyday life, and practical are how you behave and work in real life. Depending on the nature of topics, worksheets, home assignment and/or activity are included. The practice sessions (tutorials) would also provide support to a student in performing actions commensurate to his/her beliefs. It is intended that this would lead to development of commitment, namely behaving and working based on basic human values.

It is recommended that this content be placed before the student as it is, in the form of a basic foundation course, without including anything else or excluding any part of this content. Additional content may

be offered in separate, higher courses. This course is to be taught by faculty from every teaching department, not exclusively by any one department.

Teacher preparation with a minimum exposure to at least one 8-day Faculty Development Program on Universal Human Values is deemed essential.

Text Books:

a. The Textbook

R R Gaur, R Asthana, G P Bagaria, *A Foundation Course in Human Values and Professional Ethics*, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-47-1

b. The Teacher's Manual

R R Gaur, R Asthana, G P Bagaria, *Teachers' Manual for A Foundation Course in Human Values and Professional Ethics*, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2

Reference Books:

1. *Jeevan Vidya: Ek Parichaya*, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.
2. *Human Values*, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3. *The Story of Stuff* (Book).
4. *The Story of My Experiments with Truth* - by Mohandas Karamchand Gandhi
5. *Small is Beautiful* - E. F Schumacher.
6. *Slow is Beautiful* - Cecile Andrews
7. *Economy of Permanence* - J C Kumarappa
8. *Bharat Mein Angreji Raj* – Pandit Sunderlal
9. *Rediscovering India* - by Dharampal
10. *Hind Swaraj or Indian Home Rule* - by Mohandas K. Gandhi
11. *India Wins Freedom* - Maulana Abdul Kalam Azad
12. *Vivekananda* - Romain Rolland (English)
13. *Gandhi* - Romain Rolland (English).

Web References:

1. <https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%201-Introduction%20to%20Value%20Education.pdf>
2. <https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%202-Harmony%20in%20the%20Human%20Being.pdf>
3. <https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%203-Harmony%20in%20the%20Family.pdf>
4. <https://fdp-si.aicte-india.org/UHV%201%20Teaching%20Material/D3-S2%20Respect%20July%2023.pdf>
5. <https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%205-Harmony%20in%20the%20Nature%20and%20Existence.pdf>

6. <https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-days%20FDP-SI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%203D%20D3-S2A%20Und%20Nature-Existence.pdf>
7. <https://fdp-si.aicte-india.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Ethics%20v1.pdf>
8. <https://www.studocu.com/in/document/kiet-group-of-institutions/universal-human-values/chapter-5-holistic-understanding-of-harmony-on-professional-ethics/62490385>
https://onlinecourses.swayam2.ac.in/aic22_ge23/preview



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RG 23 Regulations

ARTIFICIAL INTELLIGENCE

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3301T	3:0:0:3	3	CIE: 30 SEE:70	3 Hours	ESC

Course Objectives:

The student should be made to study the concepts of Artificial Intelligence.

- The student should be made to learn the methods of solving problems using Artificial Intelligence.
- The student should be made to introduce the concepts of Expert Systems.
- To understand the applications of AI, namely game playing, theorem proving, and machine learning.
- To learn different knowledge representation techniques

Syllabus		Total Hours:48
Module-I	INTRODUCTION	9Hrs
Introduction: AI problems, foundation of AI and history of AI intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation.		
Module-II	SEARCHING	10Hrs
Searching- Searching for solutions, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A*, AO* Algorithms, Problem reduction, Game Playing-Adversial search, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning, Evaluation functions.		
Module-III	REPRESENTATION OF KNOWLEDGE	10Hrs
Representation of Knowledge: Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules, Rules based deduction systems. Reasoning under uncertainty, review of probability, Bayes' probabilistic interferences and dempster Shafer theory.		
Module-IV	LOGIC CONCEPTS	10Hrs
Logic concepts: First order logic. Inference in first order logic, propositional vs. first order inference, unification & lifts forward chaining, Backward chaining, Resolution, Learning from observation Inductive learning, Decision trees, Explanation based learning, Statistical Learning methods, Reinforcement Learning.		
Module-V	EXPERT SYSTEMS	9Hrs
Expert Systems: Architecture of expert systems, Roles of expert systems – Knowledge Acquisition Meta knowledge Heuristics. Typical expert systems – MYCIN, DART, XCON:		

Expert systems shells.

Text Books:

1. S. Russel and P. Norvig, “Artificial Intelligence – A Modern Approach”, Second Edition, Pearson Education.
2. Kevin Night and Elaine Rich, Nair B., “Artificial Intelligence (SIE)”, Mc Graw Hill

Reference Books:

1. David Poole, Alan Mackworth, Randy Goebel, “Computational Intelligence: a logical approach”, Oxford University Press.
2. G. Luger, “Artificial Intelligence: Structures and Strategies for complex problemsolving”, Fourth Edition, Pearson Education.
3. J. Nilsson, “Artificial Intelligence: A new Synthesis”, Elsevier Publishers.
4. Artificial Intelligence, Saroj Kaushik, CENGAGE Learning.

Web References:

1. <https://ai.google/>
2. https://swayam.gov.in/nd1_noc19_me71/preview



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ADVANCED DATA STRUCTURES & ALGORITHM ANALYSIS

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0506T	3:0:0:3	3	CIE: 30 SEE:70	3 Hours	PCC
Course Objectives:					
This course will enable students to: • Provide knowledge on advance data structures frequently used in Computer Sciencedomain • Develop skills in algorithm design techniques • Understand the use of various data structures in the algorithm design					
Course Outcomes (CO):					
On completion of this course, student will be able to • Illustrate the working of the advanced tree data structures and their applications. • Understand the Graph data structure, traversals and apply them in various contexts. • Apply different algorithmic techniques for solving real world problems • Recommend appropriate data structures based on the problem being solved. • Analyze algorithms with respect to space and time complexities. • Design new algorithms					
Syllabus				Total Hours:48	
Module-I	Introduction			9Hrs	
Introduction to Algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations. AVL Trees – Creation, Insertion, Deletion operations and Applications B-Trees – Creation, Insertion, Deletion operations and Applications					
Module-II	Heap Trees (Priority Queues)			10Hrs	
Heap Trees (Priority Queues) – Min and Max Heaps, Operations and Applications Graphs – Terminology, Representations, Basic Search and Traversals, Connected Components and Bi connected Components, applications Divide and Conquer: The General Method, Quick Sort, Merge Sort, Strassen’s matrix multiplication, Convex Hull					
Module-III	Greedy Method			10Hrs	
Greedy Method: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Paths Dynamic Programming: General Method, All pairs shortest paths, Single Source Shortest Paths – General Weights (Bellman Ford Algorithm), Optimal Binary Search Trees, 0/1 Knapsack, String Editing, Travelling Salesperson problem					
Module-IV	Backtracking & Branch and Bound			10Hrs	

Backtracking: General Method, 8-Queens Problem, Sum of Subsets problem, Graph Coloring, 0/1 Knapsack Problem (Hamiltonian Cycle)
Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem

Module-V

P and NP Problems

9Hrs

NP Hard and NP Complete Problems: Basic Concepts, **Satisfiability Problem**, Cook's theorem
 NP Hard Graph Problems: Clique Decision Problem (CDP), Chromatic Number Decision Problem (CNDP), Traveling Salesperson Decision Problem (TSP)
 NP Hard Scheduling Problems: Scheduling Identical Processors, Job Shop Scheduling

Text Books:

1. Fundamentals of Data Structures in C++, Horowitz, Ellis; Sahni, Sartaj; Mehta, Dinesh 2nd Edition Universities Press

Reference Books:

1. Computer Algorithms/C++ Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran 2nd Edition University Press.(added to reference).
2. Data Structures and program design in C, Robert Kruse, Pearson Education Asia.
3. An introduction to Data Structures with applications, Trembley & Sorenson, McGrawHill.
4. The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison-Wesley, 1997.
5. Data Structures using C & C++: Langsam, Augenstein & Tanenbaum, Pearson, 1995.
6. Algorithms + Data Structures & Programs: N.Wirth, PHI.
7. Fundamentals of Data Structures in C++: Horowitz Sahni & Mehta, Galgottia Pub.
8. Data structures in Java: Thomas Standish, Pearson Education Asia.

Web References:

1. https://onlinecourses.swayam2.ac.in/cec20_cs03/preview
2. https://www.tutorialspoint.com/advanced_data_structures/index.asp
3. <http://peterindia.net/Algorithms.html>
4. [Abdul Bari, 1. Introduction to Algorithms \(youtube.com\)](#)



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OBJECT-ORIENTED PROGRAMMING THROUGH JAVA

Course Code	L: T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0507T	3:0:0:3	3	CIE: 30 SEE:70	3 Hours	PCC

Course Objectives:

This course will enable students to:

- Identify Java language components and how they work together in applications
- Learn the fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries.
- Learn how to extend Java classes with inheritance and dynamic binding and how to use exception handling in Java applications
- Understand how to design applications with threads in Java
- Understand how to use Java apis for program development

Course Outcomes (CO):

On completion of this course, student will be able to

- Analyze problems, design solutions using OOP principles, and implement them efficiently in Java
- Design and implement classes to model real-world entities, with a focus on attributes, behaviors, and relationships between objects
- Demonstrate an understanding of inheritance hierarchies and polymorphic behaviour, including method overriding and dynamic method dispatch.
- Apply Competence in handling exceptions and errors to write robust and fault-tolerant code
- Perform file input/output operations, including reading from and writing to files using Java I/O classes, graphical user interface (GUI) programming using JavaFX.
- Choose appropriate data structure of Java to solve a problem

Syllabus		Total Hours:48
Module-I	Object Oriented Programming	9Hrs
Object Oriented Programming: Basic concepts, Principles, Program Structure in Java: Introduction, Writing Simple Java Programs, Elements or Tokens in Java Programs, Java Statements, Command Line Arguments, User Input to Programs, Escape Sequences Comments, Programming Style. Data Types, Variables, and Operators :Introduction, Data Types in Java, Declaration of Variables, Data Types, Type Casting, Scope of Variable Identifier, Literal Constants, Symbolic Constants, Formatted Output with printf() Method, Static Variables and Methods, Attribute Final, Introduction to Operators, Precedence and Associativity of Operators, Assignment Operator (=), Basic Arithmetic Operators, Increment (++) and Decrement (- -) Operators, Ternary Operator, Relational Operators, Boolean Logical Operators, Bitwise Logical Operators. Control Statements: Introduction, if Expression, Nested if Expressions, if–else Expressions, Ternary Operator?:, Switch Statement, Iteration Statements, while Expression, do–while Loop, for Loop,		

Nested for Loop, For-Each for Loop, Break Statement, Continue Statement.

Module-II

Classes and Objects & Methods

10Hrs

Classes and Objects: Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Assigning One Object to Another, Access Control for Class Members, Accessing Private Members of Class, Constructor Methods for Class, Overloaded Constructor Methods, Nested Classes, Final Class and Methods, Passing Arguments by Value and by Reference, Keyword this.

Methods: Introduction, Defining Methods, Overloaded Methods, Overloaded Constructor Methods, Class Objects as Parameters in Methods, Access Control, Recursive Methods, Nesting of Methods, Overriding Methods, Attributes Final and Static.

Module-III

Arrays

10Hrs

Arrays: Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Operations on Array Elements, Assigning Array to Another Array, Dynamic Change of Array Size, Sorting of Arrays, Search for Values in Arrays, Class Arrays, Two-dimensional Arrays, Arrays of Varying Lengths, Three-dimensional Arrays, Arrays as Vectors.

Inheritance: Introduction, Process of Inheritance, Types of Inheritances, Universal Super Class- Object Class, Inhibiting Inheritance of Class Using Final, Access Control and Inheritance, Multilevel Inheritance, Application of Keyword Super, Constructor Method and Inheritance, Method Overriding, Dynamic Method Dispatch, Abstract Classes, Interfaces and Inheritance.

Interfaces: Introduction, Declaration of Interface, Implementation of Interface, Multiple Interfaces, Nested Interfaces, Inheritance of Interfaces, Default Methods in Interfaces, Static Methods in Interface, Functional Interfaces, Annotations.

Module-IV

Packages and Java Library

10Hrs

Packages and Java Library: Introduction, Defining Package, Importing Packages and Classes into Programs, Path and Class Path, Access Control, Packages in Java SE, Java.lang Package and its Classes, Class Object, Enumeration, class Math, Wrapper Classes, Auto-boxing and Auto-unboxing, Java util Classes and Interfaces, Formatter Class, Random Class, Time Package, Class Instant (java.time.Instant), Formatting for Date/Time in Java, Temporal Adjusters Class, Temporal Adjusters Class.

Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Clauses, Class Throwable, Unchecked Exceptions, Checked Exceptions.

Java I/O and File: Java I/O API, standard I/O streams, types, Byte streams, Character streams, Scanner class, Files in Java, [Serialization and Deserialization](#) (Text Book 2)

Module-V

String Handling in Java

9Hrs

Introduction, Interface Char Sequence, Class String, Methods for Extracting Characters from Strings, Comparison, Modifying, Searching; Class String Buffer.

Multithreaded Programming: Introduction, Need for Multiple Threads Multithreaded Programming for Multi-core Processor, Thread Class, Main Thread-Creation of New Threads, Thread States, Thread Priority-Synchronization, Deadlock and Race Situations, Inter-thread Communication - Suspending, Resuming, and Stopping of Threads.

Java Database Connectivity: Introduction, JDBC Architecture, Installing MySQL and MySQL Connector/J, JDBC Environment Setup, Establishing JDBC Database Connections, ResultSet Interface
Java FX GUI: Java FX Scene Builder, Java FX App Window Structure, displaying text and image, event handling, laying out nodes in scene graph, mouse events (Text Book 3)

Text Books:

1. JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2. Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
3. JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

Reference Books:

1. The complete Reference Java, 11thedition, Herbert Schildt,TMH
2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Web References:

- 1.<https://nptel.ac.in/courses/106/105/106105191/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview



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ADVANCED DATA STRUCTURES & ALGORITHM ANALYSIS LAB

(Common to CSE, AI&ML, DS, CS)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0508P	0:0:3:1.5	1.5	CIE: 30 SEE:70	3 Hours	PCC

Course Objectives:

This course will enable students to:

- Acquire practical skills in constructing and managing Data structures
- Apply the popular algorithm design methods in problem-solving scenarios

Course Outcomes (CO):

On completion of this course, student will be able to

- Design and develop programs to solve real world problems with the popular algorithm design methods
- Demonstrate an understanding of Non-Linear data structures by developing implementing the operations on AVL Trees, B-Trees, Heaps and Graphs
- Critically assess the design choices and implementation strategies of algorithms and data structures in complex applications.
- Utilize appropriate data structures and algorithms to optimize solutions for specific computational problems
- Compare the performance of different of algorithm design strategies
- Design algorithms to new real-world problems

Experiments covering the Topics:

- Operations on AVL trees, B-Trees, Heap Trees
- Graph Traversals
- Sorting techniques
- Minimum cost spanning trees
- Shortest path algorithms
- 0/1 Knapsack Problem
- Travelling Salesperson problem
- Optimal Binary Search Trees
- N-Queens Problem
- Job Sequencing

Sample Programs:

Week-1:

1. Construct an AVL tree for a given set of elements which are stored in a file. And implement insert and delete operation on the constructed tree. Write contents of tree into a new file using in-order.

Week-2:

2. Construct B-Tree an order of 5 with a set of 100 random elements stored in array. Implement searching, insertion and deletion operations.

Week-3:

3. Construct Min and Max Heap using arrays, delete any element and display the content of the Heap.

Week-4:

4. Implement BFT and DFT for given graph, when graph is represented by
 - a) Adjacency Matrix
 - b) Adjacency Lists

Week-5:

5. Write a program for finding the bi-connected components in a given graph.

Week-6:

6. Implement Quick sort and Merge sort and observe the execution time for various input sizes (Average, Worst and Best cases).

Week-7:

7. Compare the performance of Single Source Shortest Paths using Greedy method when the graph is represented by adjacency matrix and adjacency lists

Week-8:

8. Implement Job sequencing with deadlines using Greedy strategy.

Week-9:

9. Write a program to solve 0/1 Knapsack problem Using Dynamic Programming.

Week-10:

10. Implement N-Queens Problem Using Backtracking.

Week-11:

11. Use Backtracking strategy to solve 0/1 Knapsack problem.

Week-12:

12. Implement Travelling Sales Person problem using Branch and Bound approach

Reference Books:

1. Fundamentals of Data Structures in C++, Horowitz Ellis, SahniSartaj, Mehta, Dinesh, 2ndEdition, Universities Press
2. Computer Algorithms/C++ Ellis Horowitz, SartajSahni, SanguthevarRajasekaran, 2ndEdition, University Press
3. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
4. An introduction to Data Structures with applications, Trembley& Sorenson, McGraw Hill

Web References:

1. <http://cse01-iiith.vlabs.ac.in/>
2. <http://peterindia.net/Algorithms.html>
3. https://onlinecourses.swayam2.ac.in/cec20_cs03/preview



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OBJECT-ORIENTED PROGRAMMING THROUGH JAVA LAB

(Common to CSE, AI&ML, DS, CS)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0509P	0:0:3:1.5	1.5	CIE: 30 SEE:70	3 Hours	PCC

Course Objectives:

The aim of this course is to:

- Practice object-oriented programming in the Java programming language
- Implement Classes, Objects, Methods, Inheritance, Exception, Runtime Polymorphism, User defined Exception handling mechanism
- Illustrate inheritance, Exception handling mechanism, JDBC connectivity
- Construct Threads, Event Handling, implement packages, Java FX GUI

Course Outcomes (CO):

On completion of this course, student will be able to

- Demonstrate a solid understanding of Java syntax, including data types, control structures, methods, classes, objects, inheritance, polymorphism, and exception handling.
- Apply fundamental OOP principles such as encapsulation, inheritance, polymorphism, and abstraction to solve programming problems effectively.
- Familiar with commonly used Java libraries and APIs, including the Collections Framework, Java I/O, JDBC, and other utility classes.
- Develop problem-solving skills and algorithmic thinking, applying OOP concepts to design efficient solutions to various programming challenges.
- Proficiently construct graphical user interface (GUI) applications using JavaFX.
- Develop new programs for solving typical computer science problems.

Experiments covering the Topics:

- Object Oriented Programming fundamentals- data types, control structures
- Classes, methods, objects, Inheritance, polymorphism,
- Exception handling, Threads, Packages, Interfaces
- Files, I/O streams, JavaFX GUI

Sample Programs:

Week-1:

1. Develop a java program to display default value of all primitive data type of JAVA
2. Develop a java program that display the roots of a quadratic equation $ax^2+bx=0$. Calculate the discriminate D and basing on value of D, describe the nature of root.

Week-2:

3. Develop a JAVA program to search for an element in a given list of elements using binary search mechanism.
4. Develop a JAVA program to sort for an element in a given list of elements using bubble sort

Week-3:

5. Develop a JAVA program using StringBuffer to delete, remove character.

Week-4:

6. Develop a JAVA program to implement class mechanism. Create a class, methods and invoke them inside main method.
7. Develop a JAVA program implement method overloading.

Week-5:

8. Develop a JAVA program to implement constructor.
9. Develop a JAVA program to implement constructor overloading.

Week-6:

10. Develop a JAVA program to implement Single Inheritance
11. Develop a JAVA program to implement multi level Inheritance

Week-7:

12. Develop a JAVA program for abstract class to find areas of different shapes
13. Develop a JAVA program give example for “super” keyword.

Week-8:

14. Develop a JAVA program to implement Interface. What kind of Inheritance can be achieved?
15. Develop a JAVA program that implements Runtime polymorphism

Week-9:

16. Develop a JAVA program that describes exception handling mechanism
17. Develop a JAVA program Illustrating Multiple catch clauses

Week-10:

18. Develop a JAVA program for creation of Java Built-in Exceptions
19. Develop a JAVA program for creation of User Defined Exception

Week-11:

20. Develop a JAVA program that creates threads by extending Thread class. First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds, (Repeat the same by implementing Runnable)
21. Develop a program illustrating is Alive() and join ()

Week-12:

22. Develop a Program illustrating Daemon Threads.
23. Develop a JAVA program Producer Consumer Problem

Week-13:

24. Develop a JAVA program that import and use the user defined packages
25. Without writing any code, build a GUI that display text in label and image in anImageView (use JavaFX)

Week-14:

26. Build a Tip Calculator app using several JavaFX components and learn how to respond to user interactions with the GUI
27. Develop a java program that connects to a database using JDBC

Week-15:

28. Develop a java program to connect to a database using JDBC and insert values into it.
29. Develop a java program to connect to a database using JDBC and delete values from it

Text Books:

- 1 Fundamentals of Data Structures in C++, Horowitz Ellis, SahniSartaj, Mehta, Dinesh, 2ndEdition, Universities Press
- 2 Computer Algorithms/C++ Ellis Horowitz, SartajSahni, SanguthevarRajasekaran, 2ndEdition, University Press
- 3 Data Structures and program design in C, Robert Kruse, Pearson Education Asia
- 4 An introduction to Data Structures with applications, Trembley& Sorenson, McGraw Hill

Reference Books:

1. The complete Reference Java, 11th edition, Herbert Schildt, TMH
2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Web References:

1. <https://nptel.ac.in/courses/106/105/106105191/>

2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview



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PYTHON PROGRAMMING (SKILL ENHANCEMENT COURSE) (Common to CSE, AI&ML, DS, CS)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0510P	0:1:2:2	2	CIE: 30 SEE:70	3 Hours	SEC

Course Objectives:

This course will enable students to:

- Introduce core programming concepts of Python programming language.
- Demonstrate about Python data structures like Lists, Tuples, Sets and dictionaries
- Implement Functions, Modules and Regular Expressions in Python Programming and to create practical and contemporary applications using these

Course Outcomes (CO):

On completion of this course, student will be able to

- Classify data structures of Python
- Apply Python programming concepts to solve a variety of computational problems
- Understand the principles of object-oriented programming (OOP) in Python, including classes, objects, inheritance, polymorphism, and encapsulation, and apply them to design and implement Python programs
- Become proficient in using commonly used Python libraries and frameworks such as JSON, XML, NumPy, pandas
- Exhibit competence in implementing and manipulating fundamental data structures such as lists, tuples, sets, dictionaries
- Propose new solutions to computational problems

Syllabus		Total Hours:48
Module-I	History of Python Programming Language	9Hrs

History of Python Programming Language, Thrust Areas of Python, Installing Anaconda Python Distribution, Installing and Using Jupyter Notebook.

Parts of Python Programming Language: Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types, Indentation, Comments, Reading Input, Print Output, Type Conversions, the type () Function and Is Operator, Dynamic and Strongly Typed Language.

Control Flow Statements: if statement, if-else statement, if...elif...else, Nested if statement, while Loop, for Loop, continue and break Statements, Catching Exceptions Using try and except Statement.

Sample Experiments:

Week-1:

1. Develop a Python Program to find the largest element among three Numbers.
2. Develop a Python Program to display all prime numbers within an interval

Week-2: - Develop a Python Program to swap two numbers without using a temporary variable. - Demonstrate the following Operators in Python with suitable examples. - Arithmetic Operators - Relational Operators - Assignment Operators - Logical Operators - Bit wise Operators - Ternary Operator - Membership Operators - Identity Operators Week-3: - Develop a Python Program to add and multiply complex numbers - Develop a Python Program to print multiplication table of a given number.		
Module-II	Functions, Strings, Lists	10Hrs
Functions: Built-In Functions, Commonly Used Modules, Function Definition and Calling the function, return Statement and void Function, Scope and Lifetime of Variables, Default Parameters, Keyword Arguments, *args and **kwargs, Command Line Arguments. Strings: Creating and Storing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and Joining, String Methods, Formatting Strings. Lists: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods, del Statement. Sample Experiments:		
Week-4: - Develop a Python Program to define a function with multiple return values. - Develop a Python Program to define a function using default arguments. Week-5: - Develop a Python Program to find the length of the string without using any library functions. - Develop a Python Program to check if the substring is present in a given string or not. Week-6: - Develop a Python Program to perform the given operations on a list: - Addition - Insertion - Slicing - Develop a Python Program to perform any 5 built-in functions by taking any list.		
Module-III	Dictionaries	10Hrs
Dictionaries: Creating Dictionary, Accessing and Modifying key:value Pairs in Dictionaries, Built-In Functions Used on Dictionaries, Dictionary Methods, del Statement. Tuples and Sets: Creating Tuples, Basic Tuple Operations, tuple() Function, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists, Relation between Tuples and Dictionaries, Using zip() Function, Sets, Set Methods, Frozenset. Sample Experiments:		
Week-7: - Develop a Python Program to create tuples (name, age, address, college) for at least two members and concatenate the tuples and print the concatenated tuples. - Develop a Python Program to count the number of vowels in a string (No control flow allowed). Week-8: - Develop a Python Program to check if a given key exists in a dictionary or not.		

16. Develop a Python Program to add a new key-value pair to an existing dictionary.
17. Develop a Python Program to sum all the items in a given dictionary.

Module-IV	Files	10Hrs
Files: Types of Files, Creating and Reading Text Data, File Methods to Read and Write Data, Reading and Writing Binary Files, Pickle Module, Reading and Writing CSV Files, Python os and os.path Modules. Object-Oriented Programming: Classes and Objects, Creating Classes in Python, Creating Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes Vs Data Attributes, Encapsulation, Inheritance, Polymorphism. Sample Experiments: Week-9: 18. Develop a Python Program to sort words in a file and put them in another file. The output file should have only lower-case words, so any upper-case words from source must be lowered. 19. Develop a Python Program to print each line of a file in reverse order. Week-10: 20. Develop a Python Program to compute the number of characters, words and lines in a file. 21. Develop a Python Program to create, display, append, insert and reverse the order of the items in the array. Week-11: 22. Develop a Python Program to add, transpose and multiply two matrices. 23. Develop a Python Program to create a class that represents a shape. Include methods to calculate its area and perimeter. Implement subclasses for different shapes like circle, triangle, and square.		
Module-V	Introduction to Data Science	9Hrs
Introduction to Data Science: Functional Programming, JSON and XML in Python, NumPy with Python, Pandas Sample Experiments: Week-12: 24. Develop a Python Program to check whether a JSON string contains complex object or not. 25. Develop a Python Program to demonstrate NumPy arrays creation using array () function. Week-13: 26. Develop a Python Program to demonstrate use of ndim, shape, size, dtype. 27. Develop a Python Program to demonstrate basic slicing, integer and Boolean indexing. Week-14: 28. Develop a Python program to find min, max, sum, cumulative sum of array 29. Create a dictionary with at least five keys and each key represent value as a list where this list contains at least ten values and convert this dictionary as a pandas data frame and explore the data through the data frame as follows: a) Apply head () function to the pandas data frame b) Perform various data selection operations on Data Frame Week-15: 30. Select any two columns from the above data frame, and observe the change in one attribute with respect to other attribute with scatter and plot operations in matplotlib		

Text Books:

1. Gowrishankar S, Veena A., Introduction to Python Programming, CRC Press

Reference Books:

1. Python Programming, S Sridhar, J Indumathi, V M Hariharan, 2nd Edition, Pearson, 2024
2. Introduction to Programming Using Python, Y. Daniel Liang, Pearson.

Web References:

1. <https://www.coursera.org/learn/python-for-applied-data-science-ai>
2. <https://www.coursera.org/learn/python?specialization=python#syllabus>



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ENVIRONMENTAL SCIENCE (Common to CSE, AI&ML, DS, CS)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0109T	2:0:0:0	0	CIE: 30	1.5 hrs	AC
Course Objectives:					
This course will enable students to: • To make the students to get awareness on environment. • To understand the importance of protecting natural resources, ecosystems for future generations and pollution causes due to the day-to-day activities of human life • To save earth from the inventions by the engineers					
Syllabus				Total Hours:48	
Module-I	Multidisciplinary Nature of Environmental Studies			9Hrs	
Multidisciplinary Nature of Environmental Studies: – Definition, Scope and Importance – Need for Public Awareness. Natural Resources : Renewable and non-renewable resources – Natural resources and associated problems – Forest resources – Use and over – exploitation, deforestation, case studies – Timber extraction – Mining, dams and other effects on forest and tribal people – Water resources – Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems – Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies – Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. – Energy resources:					
Module-II	Ecosystems			10Hrs	
Ecosystems: Concept of an ecosystem. – Structure and function of an ecosystem – Producers, consumers and decomposers – Energy flow in the ecosystem – Ecological succession – Food chains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the following ecosystem: a. Forest ecosystem. b. Grassland ecosystem c. Desert ecosystem. d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries) Biodiversity and its Conservation : Introduction 0 Definition: genetic, species and ecosystem diversity – Bio-geographical classification of India – Value of biodiversity: consumptive use, Productive use, social, ethical, aesthetic and option values – Biodiversity at global, National and local levels – India as a mega-diversity nation – Hot-spots of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – Endangered and endemic species of India – Conservation of					

biodiversity: In-situ and Ex-situ conservation of biodiversity.

Module-III	Environmental Pollution	10Hrs
Environmental Pollution: Definition, Cause, effects and control measures of : a. Air Pollution. b. Water pollution c. Soil pollution d. Marine pollution e. Noise pollution f. Thermal pollution g. Nuclear hazards Solid Waste Management: Causes, effects and control measures of urban and industrial wastes – Role of an individual in prevention of pollution – Pollution case studies – Disaster management: floods, earthquake, cyclone and landslides.		
Module-IV	Social Issues and the Environment	10Hrs
Social Issues and the Environment: From Unsustainable to Sustainable development – Urban problems related to energy – Water conservation, rain water harvesting, watershed management – Resettlement and rehabilitation of people; its problems and concerns. Case studies – Environmental ethics: Issues and possible solutions – Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies – Wasteland reclamation. – Consumerism and waste products. – Environment Protection Act. – Air (Prevention and Control of Pollution) Act. – Water (Prevention and control of Pollution) Act – Wildlife Protection Act – Forest Conservation Act – Issues involved in enforcement of environmental legislation – Public awareness.		
Module-V	Human Population and the Environment	9Hrs
Human Population and the Environment: Population growth, variation among nations. Population explosion – Family Welfare Programmes. – Environment and human health – Human Rights – Value Education – HIV/AIDS – Women and Child Welfare – Role of information Technology in Environment and human health – Case studies. Field Work: Visit to a local area to document environmental assets River/forest grassland/hill/mountain – Visit to a local polluted site-Urban/Rural/Industrial/Agricultural Study of common plants, insects, and birds – river, hill slopes, etc.		

Text Books:

1. Textbook of Environmental Studies for Undergraduate Courses Erach Bharucha for University Grants Commission, Universities Press.
2. Palaniswamy, "Environmental Studies", Pearson education
3. S.Azeem Unnisa, "Environmental Studies" Academic Publishing Company
4. K.Raghavan Nambiar, "Text book of Environmental Studies for Undergraduate Courses as per UGC model syllabus", Scitech Publications (India), Pvt. Ltd.

Reference Books:

1. Deeksha Dave and E.Sai Baba Reddy, "Textbook of Environmental Science", Cengage Publications.
2. M.Anji Reddy, "Text book of Environmental Sciences and Technology", BSPublication.
3. J.P.Sharma, Comprehensive Environmental studies, Laxmi publications.
4. J. Glynn Henry and Gary W. Heinke, "Environmental Sciences and Engineering", Prentice hall of India Private limited
5. G.R.Chatwal, "A Text Book of Environmental Studies" Himalaya Publishing House
6. Gilbert M. Masters and Wendell P. Ela, "Introduction to Environmental Engineering and Science, Prentice hall of India Private limited.



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Semester-4 (Theory-6, Lab-3, SC-1, MC-1)							
Sl. No.	Category	Course Code	Course Title	Hours per week			Credits
				L	T	P	
1	MEC	23A0020T	Optimization Techniques	2	0	0	2
2	BSC	23A0017T	Probability & Statistics	3	0	0	3
3	PCC	23A3302T	Machine Learning	3	0	0	3
4	PCC	23A0512T	Database Management Systems	3	0	0	3
5	PCC	23A0406T	Digital Logic and Computer Organization	3	0	0	3
6	PCC(Lab)	23A3303P	Machine Learning Lab	0	0	3	1.5
7	PCC(Lab)	23A0515P	Database Management Systems Lab	0	0	3	1.5
8	SEC	23A0516P	Full Stack Development-1	0	1	2	2
9	ESC	23A0413T	Design Thinking & Innovation	0	1	2	2
Total credits							21
Mandatory Community Service Project Internship of 08 weeks duration during summer vacation							

Category	Credits
Basic Science Course (BSC)	3
Professional Core Courses (PCC)	12
Engineering Science Courses (ESC)	2
Skill Oriented Course (SEC)	2
Management Elective Course (MEC)	2
Total	21



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OPTIMIZATION TECHNIQUES

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0020T	2:0:0:2	2	CIE: 30 SEE:70	3 Hours	MEC

Course Objectives:

The objectives of the course are

- To provide the basic knowledge about Optimization, importance, application areas of the industry, Linear Programming.
- To impart different optimization models under typical situations in the business organization like transportation, assignment.
- To understand the process of sequencing in a typical industry.
- To describe different game strategies under cut-throat competitive business environment
- To develop networks of activities of projects and to find out optimal modes of completing projects using network modelling evaluation techniques

Course Outcomes(CO):

- Understanding Optimization and Formulation of Linear Programming Models
- Formulate and Solve Transportation & Assignment Models
- Sequencing of operations and optimizing
- Discuss the game theory and strategies
- Developing networks of activities and finding optimal mode of projects evaluation

Syllabus		Total Hours:48
Module-I	Introduction	9Hrs

Meaning, Nature, Scope & Significance of Optimization - Typical applications. The Linear Programming Problem – Introduction, Formulation of Linear Programming problem, Limitations of L.P.P, Graphical method, Simplex method: Maximization and Minimization model(exclude Duality problems), Big-M method and Two Phase method.

Module-II	Transportation Problem	10Hrs
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Introduction, Transportation Model, Finding initial basic feasible solutions, Moving towards optimality, Unbalanced Transportation problems, Transportation problems with maximization, Degeneracy.

Assignment Problem – Introduction, Mathematical formulation of the problem, Solution of an Assignment problem, Hungarian Algorithm, Multiple Solution, Unbalanced Assignment problems, Maximization in Assignment Model.

Module-III	Sequencing	10Hrs
Job sequencing, Johnsons Algorithm for n Jobs and Two machines, n Jobs and Three Machines, n jobs through m machines, Two jobs and m Machines Problems		
Module-IV	Game Theory	10Hrs
Concepts, Definitions and Terminology, Two Person Zero Sum Games, Pure Strategy Games (with Saddle Point), Principal of Dominance, Mixed Strategy Games (Game without Saddle Point), Significance of Game Theory in Managerial Application.		
Module-V	Project Management	9Hrs
Network Analysis – Definition –objectives -Rules for constructing network diagram- Determining Critical Path – Earliest & Latest Times – Floats - Application of CPM and PERT techniques in Project Planning and Control – PERT Vs CPM. (exclude Project Crashing).		
Text Books: 1. Operations Research / R.Pannarselvam, PHI Publications. 2. Operations Research / S.D.Sharma-Kedarnath 3. Operations Research /A.M.Natarajan,P.Balasubramani,A. Tamilarasi/Pearson Education. 4. Engineering Optimization: Theory and practice / S.S.Rao, New Age International (P) Limited.		
Reference Books: 1. Quantitative Techniques in Management / ND Vohra, Tata McGraw Hill, 4th Edition, 2011. 2. Introduction to O.R/Hiller & Libermann (TMH). 3. Operations Research: Methods & Problems / Maurice Saseini, Arthur Yaspan & Lawrence Friedman. Pearson 4. Quantitative Analysis For Management/ Barry Render, Ralph M. Stair, Jr and Michael E.Hanna/ 5. Operations Research / Wagner/ PHI Publications.		
Online Learning Sources: https://onlinecourses.swayam2.ac.in/cec20_ma10/preview https://onlinecourses.nptel.ac.in/noc20_ma23/preview https://onlinecourses.nptel.ac.in/noc19_ma29/preview		



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PROBABILITY & STATISTICS

(Common to CSE, AI&ML, DS, CS)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0017T	3:0:0:3	3	CIE: 30 SEE:70	3 Hours	BSC

Course Objectives:

This course will enable students to:

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Course Outcomes(CO):

On completion of this course, student will be able to

- Acquire knowledge in finding the analysis of the data quantitatively or categorically and various statistical elementary tools.
- Develop skills in designing mathematical models involving probability, random variables and the critical thinking in the theory of probability and its applications in real life problems.
- Apply the theoretical probability distributions like binomial, Poisson, and Normal in the relevant application areas.
- Analyze to test various hypotheses included in theory and types of errors for large samples.
- Apply the different testing tools like t-test, F-test, chi-square test to analyze the relevant real life problems.

Syllabus		Total Hours:48
Module-I	Descriptive statistics	9Hrs
Statistics Introduction, Population vs Sample, Collection of data, primary and secondary data, Measures of Central tendency, Measures of Variability (spread or variance) Skewness, Kurtosis, correlation, correlation coefficient, rank correlation, regression coefficients, method of least squares, regression lines.		
Module-II	Probability	10Hrs
Probability, probability axioms, addition law and multiplicative law of probability, conditional probability, Baye's theorem, random variables (discrete and continuous), probability density functions, properties, mathematical expectation.		
Module-III	Probability distributions	10Hrs
Probability distributions: Binomial, Poisson and Normal-their properties (Chebyshevs inequality). Approximation of the binomial distribution to normal distribution.		

Module-IV	Estimation and Testing of hypothesis, large sample tests	10Hrs
Estimation-parameters, statistics, sampling distribution, point estimation, Formulation of null hypothesis, alternative hypothesis, the critical and acceptance regions, level of significance, two types of errors and power of the test. Large Sample Tests: Test for single proportion, difference of proportions, test for single mean and difference of means. Confidence interval for parameters in one sample and two sample problems		
Module-V	Small sample tests	9Hrs
Student t-distribution (test for single mean, two means and paired t-test), testing of equality of variances (F-test), χ^2 - test for goodness of fit, χ^2 - test for independence of attributes.		
Text Books: 1. Miller and Freunds, Probability and Statistics for Engineers, 7/e, Pearson, 2008. 2. S.C. Gupta and V.K. Kapoor, Fundamentals of Mathematical Statistics, 11/e, Sultan Chand & Sons Publications, 2012.		
Reference Books: 1. S. Ross, a First Course in Probability, Pearson Education India, 2002. 2. W. Feller, an Introduction to Probability Theory and its Applications, 1/e, Wiley, 1968. 3. B. V. Ramana, Higher Engineering Mathematics, Mc Graw Hill Education.		
Web References: 1. https://onlinecourses.nptel.ac.in/noc21_ma74/preview 2. https://onlinecourses.nptel.ac.in/noc22_mg31/preview		



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MACHINE LEARNING

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3302T	3:0:0:3	3	CIE: 30 SEE:70	3 Hours	PCC

Course Objectives:

The objectives of the course are

- Define machine learning and its different types (supervised and unsupervised) and understand their applications.
- Apply supervised learning algorithms including decision trees and k-nearestneighbors (k-NN).
- Implement unsupervised learning techniques, such as K-means clustering.

Course Outcomes(CO):

- CO1: Identify machine learning techniques suitable for a given problem. (L3)
- CO2: Solve real-world problems using various machine learning techniques. (L3)
- CO3: Apply Dimensionality reduction techniques for data preprocessing. (L3)
- CO4: Explain what is learning and why it is essential in the design of intelligent machines. (L2)
- CO5: Evaluate Advanced learning models for language, vision, speech, decisionmaking etc. (L5)

Syllabus

Total Hours:48

Module-I

Introduction to Machine Learning

9Hrs

Evolution of Machine Learning, Paradigms for ML, Learning by Rote, Learning by Induction, Reinforcement Learning, Types of Data, Matching, Stages in Machine Learning, Data Acquisition, Feature Engineering, Data Representation, Model Selection, Model Learning, Model Evaluation, Model Prediction, Search and Learning, Data Sets.

Module-II

Nearest Neighbor-Based Models

10Hrs

Introduction to Proximity Measures, Distance Measures, Non-Metric Similarity Functions, Proximity Between Binary Patterns, Different Classification Algorithms Based on the Distance Measures ,K-Nearest Neighbor Classifier, Radius Distance Nearest Neighbor Algorithm, KNN Regression, Performance of Classifiers, Performance of Regression Algorithms.

Module-III

Models Based on Decision Trees

10Hrs

Decision Trees for Classification, Impurity Measures, Properties, Regression Based on Decision Trees, Bias-Variance Trade-off, Random Forests for Classification and Regression.

The Bayes Classifier: Introduction to the Bayes Classifier, Bayes' Rule and Inference, The Bayes Classifier and its Optimality, Multi-Class Classification | Class Conditional Independence and Naive Bayes Classifier (NBC)

Module-IV	Linear Discriminants for Machine Learning	10Hrs
Introduction to Linear Discriminants, Linear Discriminants for Classification, Perceptron Classifier, Perceptron Learning Algorithm, Support Vector Machines, Linearly Non-Separable Case, Non-linear SVM, Kernel Trick, Logistic Regression, Linear Regression, Multi-Layer Perceptrons (MLPs), Backpropagation for Training an MLP.		
Module-V	Clustering	9Hrs
Introduction to Clustering, Partitioning of Data, Matrix Factorization Clustering of Patterns, Divisive Clustering, Agglomerative Clustering, Partitional Clustering, K-Means Clustering, Soft Partitioning, Soft Clustering, Fuzzy C-Means Clustering, Rough Clustering, Rough K-Means Clustering Algorithm, Expectation Maximization-Based Clustering, Spectral Clustering.		
Text Books: 1. Machine Learning Theory and Practice”, M N Murthy, V S Ananthanarayana, Universities Press (India), 2024		
Reference Books: 1. Machine Learning”, Tom M. Mitchell, McGraw-Hill Publication, 2017 2. “Machine Learning in Action”, Peter Harrington, DreamTech 2. 3. “Introduction to Data Mining”, Pang-Ning Tan, Michel Stenbach, Vipin Kumar, 7th Edition, 2019.		



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DATABASE MANAGEMENT SYSTEMS

(Common to CSE, AI&ML, DS, CS)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0512T	3:0:0:3	3	CIE: 30 SEE:70	3 Hours	PCC

Course Objectives:

This course will enable students to:

- Introduce database management systems and to give a good formal foundation on the relational model of data and usage of Relational Algebra
- Introduce the concepts of basic SQL as a universal Database language
- Demonstrate the principles behind systematic database design approaches by covering conceptual design, logical design through normalization
- Provide an overview of physical design of a database system, by discussing Database indexing techniques and storage techniques

Course Outcomes(CO):

On completion of this course, student will be able to

- Understand the basic concepts of database management systems (L2)
- Analyze a given database application scenario to use ER model for conceptual design of the database (L4)
- Utilize SQL proficiently to address diverse query challenges (L3).
- Employ normalization methods to enhance database structure (L3)
- Assess and implement transaction processing, concurrency control and database recovery protocols in databases. (L4)

Syllabus		Total Hours:48
Module-I	Introduction	9Hrs
Introduction: Database system, Characteristics (Database Vs File System), Database Users, Advantages of Database systems, Database applications. Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture for data independence; Database system structure, environment, Centralized and Client Server architecture for the database. Entity Relationship Model: Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, specialization, generalization using ER Diagrams.		
Module-II	Relational Model	10Hrs
Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra, Relational Calculus. BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update).		

Module-III	SQL	10Hrs
SQL: Basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions(Date and Time, Numeric, String conversion).Creating tables with relationship, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, ordering, implementation of different types of joins, view(updatable and non-updatable), relational set operations.		
Module-IV	Schema Refinement	10Hrs
Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependency Lossless join and dependency preserving decomposition, (1NF, 2NF and 3 NF), concept of surrogate key, Boyce-Codd normal form(BCNF), MVD, Fourth normal form(4NF), Fifth Normal Form (5NF).DeNormalization		
Module-V	Transaction Concept	9Hrs
Transaction Concept: Transaction State, ACID properties, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for Serializability, lock based, time stamp based, optimistic, concurrency protocols, Deadlocks, Failure Classification, Storage, Recovery and Atomicity, Recovery algorithm. Introduction to Indexing Techniques: B+ Trees, operations on B+Trees, Hash Based Indexing:		
Text Books: 1. Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke,TMH (For Chapters 2, 3, 4) 2. Database System Concepts,5th edition, Silberschatz, Korth, Sudarsan,TMH (ForChapter 1 and Chapter 5)		
Reference Books: 1. Introduction to Database Systems, 8thedition, C J Date, Pearson. 2. Database Management System, 6th edition, RamezElmasri, Shamkant B. Navathe, Pearson 3. Database Principles Fundamentals of Design Implementation and Management,Corlos Coronel, Steven Morris, Peter Robb, Cengage Learning.		
Web References: 1. https://nptel.ac.in/courses/106/105/106105175/ 2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview		



GEETHANJALI INSTITUTE OF SCIENCE & TECHNOLOGY

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3rd Mile, Bombay Highway, Gangavaram (V), Kovur(M), SPSR Nellore (Dt), Andhra Pradesh, India- 524137

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DIGITAL LOGIC & COMPUTER ORGANIZATION

(Common to CSE, AI&ML, DS, CS)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0406T	3:0:0:3	3	CIE: 30 SEE:70	3 Hours	PCC

Course Objectives:

This course will enable students to:

- provide students with a comprehensive understanding of digital logic design principles and computer organization fundamentals
- Describe memory hierarchy concepts
- Explain input/output (I/O) systems and their interaction with the CPU, memory, and peripheral devices

Course Outcomes (CO):

On completion of this course, student will be able to

- Differentiate between combinational and sequential circuits based on their characteristics and functionalities.
- Demonstrate an understanding of computer functional units.
- Analyze the design and operation of processors, including instruction execution, pipelining, and control unit mechanisms, to comprehend their role in computer systems.
- Describe memory hierarchy concepts, including cache memory, virtual memory, and secondary storage, and evaluate their impact on system performance and scalability.
- Explain input/output (I/O) systems and their interaction with the CPU, memory, and peripheral devices, including interrupts, DMA, and I/O mapping techniques.
- Design Sequential and Combinational Circuits

Syllabus		Total Hours:48
Module-I	Data Representation	9Hrs
Data Representation: Binary Numbers, Fixed Point Representation. Floating Point Representation. Number base conversions, Octal and Hexadecimal Numbers, components, Signed binary numbers, Binary codes Digital Logic Circuits-I: Basic Logic Functions, Logic gates, universal logic gates, Minimization of Logic expressions. K-Map Simplification, Combinational Circuits, Decoders, Multiplexers		
Module-II	Digital Logic Circuits	10Hrs
Digital Logic Circuits-II: Sequential Circuits, Flip-Flops, Binary counters, Registers, Shift Registers, Ripple counters Basic Structure of Computers: Computer Types, Functional units, Basic operational concepts, Bus structures, Software, Performance, multiprocessors and multi computers, Computer Generations, Von-Neumann Architecture		

Module-III	Computer Arithmetic	10Hrs
Computer Arithmetic: Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Positive Numbers, Signed-operand Multiplication, Fast Multiplication, Integer Division, Floating-Point Numbers and Operations. Processor Organization: Fundamental Concepts, Execution of a Complete Instruction, Multiple-Bus Organization, Hardwired Control and Multi programmed Control		
Module-IV	The Memory Organization	10Hrs
Basic Concepts, Semiconductor RAM Memories, Read-Only Memories, Speed, Size and Cost, Cache Memories, Performance Considerations, Virtual Memories, Memory Management Requirements, Secondary Storage		
Module-V	Input /Output Organization	9Hrs
Accessing I/O Devices, Interrupts, Processor Examples, Direct Memory Access, Buses, Interface Circuits, Standard I/O Interfaces, Arbitration		
Text Books: 1. Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, 6th edition, McGraw Hill, 2023. 2. Digital Design, 6th Edition, M. Morris Mano, Pearson Education, 2018. 3. Computer Organization and Architecture, William Stallings, 11th Edition, Pearson, 2022.		
Reference Books: 1. Computer Systems Architecture, M. Morris Mano, 3rd Edition, Pearson, 2017. 2. Computer Organization and Design, David A. Paterson, John L. Hennessy, Elsevier, 2004. 3. Fundamentals of Logic Design, Roth, 5th Edition, Thomson, 2003.		
Web References: https://nptel.ac.in/courses/106/103/106103068/		



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MACHINE LEARNING LAB					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3303P	0:0:3:1.5	1.5	CIE: 30 SEE:70	3 Hours	PCC
Course Objectives:					
• To learn about computing central tendency measures and Data preprocessing techniques • To learn about classification and regression algorithms • To apply different clustering algorithms for a problem.					
Experiments:				Total Hours:48	
WEEK-1 1. Compute Central Tendency Measures: Mean, Median, Mode Measure of Dispersion: Variance, Standard Deviation.					
WEEK-2 2. Apply the following Pre-processing techniques for a given dataset. a. Attribute selection b. Handling Missing Values c. Discretization d. Elimination of Outliers.					
WEEK-3 3. Apply KNN algorithm for classification and regression					
WEEK-4 4. Demonstrate decision tree algorithm for a classification problem and perform parameter tuning for better results.					
WEEK-5 5. Demonstrate decision tree algorithm for a regression problem.					
WEEK-6 6. Apply Random Forest algorithm for classification and regression.					
WEEK-7 7. Demonstrate Naïve Bayes Classification algorithm.					
WEEK-8 8. Apply Support Vector algorithm for classification.					
WEEK-9 9. Demonstrate simple linear regression algorithm for a regression problem.					
WEEK-10 10. Apply Logistic regression algorithm for a classification problem.					
WEEK-11 11. Demonstrate Multi-layer Perceptron algorithm for a classification problem					
WEEK-12 12. Implement the K-means algorithm and apply it to the data you selected. Evaluate performance by measuring the sum of the Euclidean distance of each example from its class center. Test the performance of the algorithm as a function of the parameters K.					

WEEK-13

13. Demonstrate the use of Fuzzy C-Means Clustering

WEEK-14

14. Demonstrate the use of Expectation Maximization based clustering algorithm



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DATABASE MANAGEMENT SYSTEMS LAB					
(Common to CSE, AI&ML, DS, CS)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0515P	0:0:3:1.5	1.5	CIE: 30 SEE:70	3 Hours	PCC
Course Objectives:					
This course will enable students to: • Populate and query a database using SQL DDL/DML Commands • Declare and enforce integrity constraints on a database • Writing Queries using advanced concepts of SQL • Programming PL/SQL including procedures, functions, cursors and triggers.					
Course Outcomes(CO):					
On completion of this course, student will be able to • Utilizing Data Definition Language (DDL), Data Manipulation Language (DML), and Data Control Language (DCL) commands effectively within a database environment (L3) • Constructing and execute queries to manipulate and retrieve data from databases. (L3) • Develop application programs using PL/SQL. (L3) • Analyze requirements and design custom Procedures, Functions, Cursors, and Triggers, leveraging their capabilities to automate tasks and optimize database functionality (L4) • Establish database connectivity through JDBC (Java Database Connectivity) (L3)					
Experiments:				Total Hours:48	
Week-1: 1. Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.					
Week-2: 2. Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSET, Constraints. Example:- Select the roll number and name of the student who secured fourth rank in the class.					
Week-3: 3. Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.					
Week-4: 4. Queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)					
Week-5 5. i. Create a simple PL/SQL program which includes declaration section, executable					

section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found)

- ii. Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.

Week-6:

6. Develop a PL/SQL program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.

Week-7:

7. Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, USER defined Exceptions, RAISE- APPLICATION ERROR.

Week-8:

8. Program development using creation of procedures, passing parameters IN and OUT of PROCEDURES.

Week-9:

9. Program development using creation of stored functions, invoke functions in SQL Statements and write complex functions.

Week-10:

10. Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.

Week-11:

11. Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers

Week-12:

12. Create a table and perform the search operation on table using indexing and non-indexing techniques.

Week-13:

13. Develop a Java program that connects to a database using JDBC

Week-14:

14. Develop a Java program to connect to a database using JDBC and insert values into it

Week-15:

15. Develop a Java program to connect to a database using JDBC and delete values from it

Experiments covering the topics:

- DDL, DML, DCL commands
- Queries, nested queries, built-in functions,
- PL/SQL programming- control structures
- Procedures, Functions, Cursors, Triggers,
- Database connectivity- ODBC/JDBC

Text Books:

1. Oracle: The Complete Reference by Oracle Press
2. Nilesh Shah, "Database Systems Using Oracle", PHI, 2007
3. Rick F Vander Lans, "Introduction to SQL", Fourth Edition, Pearson Education, 2007



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FULL STACK DEVELOPMENT – 1

(Skill Enhancement Course)

(Common to CSE, AI&ML, DS, CS)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0516P	0:1:2:2	2	CIE: 30 SEE:70	3 Hours	SEC

Course Objectives:

This course will enable students to:

- Make use of HTML elements and their attributes for designing static web pages
- Build a web page by applying appropriate CSS styles to HTML elements
- Experiment with JavaScript to develop dynamic web pages and validate forms

Course Outcomes(CO):

On completion of this course, student will be able to

- CO1: Design Websites. (L6)
- CO2: Apply Styling to web pages. (L4)
- CO3: Make Web pages interactive. (L6)
- CO4: Design Forms for applications. (L6)
- CO5: Choose Control Structure based on the logic to be implemented. (L3)
- CO6: Understand HTML tags, Attributes and CSS properties (L2)

Experiments:

Total Hours:48

1. Lists, Links and Images

Week-1:

- a. Develop a HTML program, to explain the working of lists.

Note: It should have an ordered list, unordered list, nested lists and ordered list in an unordered list and definition lists.

- b. Develop a HTML program, to explain the working of hyperlinks using <a> tag and href, target Attributes.

Week-2:

- c. Create a HTML document that has your image and your friend's image with a specific height and width. Also when clicked on the images it should navigate to their respective profiles.

- d. Develop a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like to 100*100 pixels. Each thumbnail image is also a link to a full sized version of the image. Create an image gallery using this technique

2. HTML Tables, Forms and Frames

Week-3:

- a. Develop a HTML program, to explain the working of tables. (use tags: <table>, <tr>, <th>, <td> and attributes: border, rowspan, colspan)

- b. Develop a HTML program, to explain the working of tables by preparing a timetable. (Note: Use <caption> tag to set the caption to the table & also use cell spacing, cell padding, border, rowspan, colspan etc.).

Week-4:

c. Develop a HTML program, to explain the working of forms by designing Registration form. (Note: Include text field, password field, number field, date of birth field, checkboxes, radio buttons, list boxes using <select>&<option> tags, <text area> and two buttons ie: submit and reset. Use tables to provide a better view).

d. Develop a HTML program, to explain the working of frames, such that page is to be divided into 3 parts on either direction. (Note: first frame image, second frame paragraph, third frame ☐ hyperlink. And also make sure of using “no frame” attribute such that frames to be fixed).

3. HTML 5 and Cascading Style Sheets, Types of CSS**Week-5:**

- a. Develop a HTML program, that makes use of <article>, <aside>, <figure>, <figcaption>, <footer>, <header>, <main>, <nav>, <section>, <div>, tags.
- b. Develop a HTML program, to embed audio and video into HTML web page.
- c. Develop a program to apply different types (or levels of styles or style specification formats) - **inline, internal, external styles to HTML elements. (identify selector, property and value).**

4. Selector forms**Week-6:**

- a. Develop a HTML program to apply different types of selector forms
 - Simple selector (element, id, class, group, universal)
 - Combinator selector (descendant, child, adjacent sibling, general sibling)
 - Pseudo-class selector
 - Pseudo-element selector
 - Attribute selector

5. CSS with Color, Background, Font, Text and CSS Box Model**Week-7:**

- a. Develop a HTML program to demonstrate the various ways you can reference a color in CSS.
- b. Write a CSS rule that places a background image halfway down the page, tilting it horizontally. The image should remain in place when the user scrolls up or down.

Week-8:

- c. Develop a HTML program using the following terms related to CSS font and text:
 - i. font-size
 - ii. font-weight
 - iii. font-style
 - iv. text-decoration
 - v. text-transformation
 - vi. text-alignment
- d. Develop a HTML program, to explain the importance of CSS Box model using
 - i. Content
 - ii. Border
 - iii. Margin
 - iv. padding

6. Applying JavaScript - internal and external, I/O, Type Conversion**Week-9:**

- a. Develop a program to embed internal and external JavaScript in a web page.

- b. Develop a program to explain the different ways for displaying output using JavaScript.
- c. Develop a program to explain the different ways for taking input using Java script.
- d. Create a webpage which uses prompt dialogue box to ask a voter for his name and age. Display the information in table format along with either the voter can vote or not

7. JavaScript Pre-defined and User-defined Objects

Week-10:

- a. Develop a program using document object properties and methods in Java script.
- b. Develop a program using window object properties and methods in Java script.

Week-11:

- c. Develop a program using array object properties and methods in JavaScript.
- d. Develop a program using math object properties and methods in JavaScript.
- e. Develop a program using string object properties and methods in JavaScript.

Week-12:

- f. Develop a program using regex object properties and methods in JavaScript.
- g. Develop a program using date object properties and methods in JavaScript.
- h. Develop a program to explain user-defined object by using properties, methods, accessors, constructors and display in JavaScript.

8. JavaScript Conditional Statements and Loops

Week-13:

- a. Develop a program which asks the user to enter three integers, obtains the numbers from the user and outputs HTML text that displays the larger number followed by the words “LARGER NUMBER” in an information message dialog. If the numbers are equal, output HTML text as “EQUAL NUMBERS”.
- b. Develop a program to display week days using switch case in JavaScript.
- c. Develop a program to print 1 to 10 numbers using for, while and do-while loops in JavaScript.

Week-14:

- d. Develop a program to print data in object using for-in, for-each and for-of loops in JavaScript
- e. Develop a program to determine whether a given number is an ‘ARMSTRONG NUMBER’ or not in JavaScript. [Eg: 153 is an Armstrong number, since sum of the cube of the digits is equal to the number i.e., $13^3 + 53^3 + 33^3 = 153$]
- f. Develop a program to display the denomination of the amount deposited in the bank in terms of 100's, 50's, 20's, 10's, 5's, 2's & 1's. (Eg: If deposited amount is Rs.163, the output should be 1-100's, 1-50's, 1-10's, 1-2's & 1-1's)

9. Javascript Functions and Events

Week-15:

- a. Design a appropriate function should be called to display
 - Factorial of that number
 - Fibonacci series up to that number
 - Prime numbers up to that number
 - Is it palindrome or not
- b. Design a HTML page having a text box and four buttons named Factorial, Fibonacci, Prime,

and Palindrome. When a button is pressed an appropriate function should be called to display
Factorial of that number
Fibonacci series up to that number
Prime numbers up to that number
Is it palindrome or not

- c. Develop a program to validate the following fields in a registration page
- Name (start with alphabet and followed by alphanumeric and the length should not be less than 6 characters)
 - Mobile (only numbers and length 10 digits)
 - E-mail (should contain format like xxxxxxx@xxxxxx.xxx)

Experiments covering the Topics:

- Lists, Links and Images
- HTML Tables, Forms and Frames
- HTML 5 and Cascading Style Sheets, Types of CSS
- Selector forms
- CSS with Color, Background, Font, Text and CSS Box Model
- Applying JavaScript - internal and external, I/O, Type Conversion
- JavaScript Conditional Statements and Loops, Pre-defined and User-defined Objects
- JavaScript Functions and Events
- Node.js

Text Books:

1. Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2013.
2. Web Programming with HTML5, CSS and JavaScript, John Dean, Jones & Bartlett Learning, 2019 (Chapters 1-11).
3. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O'Reilly.

Reference Books:

Web References:

1. <https://www.w3schools.com/html>
2. <https://www.w3schools.com/css>
3. <https://www.w3schools.com/js/>
4. <https://www.w3schools.com/nodejs>
5. <https://www.w3schools.com/typescript>



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DESIGN THINKING FOR INNOVATION

(Common to CSE, AI&ML, DS, CS)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0413T	1:0:2:2	2	CIE: 30 SEE:70	3 Hours	ESC
Course Objectives:					
This course will enable students to: The objective of this course is to familiarize students with design thinking process as a toolfor breakthrough innovation. It aims to equip students with design thinking skills and ignite the minds to create innovative ideas, develop solutions for real-time problems.					
Course Outcomes (CO):					
On completion of this course, student will be able to • Define the concepts related to design thinking. (L1, L2) • Explain the fundamentals of Design Thinking and innovation (L1, L2) • Apply the design thinking techniques for solving problems in various sectors. (L3) • Analyse to work in a multidisciplinary environment (L4) • Evaluate the value of creativity (L5) □ Formulate specific problem statements of real time issues (L3, L6)					
Syllabus				Total Hours:48	
Module-I	Introduction to Design Thinking			9Hrs	
Introduction to elements and principles of Design, basics of design-dot, line, shape, form as fundamental design components. Principles of design. Introduction to design thinking, history of Design Thinking, New materials in Industry.					
Module-II	Design Thinking Process			10Hrs	
Design thinking process (empathize, analyze, idea & prototype), implementing the process in driving inventions, design thinking in social innovations. Tools of design thinking - person, costumer, journey map, brainstorming, product development Activity: Every student presents their idea in three minutes, Every student can present design process in the form of flow diagram or flow chart etc. Every student should explain about product development.					
Module-III	Innovation			10Hrs	
Art of innovation, Difference between innovation and creativity, role of creativity and innovation in organizations- Creativity to Innovation- Teams for innovation- Measuring the impact and value of creativity. Activity: Debate on innovation and creativity, Flow and planning from idea to innovation, Debate on value-based innovation.					
Module-IV	Product Design			10Hrs	

Problem formation, introduction to product design, Product strategies, Product value, Product planning, product specifications- Innovation towards product design- Case studies

Activity: Importance of modelling, how to set specifications, Explaining their own product design.

Module-V	Design Thinking in Business Processes	9Hrs
Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining Relevance, Extreme competition, Standardization. Design thinking to meet corporate needs- Design thinking for Startups- Defining and testing Business Models and Business Cases-Developing & testing prototypes. Activity: How to market our own product, About maintenance, Reliability and plan for startup.		
Text Books: 1. Tim Brown, Change by design, Harper Bollins (2009) 2. Idris Mootee, Design Thinking for Strategic Innovation, 2013, John Wiley & Sons.		
Reference Books: 1. David Lee, Design Thinking in the Classroom, Ulysses press 2. Shruti N Shetty, Design the Future, Norton Press 3. William Lidwell, Universal Principles of Design- Kritin Holden, Jill Butter. 4. Chesbrough, H., The Era of Open Innovation – 2013		
Web References: https://nptel.ac.in/courses/110/106/110106124/ https://nptel.ac.in/courses/109/104/109104109/ https://swayam.gov.in/nd1_noc19_mg60/preview		



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COMMUNITY SERVICE PROJECT

Experiential learning through community engagement

(Common to CSE, AI&ML, DS, CS)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3304P	0:0:0:0	2	CIE: 30 SEE:70	3 Hours	PCC

Introduction

- Community Service Project is an experiential learning strategy that integrates meaningful community service with instruction, participation, learning and community development.
- Community Service Project involves students in community development and service activities and applies the experience to personal and academic development.
- Community Service Project is meant to link the community with the college for mutual benefit. The community will benefit with the focused contribution of the college students for the village/ local development. The college finds an opportunity to develop social sensibility and responsibility among students and emerge as a socially responsible institution.

Objective

Community Service Project should be an integral part of the curriculum, as an alternative to the 2 months of Summer Internships / Apprenticeships / On the Job Training, whenever there is an exigency when students cannot pursue their summer internships. The specific objectives are;

- To sensitize the students to the living conditions of the people who are around them,
- To help students to realize the stark realities of society.
- To bring about an attitudinal change in the students and help them to develop societal consciousness, sensibility, responsibility and accountability
- To make students aware of their inner strength and help them to find new /out of box solutions to social problems.
- To make students socially responsible citizens who are sensitive to the needs of the disadvantaged sections.
- To help students to initiate developmental activities in the community in coordination with public and government authorities.
- To develop a holistic life perspective among the students by making them study culture, traditions, habits, lifestyles, resource utilization, wastages and its management, social problems, public administration system and the roles and responsibilities of different persons across different social systems.

Implementation of Community Service Project

- Every student should put in 6 weeks for the Community Service Project during the summer vacation.
- Each class/section should be assigned with a mentor.

- Specific Departments could concentrate on their major areas of concern. For example, Dept. of Computer Science can take up activities related to Computer Literacy to different sections of people like - youth, women, housewives, etc
- A logbook must be maintained by each of the students, where the activities undertaken/involved to be recorded.
- The logbook has to be countersigned by the concerned mentor/faculty in charge.
- An evaluation to be done based on the active participation of the student and grade could be awarded by the mentor/faculty member.
- The final evaluation to be reflected in the grade memo of the student.
- The Community Service Project should be different from the regular programs of NSS/NCC/Green Corps/Red Ribbon Club, etc.
- Minor project reports should be submitted by each student. An internal Viva shall also be conducted by a committee constituted by the principal of the college.
- Award of marks shall be made as per the guidelines of Internship/apprentice/ on the job training.

Procedure

- A group of students or even a single student could be assigned for a particular habitation or village or municipal ward, as far as possible, in the near vicinity of their place of stay, to enable them to commute from their residence and return back by evening or so.
- The Community Service Project is a twofold one –
 - First, the student/s could conduct a survey of the habitation, if necessary, in terms of their own domain or subject area. Or it can even be a general survey, incorporating all the different areas. A common survey format could be designed. This should not be viewed as a duplication of work by the Village or Ward volunteers, rather, it could be another primary source of data.
 - Secondly, the student/s could take up a social activity, concerning their domain or subject area. The different areas, could be like –
 - Agriculture
 - Health
 - Marketing and Cooperation
 - Animal Husbandry
 - Horticulture
 - Fisheries
 - Sericulture
 - Revenue and Survey
 - Natural Disaster Management
 - Irrigation
 - Law & Order
 - Excise and Prohibition
 - Mines and Geology
 - Energy
 - Internet
 - Free Electricity
 - Drinking Water

EXPECTED OUTCOMES BENEFITS OF COMMUNITY SERVICE PROJECT TO STUDENTS
Learning Outcomes • Positive impact on students' academic learning • Improves students' ability to apply what they have learned in "the real world" • Positive impact on academic outcomes such as demonstrated complexity of understanding, problem analysis, problem-solving, critical thinking, and cognitive development. • Improved ability to understand complexity and ambiguity Personal Outcomes Greater sense of personal efficacy, personal identity, spiritual growth, and moral development Greater interpersonal development, particularly the ability to work well with others, and build leadership and communication skills. Social Outcomes • Reduced stereotypes and greater inter-cultural understanding • Improved social responsibility and citizenship skills • Greater involvement in community service after graduation Career Development • Connections with professionals and community members for learning and career opportunities • Greater academic learning, leadership skills, and personal efficacy can lead to greater opportunity. Relationship with the Institution • Stronger relationships with faculty • Greater satisfaction with college • Improved graduation rates
BENEFITS OF COMMUNITY SERVICE PROJECT TO FACULTY MEMBERS • Satisfaction with the quality of student learning • New avenues for research and publication via new relationships between faculty and community • Providing networking opportunities with engaged faculty in other disciplines or institutions • A stronger commitment to one's research.
BENEFITS OF COMMUNITY SERVICE PROJECT TO COLLEGES AND UNIVERSITIES • Improved institutional commitment. • Improved student retention • Enhanced community relations

BENEFITS OF COMMUNITY SERVICE PROJECT TO COMMUNITY

- Satisfaction with student participation
- Valuable human resources needed to achieve community goals.
- New energy, enthusiasm and perspectives applied to community work.
- Enhanced community-university relations.

SUGGESTIVE LIST OF PROGRAMMES UNDER COMMUNITY SERVICE PROJECT

The following the recommended list of projects for Engineering students. The lists are not exhaustive and open for additions, deletions, and modifications. Colleges are expected to focus on specific local issues for this kind of project. The students are expected to carry out these projects with involvement, commitment, responsibility, and accountability. The mentors of a group of students should take the responsibility of motivating, facilitating, and guiding the students. They have to interact with local leadership and people and appraise the objectives and benefits of this kind of project. The project reports shall be placed in the college website for reference. Systematic, Factual, methodical and honest reporting should be ensured.

For Engineering Students

1. Water facilities and drinking water availability
2. Health and hygiene
3. Stress levels and coping mechanisms
4. Health intervention programmes
5. Horticulture
6. Herbal plants
7. Botanical survey
8. Zoological survey
9. Marine products
10. Aqua culture
11. Inland fisheries
12. Animals and species
13. Nutrition
14. Traditional health care methods
15. Food habits
16. Air pollution
17. Water pollution
18. Plantation
19. Soil protection
20. Renewable energy
21. Plant diseases
22. Yoga awareness and practice
23. Health care awareness programmes and their impact
24. Use of chemicals on fruits and vegetables
25. Organic farming
26. Crop rotation

27. Floury culture
28. Access to safe drinking water
29. Geographical survey
30. Geological survey
31. Sericulture
32. Study of species
33. Food adulteration
34. Incidence of Diabetes and other chronic diseases
35. Human genetics
36. Blood groups and blood levels
37. Internet Usage in Villages
38. Android Phone usage by different people
39. Utilisation of free electricity to farmers and related issues
40. Gender ration in schooling level- observation.

Complimenting the community service project the students may be involved to take up some awareness campaigns on social issues/special groups. The suggested list of programs

Programs for School Children

1. Reading Skill Program (Reading Competition)
2. Preparation of Study Materials for the next class.
3. Personality / Leadership Development
4. Career Guidance for X class students
5. Screening Documentary and other educational films
6. Awareness Program on Good Touch and Bad Touch (Sexual abuse)
7. Awareness Program on Socially relevant themes.

Programs for Women Empowerment

1. Government Guidelines and Policy Guidelines
2. Women's Rights
3. Domestic Violence
4. Prevention and Control of Cancer
5. Promotion of Social Entrepreneurship

General Camps

1. General Medical camps
2. Eye Camps
3. Dental Camps
4. Importance of protected drinking water
5. ODF awareness camp
6. Swatch Bharath
7. AIDS awareness camp
8. Anti Plastic Awareness
9. Programs on Environment
10. Health and Hygiene

11. Hand wash programmes
12. Commemoration and Celebration of important days

Programs for Youth Empowerment

1. Leadership
2. Anti-alcoholism and Drug addiction
3. Anti-tobacco
4. Awareness on Competitive Examinations
5. Personality Development

Common Programs

1. Awareness on RTI
2. Health intervention programmes
3. Yoga
4. Tree plantation
5. Programs in consonance with the Govt. Departments like –
 - i. Agriculture
 - ii. Health
 - iii. Marketing and Cooperation
 - iv. Animal Husbandry
 - v. Horticulture
 - vi. Fisheries
 - vii. Sericulture
 - viii. Revenue and Survey
 - ix. Natural Disaster Management
 - x. Irrigation
 - xi. Law & Order
 - xii. Excise and Prohibition
 - xiii. Mines and Geology
 - xiv. Energy

Role of Students:

- Students may not have the expertise to conduct all the programmes on their own. The students then can play a facilitator role.
 - For conducting special camps like Health related, they will be coordinating with the Governmental agencies.
 - As and when required the College faculty themselves act as Resource Persons.
 - Students can work in close association with Non-Governmental Organizations like Lions Club, Rotary Club, etc or with any NGO actively working in that habitation.
 - And also, with the Governmental Departments. If the program is rolled out, the District Administration could be roped in for the successful deployment of the program.
- An in-house training and induction program could be arranged for the faculty and participating students, to expose them to the methodology of Service Learning.

Timeline for the Community Service Project Activity

Duration: 8 weeks

1. Preliminary Survey (One Week)

- A preliminary survey including the socio-economic conditions of the allotted habitation to be conducted.
- A survey form based on the type of habitation to be prepared before visiting the habitation with the help of social sciences faculty. (However, a template could be designed for different habitations, rural/urban.
- The Governmental agencies, like revenue administration, corporation and municipal authorities and village secretariats could be aligned for the survey.

2. Community Awareness Campaigns (One Week)

- Based on the survey and the specific requirements of the habitation, different awareness campaigns and programmes to be conducted, spread over two weeks of time. The list of activities suggested could be taken into consideration.

3. Community Immersion Programme (Three Weeks)

Along with the Community Awareness Programmes, **the student batch can also work with any one of the below-listed governmental agencies and work in tandem with them. This community involvement programme will involve the students in exposing themselves to experiential learning about the community and its dynamics. Programs could be in consonance with the Govt. Departments.**

4. Community Exit Report (One Week)

- During the last week of the Community Service Project, a detailed report of the outcome of the 8 weeks' works to be drafted and a copy shall be submitted to the local administration. This report will be a basis for the next batch of students visiting that habitation. The same report submitted to the teacher-mentor will be evaluated by the mentor and suitable marks are awarded for onward submission to the University. Throughout the Community Service Project, a daily logbook need to be maintained by the students batch, which should be countersigned by the governmental agency representative and the teacher-mentor, who is required to periodically visit the students and guide them.



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Semester-5 (Theory-5, Lab-2, SEC-1, AC-1)							
Sl. No.	Category	Course Code	Course Title	Hours per week			Credits
				L	T	P	
1	PCC	23A3304T	Natural Language Processing	3	0	0	3
2	PCC	23A3305T	System Software Programming	3	0	0	3
3	PCC	23A3306T	Computer Vision & Image Processing	3	0	0	3
4	PCC	23A0522T	Introduction to Quantum Technologies and Applications	3	0	0	3
5	PEC	23A3307Ta 23A3307Tb 23A3307Tc 23A3307Td	Professional Elective-I 1. Data Visualization 2. Soft computing 3. Exploratory Data Analysis with Python 4. Computational Intelligence	3	0	0	3
6	OEC		Open Elective-I	3	0	0	3
7	PCC(Lab)	23A3308P	Computer Vision & Machine Learning Lab	0	0	3	1.5
8	PCC(Lab)	23A3309P	AI & System Programming Lab	0	0	3	1.5
9	SEC	23A0527P	Skill Enhancement Course Full Stack Development-II	0	1	2	2
10	AC	23A0420	Tinkering Lab	0	0	2	1
11	PCC	23A3310	Evaluation of Community Service Internship	-	-	-	2
Total				18	1	10	26

Category	Credits
Professional Core Courses(PCC)	17
Professional Elective Courses(PEC)	3
Open Elective Courses(OEC)	3
Skill Enhancement Course(SEC)	2
Audit Course(AC)	1
Total	26



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Natural Language Processing (Professional Core)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3304T	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PCC
Course Objectives:					
This course will enable students:					
- To provide a strong foundation in the principles and techniques of Natural Language Processing (NLP). - To introduce classical and deep learning-based approaches to NLP tasks. - To enable students to build and evaluate models for various NLP applications such as text classification, sentiment analysis, and machine translation. - To expose students to modern tools and libraries used in NLP such as NLTK, SpaCy, and Hugging Face Transformers. - To provide insights into the challenges of multilingual NLP and ethical concerns.					
Course Outcomes (CO):					
On completion of this course, student will be able to:					
- Understand the fundamentals and challenges of natural language understanding. - Apply linguistic preprocessing techniques such as tokenization, stemming, POS tagging, and parsing. - Implement NLP algorithms for tasks like classification, translation, and information retrieval. - Develop deep learning models using RNNs, LSTMs, and Transformer-based architectures for NLP. - Use NLP tools and libraries to analyze and interpret natural language data in real-world scenarios					
Syllabus					Total Hours:48
Module-I	Fundamentals of Natural Language Processing			10 Hrs	
Introduction to NLP: Definitions, Applications, Challenges, Linguistic Essentials: Syntax, Semantics, Pragmatics, Text Processing: Tokenization, Lemmatization, Stemming, Stop word Removal, Normalization, and N-gram Generation, POS Tagging and Named Entity Recognition, NLP Libraries: NLTK, SpaCy Overview. Hugging Face Transformers Library					
Module-II	Text Representation and Statistical NLP			10 Hrs	
Bag of Words and TF-IDF, Language Modeling: Unigrams, Bigrams, N-gram Models, Word Embeddings: Word2Vec, GloVe, FastText, Cosine Similarity and Distance Measures, Text Classification using Naive Bayes and SVM, Evaluation Metrics: Accuracy, Precision, Recall, FI.					
Module-III	Deep Learning for NLP			9 Hrs	
Neural Network Basics for NLP, Recurrent Neural Networks (RNNs) and Limitations, LSTM and GRU Networks, Sequence Labeling: POS Tagging, NER using Bi-LSTM, Text Classification using CNNs and RNNs, Model Evaluation and Hyper parameter Tuning.					

Module-IV	Transformers and Advanced NLP	10 Hrs
Attention Mechanism and Self-Attention, Transformer Architecture: Encoder-Decoder Models, Pretrained Language Models: BERT, RoBERTa, GPT, Fine-tuning Transformers for Text Classification, Question Answering and Text Summarization using Transformers, Sentiment Analysis and Zero-shot Classification.		
Module-V	Applications, Ethics, and Multilingual NLP	9 Hrs
Machine Translation: Rule-based vs Neural MT, Chatbots and Conversational AI, Information Retrieval and Question Answering, Speech-to-Text and Text-to-Speech Overview, Multilingual NLP and Low-Resource Languages, Bias, Fairness, and Ethics in NLP.		
Text Books: 1. Daniel Jurafsky and James H. Martin, Speech and Language Processing, Pearson Education. 2. Steven Bird, Ewan Klein, Edward Loper, Natural Language Processing with Python, O'ReillyMedia. 3. Yoav Goldberg, Neural Network Methods in NLP, Morgan & Claypool.		
Reference Books: 1. Jacob Eisenstein, Introduction to Natural Language Processing, MIT Press. 2. Delip Rao and Brian McMahan, Natural Language Processing with PyTorch, O'Reilly. 3. ThushanGanegedara, Transformers for Natural Language Processing, Packt Publishing.		



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System Software Programming (Professional Core)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3305T	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PCC
Course Objectives:					
This course will enable students:					
- To understand the architecture and design of system software including compilers, assemblers, linkers, loaders, and macro processors. - To gain in-depth knowledge of programming tools, shell environments, and low-level system utilities. - To apply principles of system programming in Unix/Linux environments. - To explore process creation, inter-process communication, signal handling, and multithreading using C/C++. - To enable development of foundational tools like simple compilers, parsers, and loaders					
Course Outcomes (CO):					
On completion of this course, student will be able to:					
1. Explain the architecture and functions of system software like assemblers, loaders, linkers, and macro processors. (BTL 2 – Understand) 2. Apply scanning and parsing techniques for programming language processing. (BTL 3 – Apply) 3. Develop and analyze assembly-level programs and understand compilation techniques. (BTL 4 – Analyze) 4. Implement Unix/Linux system programming tasks such as process creation, pipes, signals, and thread management. (BTL 4 – Analyze) 5. Demonstrate hands-on experience in shell scripting, debugging, and low-level system tools. (BTL 3 – Apply)					
Syllabus					TotalHours:48
Module-I	Language Processors and Assemblers				10 Hrs
Language processing system overview, Phases of compilation and data structures, Assemblers – features, single pass and two-pass assembler, Intermediate code generation, Literal and symbol tables, Relocation and linking concepts					
Module-II	Macro Processors and Loaders				9 Hrs
Macro instruction and features, Nested macros and macro expansion, Macro processing in two-pass assemblers, Design of macro processors, Loaders: absolute, relocating, and linking, Dynamic loading and linking, bootstrap loader					
Module-III	Scanning, Parsing, and Compilers				10 Hrs
Language grammars and ambiguity, Lexical analysis – regular expressions, token generation, Syntax analysis – parsing techniques (top-down, bottom-up), Semantic analysis and intermediate code generation, Code optimization techniques – constant folding, dead code elimination					

Module-IV	Linkers, Debuggers, and Shell Programming	9 Hrs
Symbol resolution and relocation, Linking (static vs dynamic), relocation records, Debugging techniques and breakpoints, Unix/Linux shell environment, Shell commands, variables, redirection, pipes, control statements, Shell script functions and script-based automation		
Module-V	Unix/Linux System Programming	10 Hrs
Introduction to system-level programming in C, File I/O system calls (open, read, write, close), Process creation using fork(), exec(), wait(), Inter-process communication (pipes, FIFO), Signal handling and POSIX threads (pthread_create, pthread_join), Synchronization, Case studies: background processes, daemon creation, mini shell		
Text Books: 1. Leland L. Beck, D. Manjula, System Software: An Introduction to Systems Programming, 3rd Edition, Pearson. 2. Silberschatz, Galvin, Gagne, Operating System Concepts, 10th Edition, Wiley (selectively for system calls & programming).		
Reference Books: 1. D.M. Dhamdhare, System Programming and Operating Systems, McGraw Hill. 2. Neil Matthew, Richard Stones, Beginning Linux Programming, Wrox. 3. Andrew S. Tanenbaum, Modern Operating Systems, Pearson Education. 4. Yashwant Kanetkar, Unix Shell Programming, BPB Publications.		



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Computer Vision and Image Processing (Professional Core)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3306T	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PCC
Course Objectives:					
This course will enable students:					
- Introduce fundamental concepts of image processing and computer vision. - Develop proficiency in applying algorithms for image analysis and interpretation. - Explore techniques for feature extraction, object recognition, and scene understanding. - Understand the integration of machine learning methods in computer vision applications.					
Course Outcomes (CO):					
On completion of this course, student will be able to:					
- Understand image formation, representation, and apply basic image processing and frequency domain techniques for image enhancement and restoration. - Apply edge detection, segmentation, morphological, and texture analysis techniques for extracting features from images. - Analyze 3D vision and motion using techniques like stereo vision, optical flow, and camera calibration for scene understanding and depth estimation. - Evaluate object recognition approaches and machine learning models including traditional and deep learning techniques used in computer vision. - Implement advanced computer vision applications such as image compression, face recognition, and medical image analysis using case studies.					
Syllabus					Total Hours:48
Module-I	Introduction to Computer Vision and Image Processing				10 Hrs
Overview of Computer Vision and Image Processing: Definitions and scope, Historical development and applications, Image Formation and Representation: Image acquisition methods, Sampling and quantization, Color spaces and models, Fundamentals of Image Processing: Point operations (brightness and contrast adjustments), Histogram processing, Spatial filtering techniques Fourier Transform and Frequency Domain Processing: Discrete Fourier Transform (DFT), Filtering in the frequency domain, Image restoration concept,Lucy Richardson Algorithm.					
Module-II	Image Analysis Techniques				9 Hrs
Edge Detection and Feature Extraction: Gradient operators (Sobel, Prewitt), Canny edge detector, Corner and interest point detection, Image Segmentation: Thresholding methods, Region-based segmentation, Clustering techniques (K-means, Mean-Shift), Morphological Image Processing: Erosion and dilation, Opening and closing operations, Applications in shape analysis, Texture Analysis, Statistical methods (co-occurrence matrices), Transform-based methods (Gabor filters), Applications in pattern recognition.					

Module-III	3D Vision and Motion Analysis	9 Hrs
Stereo Vision: Epipolar geometry, Disparity mapping, Depth estimation techniques, Structure from Motion (SfM): Feature tracking across frames, 3D reconstruction from motion, Applications in scene understanding, Optical Flow and Motion Analysis: Lucas-Kanade method, Horn-Schunck method, Motion segmentation, Camera Calibration and 3D Reconstruction: Intrinsic and extrinsic parameters, Calibration techniques, 3D point cloud generation.		
Module-IV	Object Recognition and Machine Learning in Vision	10 Hrs
Feature Descriptors and Matching: Scale-Invariant Feature Transform (SIFT), Speeded-Up Robust Features (SURF), Feature matching algorithms, Object Detection and Recognition: Template matching, Deformable part models, Convolutional Neural Networks (CNNs), Introduction to Machine Learning for Vision: Supervised and unsupervised learning, Support Vector Machines (SVMs), Decision trees and random forests, Deep Learning Architectures: Auto encoders, Recurrent Neural Networks (RNNs), Generative Adversarial Networks (GANs)		
Module-V	Applications and Advanced Topics	10 Hrs
Image Compression: Lossy and lossless compression techniques, Standards (e.g., JPEG, PNG), Morphological Image Processing: Dilation, erosion, opening, and closing Operations, Applications in shape analysis, Case Studies: Face recognition systems, Automated visual inspection, Medical image analysis.		
Textbooks: 1. Gonzalez, R. C., & Woods, R. E. (2008). Digital Image Processing (3rd ed.). Pearson Prentice Hall. Stony Brook University 2. Szeliski, R. (2010). Computer Vision: Algorithms and Applications. Springer.		
Reference Books: 1. Forsyth, D. A., & Ponce, J. (2002). Computer Vision: A Modern Approach. Prentice Hall. 2. Shapiro, L. G., & Stockman, G. C. (2001). Computer Vision. Prentice Hall.		
Online Learning Resources: 1. Coursera: Introduction to Computer Vision and Image Processing. LinkCoursera 2. Stanford University: CS231n: Deep Learning for Computer Vision. Linkcs231n.stanford.edu 3. MIT OpenCourseWare: Introduction to Computer Vision. Link		



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INTRODUCTION TO QUANTUM TECHNOLOGIES AND APPLICATIONS

(Professional Core)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0522T	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PCC

Course Objectives:

This course will enable students:

- Introduce fundamental quantum concepts like superposition and entanglement.
- Understand theoretical structure of qubits and quantum information.
- Explore conceptual challenges in building quantum computers.
- Explain principles of quantum communication and computing.
- Examine real-world applications and the future of quantum technologies.

Course Outcomes (CO):

On completion of this course, student will be able to:

- Explain core quantum principles in a non-mathematical manner.
- Compare classical and quantum information systems.
- Identify theoretical issues in building quantum computers.
- Discuss quantum communication and computing concepts.
- Recognize applications, industry trends, and career paths in quantum technology.

Syllabus		Total Hours:48
Module-I	Introduction to Quantum Theory and Technologies	10 Hrs

The transition from classical to quantum physics, Fundamental principles explained conceptually: Superposition, Entanglement, Uncertainty Principle, Wave-particle duality, Classical vs Quantum mechanics – theoretical comparison, Quantum states and measurement: nature of observation, Overview of quantum systems: electrons, photons, atoms, The concept of quantization: discrete energy levels, Why quantum? Strategic, scientific, and technological significance, A snapshot of quantum technologies: Computing, Communication, and Sensing, National and global quantum missions: India's Quantum Mission, EU, USA, China

Module-II	Theoretical Structure of Quantum Information Systems	9 Hrs
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What is a qubit? Conceptual understanding using spin and polarization, Comparison: classical bits vs quantum bits, Quantum systems: trapped ions, superconducting circuits, photons (non-engineering view), Quantum coherence and decoherence – intuitive explanation, Theoretical concepts: Hilbert spaces, quantum states, operators – only interpreted in abstract, The role of entanglement and non-locality in systems, Quantum information vs classical information: principles and differences, Philosophical implications: randomness, determinism, and observer role.

Module-III	Building a Quantum Computer – Theoretical Challenges and Requirements	10 Hrs
What is required to build a quantum computer (conceptual overview)?,Fragility of quantum systems: decoherence, noise, and control, Conditions for a functional quantum system: Isolation, Error management, Scalability, Stability, Theoretical barriers: Why maintaining entanglement is difficult, Error correction as a theoretical necessity, Quantum hardware platforms (brief conceptual comparison),Superconducting circuits, Trapped ions, Photonics, Vision vs reality: what's working and what remains elusive, The role of quantum software in managing theoretical complexities		
Module-IV	Quantum Communication and Computing – Theoretical Perspective	9 Hrs
Quantum vs Classical Information, Basics of Quantum Communication, Quantum Key Distribution (QKD),Role of Entanglement in Communication,The Idea of the Quantum Internet – Secure Global Networking,Introduction to Quantum Computing,Quantum Parallelism (Many States at Once),Classical vs Quantum Gates, Challenges: Decoherence and Error Correction,Real-World Importance and Future Potential.		
Module-V	Applications, Use Cases, and the Quantum Future	10 Hrs
Real-world application domains: Healthcare (drug discovery),Material science, Logistics and optimization, Quantum sensing and precision timing, Industrial case studies: IBM, Google, Microsoft, PsiQuantum,Ethical, societal, and policy considerations, Challenges to adoption: cost, skills, standardization,Emerging careers in quantum: roles, skillsets, and preparation pathways,Educational and research landscape – India's opportunity in the global quantum race		
Textbooks: 1. Michael A. Nielsen, Isaac L. Chuang, <i>Quantum Computation and Quantum Information</i>, Cambridge University Press, 10th Anniversary Edition, 2010. 2. Eleanor Rieffel and Wolfgang Polak, <i>Quantum Computing: A Gentle Introduction</i>, MIT Press, 2011. 3. Chris Bernhardt, <i>Quantum Computing for Everyone</i>, MIT Press, 2019.		
Reference Books: 1. David McMahon, <i>Quantum Computing Explained</i>, Wiley, 2008. 2. Phillip Kaye, Raymond Laflamme, Michele Mosca, <i>An Introduction to Quantum Computing</i>, Oxford University Press, 2007. 3. Scott Aaronson, <i>Quantum Computing Since Democritus</i>, Cambridge University Press, 2013. 4. Alastair I.M. Rae, <i>Quantum Physics: A Beginner's Guide</i>, Oneworld Publications, Revised Edition, 2005. 5. Eleanor G. Rieffel, Wolfgang H. Polak, <i>Quantum Computing: A Gentle Introduction</i>, MIT Press, 2011. 6. Leonard Susskind, Art Friedman, <i>Quantum Mechanics: The Theoretical Minimum</i>, Basic Books, 2014. 7. Bruce Rosenblum, Fred Kuttner, <i>Quantum Enigma: Physics Encounters Consciousness</i>, Oxford University Press, 2nd Edition, 2011. 8. Giuliano Benenti, Giulio Casati, Giuliano Strini, <i>Principles of Quantum Computation and</i>		

Information, Volume I: Basic Concepts, World Scientific Publishing, 2004.

9. **K.B. Whaley et al.**, *Quantum Technologies and Industrial Applications: European Roadmap and Strategy Document*, Quantum Flagship, European Commission, 2020.
10. **Department of Science & Technology (DST), Government of India**, *National Mission on Quantum Technologies & Applications – Official Reports and Whitepapers*, MeitY/DST Publications, 2020 onward.

Online Learning Resources:

- [IBM Quantum Experience and Qiskit Tutorials](#)
- [Coursera – Quantum Mechanics and Quantum Computation by UC Berkeley](#)
- [edX – The Quantum Internet and Quantum Computers](#)
- [YouTube – Quantum Computing for the Determined by Michael Nielsen](#)
- [Qiskit Textbook – IBM Quantum](#)



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Data Visualization (Professional Elective-I)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3307Ta	3:0:0:3	3	CIE:30 SEE:70	3Hours	PEC
Course Objectives:					
This course will enable students:					
- To understand the principles, techniques, and tools of data visualization. - To develop the ability to transform data into visual insights using different types of charts and plots. - To introduce the cognitive and perceptual foundations of effective data visualization. - To apply tools and programming environments (like Python, Tableau, or Power BI) for creating interactive and dynamic visualizations. - To analyze real-world datasets and effectively communicate data-driven findings visually.					
Course Outcomes (CO):					
After completion of the course, students will be able to:					
- Interpret different types of data and recognize the appropriate visualization methods. (Understand, Analyze) - Design effective and interactive data visualizations using various tools. (Apply, Create) - Apply visual encoding and perceptual principles in presenting complex data. (Apply, Evaluate) - Analyze and visualize real-world data sets using Python libraries and dashboards. (Analyze, Evaluate) - Create visual stories and dashboards for effective communication of insights. (Create, Apply)					
Syllabus					Total Hours:48
Module-I	Introduction to Data Visualization & Perception				9 Hrs
Introduction to Data Visualization, Importance and Scope of Data Visualization, Data Types and Sources, Visual Perception: Pre-attentive Processing, Gestalt Principles, Data-Ink Ratio, Data Density, Lie Factor, Visualization Process and Design Principles, Tools Overview: Tableau, Power BI, Python Libraries.					
Module-II	Visualization Techniques for Categorical & Quantitative Data				9 Hrs
Charts for Categorical Data: Bar Charts, Pie Charts, Column Charts, Charts for Quantitative Data: Histograms, Line Charts, Box plots, Violin Plots, Scatter Plots, Bubble Charts, Heat maps, Choosing the Right Chart Type, Best Practices in Labeling, Coloring, and Scaling.					
Module-III	Multidimensional, Temporal and Hierarchical Data Visualization				10 Hrs
Visualizing Multivariate Data: Parallel Coordinates, Radar Charts, Time-Series Visualization: Time Plots, Animation over Time, Geographic Data Visualization: Maps, Choropleths, Hierarchical Data: Treemaps, Sunburst Charts, Network and Graph Visualization.					

Module-IV	Data Visualization Using Python and Dashboards	9 Hrs
Introduction to Matplotlib, Seaborn, and Plotly, Creating Static and Interactive Charts, Pandas Visualization Capabilities, Dashboards with Dash, Streamlit, Power BI, Case Studies: Real-world Data visualization.		
Module-V	Storytelling with Data and Ethical Visualization	10 Hrs
Storytelling and Narrative Techniques in Visualization, Dashboards and Reporting, Misleading Visualizations and Bias, Ethical Principles in Data Visualization, Final Project: Create a Storytelling Dashboard with Real Data.		
Text Books: 1. Tamara Munzner, Visualization Analysis and Design, CRC Press, 2014. 2. Nathan Yau, Data Points: Visualization That Means Something, Wiley, 2013.		
Reference Books: 1. Alberto Cairo, The Truthful Art: Data, Charts, and Maps for Communication, New Riders, 2016. 2. Cole Nussbaumer-Knaflic, Storytelling with Data: A Data Visualization Guide for Business Professionals, Wiley, 2015. 3. Claus O. Wilke, Fundamentals of Data Visualization, O'Reilly, 2019. 4. Rohan Chopra, Hands-On Data Visualization with Bokeh, Packt Publishing, 2019.		
Web References: 1. NPTEL: Data Visualization - IIT Madras 2. Coursera: Data Visualization with Python by IBM		



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Soft Computing (Professional Elective-I)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3307Tb	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students to:					
- Understand the concepts of soft computing techniques and how they differ from traditional AI techniques. - Introduce the fundamentals of fuzzy logic and fuzzy systems. - Familiarize with artificial neural networks and their architectures. - Learn genetic algorithms and their role in optimization. - Explore hybrid systems integrating fuzzy logic, neural networks, and genetic algorithms.					
Course Outcomes (CO):					
After completion of the course, students will be able to:					
- Understand the components and applications of soft computing. - Apply fuzzy logic concepts to real-world problems. - Build and train various neural network models. - Implement genetic algorithms for problem-solving and optimization. - Design hybrid systems using soft computing techniques.					
Syllabus					Total Hours:48
Module-I	Introduction to Soft Computing and Fuzzy Logic				10 Hrs
Introduction to Soft Computing: Definition, Components, Differences with Hard Computing, Applications of Soft Computing, Fuzzy Logic: Crisp Sets vs Fuzzy Sets, Membership Functions, Fuzzy Set Operations, Fuzzy Rules and Fuzzy Reasoning, Fuzzy Inference Systems: Mamdani and Sugeno Models, Defuzzification Techniques.					
Module-II	Artificial Neural Networks – I				9 Hrs
Introduction to Neural Networks: Biological Neurons vs Artificial Neurons, Architecture of Neural Networks: Feed forward, Feedback, Learning Rules: Hebbian, Delta, Perceptron Learning Rule, Single Layer Perceptron and its Limitations, Multi-Layer Perceptron: Backpropagation Algorithm, Applications of Neural Networks					
Module-III	Artificial Neural Networks – II				10 Hrs
Hopfield Networks and Associative Memories, Radial Basis Function Networks, Self-Organizing Maps (SOM), Recurrent Neural Networks (RNNs) – Basic Concepts, Convolutional Neural Networks (CNNs) – Overview and Applications, Practical Use Cases in Image and Pattern Recognition					
Module-IV	Genetic Algorithms and Optimization				10 Hrs
Introduction to Genetic Algorithms, GA Operators: Selection, Crossover, Mutation, Fitness Function and Evaluation, Schema Theorem, Elitism, Applications in Function Optimization, Scheduling, and Robotics, Introduction to Particle Swarm Optimization (PSO), Applications of Particle Swarm Optimization.					
Module-V	Hybrid Systems and Advanced Topics				9Hrs
Hybrid Systems: Neuro-Fuzzy Systems, Fuzzy-GA, GA-ANN, ANFIS: Architecture and Learning, Case Studies on Hybrid Systems, Introduction to Deep Learning in Soft Computing, Real-World					

Applications: Forecasting, Control Systems, Medical Diagnosis, Image Processing.

Textbooks:

1. S. N. Sivanandam, S. N. Deepa, —Principles of Soft Computing, Wiley India, 3rd Edition
2. Timothy J. Ross, —Fuzzy Logic with Engineering Applications, Wiley, 4th Edition
3. S. Rajasekaran and G. A. Vijayalakshmi Pai, —Neural Networks, Fuzzy Logic and Genetic Algorithm: Synthesis and Applications, PHI

Reference Books:

1. Laurene Fausett, —Fundamentals of Neural Networks: Architectures, Algorithms and Applications, Pearson
2. David E. Goldberg, —Genetic Algorithms in Search, Optimization and Machine Learning, Pearson
3. Simon Haykin, —Neural Networks and Learning Machines, Pearson, 3rd Edition
4. Bart Kosko, —Neural Networks and Fuzzy Systems, Prentice Hall

Online Learning Resources:

1. NPTEL – Soft Computing by Prof. S. Sengupta (IIT Kharagpur)
2. Coursera – Neural Networks and Deep Learning (Andrew Ng)



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Exploratory Data Analysis with Python (Professional Elective-I)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3307Tc	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
- To introduce the principles and practices of Exploratory Data Analysis (EDA) using Python. - To teach techniques for data cleaning, preprocessing, transformation, and visualization. - To apply statistical techniques and visual methods to discover patterns and relationships. - To gain experience using popular Python libraries such as NumPy, Pandas, Matplotlib, and Seaborn. - To prepare datasets for further machine learning and predictive modeling.					
Course Outcomes (CO):					
After completion of the course, students will be able to:					
- Understand and apply key concepts of EDA and data preprocessing. - Perform exploratory analysis using Python libraries and interpret results. - Handle missing data, outliers, and categorical features effectively. - Create meaningful visualizations to support data-driven insights. - Use EDA as a foundation for data science workflows.					
Syllabus				Total Hours:48	
Module-I	Introduction to EDA and Python Environment			10 Hrs	
Introduction to Data Science and EDA, Importance of EDA in Data Science Life Cycle, Setting up Python Environment: Jupyter, Anaconda, VS Code, Introduction to NumPy and Pandas: Arrays, Series, Data Frames, Data loading, viewing, basic operations (info, describe, shape)					
Module-II	Data Wrangling and Preprocessing			9 Hrs	
Handling Missing Data (mean, median, drop, interpolation), Dealing with Duplicates, Outliers, and Anomalies, Encoding Categorical Variables (Label, One-hot), Data Transformation: Scaling, Normalization, Binning, Data Types Conversion and Data Type Casting.					
Module-III	Univariate and Bivariate Analysis			9 Hrs	
Measures of Central Tendency and Dispersion, Distribution Plots: Histograms, Boxplots, KDE, Bar Charts, Count Plots, Pie Charts, Bivariate Analysis: Scatter Plots, Pair Plots, Heatmaps, Correlation and Covariance Analysis.					
Module-IV	Data Visualization Techniques			10 Hrs	
Visualization with Matplotlib and Seaborn, Customizing Plots: Titles, Legends, Labels, Themes, Advanced Visuals: Violin Plots, Strip Plots, Swarm Plots, Multivariate Visualization and Subplots, Plotly, Bokeh and Interactive Visualizations (basic overview).					
Module-V	EDA Case Studies and Real-Time Datasets			10 Hrs	
Step-by-step EDA on Sample Datasets (Titanic, Iris, Sales, etc.), Outlier Detection Techniques, Feature Engineering Techniques in EDA, EDA Report Generation using Python Notebooks.					

Preparing Data for Machine Learning Models

Textbooks:

1. Jake VanderPlas, Python Data Science Handbook: Essential Tools for Working with Data, O'Reilly, 2016.
2. Wes McKinney, Python for Data Analysis, 2nd Edition, O'Reilly, 2018.

Reference Books:

1. Joel Grus, Data Science from Scratch, O'Reilly, 2019.
2. Aurelien Geron, Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, 2nd Edition, O'Reilly, 2019.
3. Allen B. Downey, Think Stats: Probability and Statistics for Programmers, O'Reilly, 2014.

Online Learning Resources:

1. NPTEL Course – Data Science for Engineers
2. Coursera – Applied Data Science with Python Specialization (University of Michigan)



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Computational Intelligence (Professional Elective-I)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3307Td	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
- Understand the concepts and foundations of computational intelligence. - Study neural networks, fuzzy logic systems, and evolutionary algorithms. - Explore hybrid systems and their applications. - Apply computational intelligence techniques to real-world problem-solving. - Analyze the effectiveness of various computational intelligence approaches.					
Course Outcomes (CO):					
- Describe and differentiate neural networks, fuzzy logic, and evolutionary computation. (Understand) - Apply neural and fuzzy systems for real-time decision-making. (Apply) - Analyze complex problems using soft computing tools. (Analyze) - Develop hybrid intelligent systems. (Create) - Evaluate and compare the performance of CI-based systems. (Evaluate)					
Syllabus					Total Hours:48
Module-I	Introduction to Computational Intelligence and Artificial Neural Networks				9 Hrs
Definition and Scope of Computational Intelligence(CI),Components of CI: Neural Networks, Fuzzy Logic, Evolutionary Computation, Biological Neuron vs. Artificial Neuron, McCulloch-Pitts Model, Perceptron, Adaline and Madaline, Multilayer Feed forward Networks, Backpropagation Algorithm, Applications of ANN in Pattern Recognition and Classification.					
Module-II	Fuzzy Logic and Fuzzy Systems				10 Hrs
Introduction to Fuzzy Logic and Fuzzy Sets, Membership Functions, Fuzzy Set Operations, Fuzzy Rules and Inference Systems, Fuzzification and Defuzzification, Fuzzy Control Systems, Fuzzy Reasoning and Approximate Reasoning					
Module-III	Evolutionary Computation Techniques				10 Hrs
Basics of Evolutionary Algorithms (EA), Genetic Algorithms (GA): Operators, Encoding, Fitness Function, Selection, Crossover and Mutation, Convergence Criteria, Genetic Programming (GP), Tools for Genetic Programming, Differential Evolution (DE), Applications of GA and GP					
Module-IV	Swarm Intelligence and Hybrid Systems				9 Hrs
Swarm Intelligence: Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), Behavior of Swarms and Collective Intelligence, Comparison of Evolutionary Algorithms and Swarm Techniques, Hybrid Systems: Neuro-Fuzzy, Fuzzy-GA, ANN-GA Systems, Case Studies in Hybrid Systems.					
Module-V	Applications of Computational Intelligence				10 Hrs
CI in Image and Signal Processing, CI for Optimization Problems and Robotics, CI in Biomedical Engineering and Finance, Intelligent Agents and Decision-Making Systems, Real-time Applications and Emerging Trends in CI.					

Text Books:

1. S.Rajasekaran and G.A.VijayalakshmiPai, Neural Networks, Fuzzy Logic and Genetic Algorithms: Synthesis and Applications, PHI Learning.
2. TimothyJ.Ross, Fuzzy Logic with Engineering Applications, WileyIndia.

Reference Books:

1. S.N.Sivanandam, S.N.Deepa, Principles of Soft Computing, Wiley India.
2. Simon Haykin, Neural Networks and Learning Machines, Pearson.
3. James Kennedy and Russell C.Eberhart, Swarm Intelligence, Morgan Kaufmann.
4. Andries P.Engelbrecht, Computational Intelligence: An Introduction,Wiley.

Online Courses:

1. NPTEL-Computational Intelligence
2. Coursera–Computational Intelligence
3. YouTube:IIT Lectures on Soft Computing and CI



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Computer Vision & Machine Learning Lab (Professional Core)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3308P	0:0:3:1.5	1.5	CIE:30 SEE:70	3 Hours	PCC
Course Objectives:					
This course will enable students to: - To impart practical knowledge of computer vision concepts using OpenCV and image processing libraries. - To implement core machine learning algorithms and evaluate model performance. - To work with real-world datasets for classification, regression, and image processing tasks. - To train, test, and validate models using Python, TensorFlow, and Scikit-learn. - To understand the integration of ML models in vision applications.					
Course Outcomes (CO):					
On completion of this course, student will be able to - Apply computer vision techniques to solve real-time image processing problems. (Apply -L3) - Train and evaluate machine learning models for classification and regression tasks. (Analyze - L4) - Design and test feature extraction techniques from images. (Create - L6) - Use OpenCV, Scikit-learn, TensorFlow/PyTorch for practical implementations. (Apply -L3) - Integrate vision-based features with ML algorithms for end-to-end solutions. (Evaluate - L5)					
List of Experiments (12 Total)					
Week-1: 1. Image preprocessing techniques: resizing, filtering, thresholding using OpenCV Week-2: 2. Edge detection using Sobel, Canny, and Laplacian operators Week-3: 3. Object detection using contour detection and bounding boxes Week-4: 4. Feature extraction using HOG, SIFT, and ORB Week-5: 5. Implement face detection using Haar cascades or DNN models Week-6: 6. Train a machine learning model (SVM / KNN) for image classification Week-7: 7. Build and evaluate a decision tree classifier using scikit-learn Week-8: 8. Implement a logistic regression model for binary classification on numerical dataset Week-9: 9. Apply PCA for feature reduction and visualization Week-10: 10. Design a simple neural network using TensorFlow/Keras for image classification Week-11: 11. Train and evaluate a CNN model for digit recognition using MNIST dataset					

Week-12:

12. Real-time emotion recognition using webcam input and pre-trained model integration

Textbooks:

1. Simon J. D. Prince, Computer Vision: Models, Learning, and Inference, Cambridge University Press.
3. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition, O'Reilly.
4. Richard Szeliski, Computer Vision: Algorithms and Applications, Springer.

Reference Books:

1. Adrian Rosebrock, Practical Python and OpenCV (PyImageSearch).
2. Ian Good fellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press.
3. Bishop C. M., Pattern Recognition and Machine Learning, Springer.

Web References:

1. <https://opencv.org>
2. <https://www.tensorflow.org/tutorials>
3. <https://www.kaggle.com/learn/intro-to-machine-learning>
4. <https://www.pyimagesearch.com>
5. NPTEL Course on Deep Learning



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AI & System Programming Lab (Professional Core)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3309P	0:0:3:1.5	1.5	CIE:30 SEE:70	3Hours	PCC
Course Objectives:					
This course will enable students: - To provide practical exposure to foundational AI algorithms and system programming. - To develop skills to write intelligent systems and low-level programs. - To integrate concepts of AI and system programming for automation and optimization.					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Implement search algorithms and logic programming using AI tools. - Construct assemblers, macro processors, and shell scripts. - Develop system utilities using C and integrate them with AI tools. - Demonstrate real-time intelligent system automation using scripting and AI logic.					
List of Experiments				Total Hours:48	
Week-1: 1. Write simple programs in Prolog for facts, rules, and queries.					
Week-2: 2. Develop a Prolog-based expert system for medical diagnosis or animal identification.					
Week-3: 3. Implement Depth-First Search (DFS) and Breadth-First Search (BFS) in Python.					
Week-4: 4. Implement A* Search Algorithm using heuristics in Python.					
Week-5: 5. Implement the Minimax algorithm for a simple game (e.g., Tic Tac Toe).					
Week-6: 6. Design and implement a two-pass assembler in C.					
Week-7: 7. Implement a Macro Processor using C for assembly language programs.					
Week-8: 8. Develop a simple Linux Shell (command interpreter) using C.					
Week-9: 9. Write shell scripts for file operations, process creation, and monitoring.					
Week-10: 10. Demonstrate inter-process communication using pipes and signals in Linux.					
Week-11: 11. Integrate AI logic (search/expert system) into a shell script or system utility for task automation.					

Week-12:

12. Develop an AI-powered system utility (e.g., Intelligent File Manager, AI Bot for CLI commands)..

Lab Software Requirements:

1. Introduction to Automata Theory, Languages and Computation, J. E. Hopcroft, R. Motwani and J. D. Ullman, 3rd Edition, Pearson, 2008.
2. Compilers Principles, Ravisethi, Pearson Education.



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Full Stack Development-II					
(Skill Enhancement course)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0527P	0:1:2:2	2	CIE:30 SEE:70	3Hours	SEC
Course Objectives:					
This course will enable students:					
- Develop a fully functioning website and deploy on a web server. - Gain Knowledge about the front end and back end Tools - Find and use code packages based on their documentation to produce working results in a project. - Create web pages that function using external data. - Implementation of web application employing efficient database access					
Course Outcomes (CO):					
On completion of this course, student will be able to:					
- To become knowledgeable about the most recent web development technologies. - Idea for creating two tier and three tier architectural web applications. - Design and Analyse real time web applications. - Constructing suitable client and server side applications. - To learn core concept of both front end and back end programming.					
Syllabus				TotalHours:48	
Module-I	UNIT-I			10 Hrs	
Web Development Basics: Web development Basics - HTML & Web servers Shell - UNIX CLI Version control - Git & Github HTML, CSS.					
Module-II	UNIT-II			9 Hrs	
Front end Development: Javascript basics OOPS Aspects of JavaScript Memory usage and Functionsin JS AJAX for data exchange with server jQuery Framework jQuery events, UI components etc.JSON data format.					
Module-III	UNIT-III			10 Hrs	
REACT JS: Introduction to React React Router and Single Page Applications React Forms, Flow Architecture and Introduction to Redux More Redux and Client-Server Communication.					

Module-IV	UNIT-IV	9 Hrs
Java Web Development: JAVA PROGRAMMING BASICS, Model View Controller (MVC) Pattern MVC Architecture using Spring RESTful API using Spring Framework Building an application using Maven.		
Module-V	UNIT-V	10 Hrs
Databases & Deployment: Relational schemas and normalization Structured Query Language (SQL) Data persistence using Spring JDBC Agile development principles and deploying application in Cloud.		
TextBooks: 1. Web Design with HTML, CSS, JavaScript and JQuery Set Book by Jon Duckett Professional JavaScript for Web Developers Book by Nicholas C. Zakas 2. Learning PHP, MySQL, JavaScript, CSS & HTML5: A Step-by-Step Guide to Creating Dynamic Websites by Robin Nixon 3. Full Stack JavaScript: Learn Backbone.js, Node.js and MongoDB. Copyright © 2015 BY AZAT MARDAN		
ReferenceBooks: 1. Full-Stack JavaScript Development by Eric Bush 2. Mastering Full Stack React Web Development Paperback – April 28, 2017 by Tomasz Dyl, Kamil Przeorski, Maciej Czarnecki		



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Tinkering Lab (Audit Course)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0420	0:0:2:1	1	CIE:30 SEE:70	3 Hours	AC
AIM:					
The aim of tinkering lab for engineering students is to provide a hands-on learning environment where students can explore, experiment, and innovate by building and testing prototypes. These labs are designed to demonstrate practical skills that complement theoretical knowledge.					
Course Objectives:					
The objectives of the course are to					
• Encourage Innovation and Creativity • Provide Hands-on Learning and Impart Skill Development • Foster Collaboration and Teamwork • Enable Interdisciplinary Learning, Prepare for Industry and Entrepreneurship • Impart Problem-Solving mind-set					
Course Outcomes (CO):					
• These labs bridge the gap between academia and industry, providing students with the practical experience. • Some students may also develop entrepreneurial skills, potentially leading to start-ups or innovation-driven careers. • Tinkering labs aim to cultivate the next generation of engineers by giving them the tools, space, and mind-set to experiment, innovate, and solve real-world challenges. • The students will be able to experiment, innovate, and solve real-world challenges.					
Syllabus				Total Hours:48	
	List of experiments				
1) Make your own parallel and series circuits using breadboard for any application of your choice.					
2) Design and 3D print a Walking Robot					
3) Design and 3D Print a Rocket					
4) Temperature & Humidity Monitoring System (DHT11 + LCD)					
5) Water Level Detection and Alert System					
6) Automatic Plant Watering System					
7) Bluetooth-Based Door Lock System					
8) Smart Dustbin Using Ultrasonic Sensor					
9) Fire Detection and Alarm System					
10) RFID-Based Attendance System					
11) Voice-Controlled Devices via Google Assistant					
12) Heart Rate Monitoring Using Pulse Sensor					
13) Soil Moisture-Based Irrigation					
14) Smart Helmet for Accident Detection					

- 15) Milk Adulteration Detection System
- 16) Water Purification via Activated Carbon
- 17) Solar Dehydrator for Food Drying
- 18) Temperature-Controlled Chemical Reactor
- 19) Ethanol Mini-Plant Using Biomass
- 20) Smart Fluid Flow Control (Solenoid + pH Sensor)
- 21) Portable Water Quality Tester
- 22) AI Crop Disease Detection
- 23) AI-based Smart Irrigation
- 24) ECG Signal Acquisition and Plotting
- 25) AI-Powered Traffic Flow Prediction
- 26) Smart Grid Simulation with Load Monitoring
- 27) Smart Campus Indoor Navigator
- 28) Weather Station Prototype
- 29) Firefighting Robot with Sensor Guidance
- 30) Facial Recognition Dustbin
- 31) Barcode-Based Lab Inventory System
- 32) Growth Chamber for Plants
- 33) Biomedical Waste Alert System
- 34) Soil Classification with AI
- 35) Smart Railway Gate
- 36) Smart Bin Locator via GPS and Load Sensors
- 37) Algae-Based Water Purifier
- 38) Contactless Attendance via Face Recognition

Web References:

1. <https://aim.gov.in/pdf/equipment-manual-pdf.pdf>
2. <https://atl.aim.gov.in/ATL-Equipment-Manual/>
3. <https://aim.gov.in/pdf/Level-1.pdf>
4. <https://aim.gov.in/pdf/Level-2.pdf>
5. <https://aim.gov.in/pdf/Level-3.pdf>



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Semester-6 (Theory-6, Lab-2, SEC-1, AC-1)							
Sl. No.	Category	Course Code	Course Title	Hours per Week			Credits
				L	T	P	
1	PCC	23A3311T	Cloud Computing for AI	3	0	0	3
2	PCC	23A3312T	Big Data Analytics & AI Applications	3	0	0	3
3	PCC	23A3313T	Full Stack AI Development	3	0	0	3
4	PEC	23A3314Ta 23A3314Tb 23A3314Tc 23A3314Td	Professional Elective-II 1. Graph Neural Networks 2. Recommender Systems 3. Predictive Analytics 4. Block chain for AI	3	0	0	3
5	PEC	23A3315Ta 23A3315Tb 23A3315Tc 23A3315Td	Professional Elective-III 1. Introduction to Quantum Computing 2. AI for Finance 3. Social Network Analysis 4. Cyber security & AI-driven Threat Detection	3	0	0	3
6	OEC		Open Elective-II	3	0	0	3
7	PCC(Lab)	23A3316P	Big Data & Cloud Computing Lab	0	0	3	1.5
8	PCC(Lab)	23A3317P	Full Stack AI Lab	0	0	3	1.5
9	SEC	23A0026T	Skill Enhancement Course Soft skills	0	1	2	2
10	AC	23A0053T	Audit Course Technical Paper Writing & IPR	2	0	0	-
11		23A3318	Workshop				
Total				20	1	08	23
Mandatory Industry Internship of 08 weeks duration during summer vacation							

Category	Credits
Professional Core Courses(PCC)	12
Professional Elective Courses(PEC)	6
Open Elective Courses(OEC)	3
Skill Enhancement Course(SEC)	2
Audit Course(AC)	-
Total	23



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Cloud Computing for AI (Professional Core)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3311T	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PCC
Course Objectives:					
This course will enable students:					
- To introduce the concepts, models, and services of cloud computing and its role in AI. - To explore the architecture and deployment of AI applications on cloud platforms. - To equip students with skills in using cloud-based tools and services for AI/ML workloads. - To understand data storage, processing, and security in cloud for AI tasks. - To apply cloud computing principles to real-world AI-based solutions					
Course Outcomes (CO):					
On completion of this course, student will be able to:					
- Explain cloud computing architecture, services, and deployment models. - Utilize cloud platforms (AWS, GCP, Azure) for training and deploying AI models. - Handle large-scale data storage and processing in the cloud environment. - Integrate AI workflows using server less and container-based architectures. - Analyze challenges in security, cost, scalability, and performance of cloud-based AI systems.					
Syllabus					Total Hours:
Module-I	Introduction to Cloud Computing and AI Integration			10 Hrs	
Basics of Cloud Computing: Characteristics, Models, and Services, Cloud Service Models: IaaS, PaaS, SaaS, Deployment Models: Public, Private, Hybrid, Community, AI and Cloud Convergence: Benefits and Challenges, Use Cases of AI in Cloud: NLP, Vision, Analytics, Overview of Cloud Providers for AI: AWS, Azure, GCP.					
Module-II	Storage, Computing, and Data Processing in the Cloud			10 Hrs	
Cloud Storage Services: S3, Blob, BigQuery, Virtualization and Elastic Computing, Distributed Computing with Hadoop and Spark, Data Ingestion and Processing Pipelines, Data Lakes and Warehousing in the Cloud, Cost Optimization for Storage and Compute Resources.					
Module-III	Cloud-based Machine Learning and Deep Learning			10 Hrs	
ML Services on AWS (SageMaker), Azure ML, GCP Vertex AI, Training and Deploying Models on Cloud, AutoML and Custom ML Model Workflows, GPUs/TPUs for Model Training, Experiment Tracking and Model Evaluation, Integration of Notebooks (Jupyter, Colab) with Cloud Storage.					
Module-IV	Advanced Cloud Concepts for AI Applications			9 Hrs	
Containers and Docker for AI Applications, Kubernetes and Cloud-native AI Workflows, Serverless Computing: AWS Lambda, Azure Functions, CI/CD Pipelines for AI Models in Cloud, Scaling AI Applications using Load Balancers and Auto-Scaling. Monitoring and Logging in Cloud for AI Workflows, VertexAI					

Module-V	Security, Ethics, and Case Studies in Cloud AI	9 Hrs
Security and Privacy in Cloud-based AI, Identity and Access Management (IAM) in Cloud, Cost Management and Billing for AI Services, Ethical Issues and Fairness in Cloud AI, Case Study: AI in Healthcare Cloud Solutions, Case Study: Real-Time Analytics in Financial Cloud Services.		
Text Books: 1. RajkumarBuyya, Christian Vecchiola, S. ThamaraiSelvi, Mastering Cloud Computing, McGraw-Hill. 2. Judith Hurwitz et al., Cloud Computing for Dummies, Wiley. 3. AurélienGéron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly.		
Reference Books: 1. Kai Hwang, Geoffrey C. Fox, Jack G. Dongarra, Distributed and Cloud Computing, Morgan Kaufmann. 2. Tomasz Kajdanowicz et al., Practical Cloud AI, Springer. 3. Mark Wilkins, AI and Machine Learning for Coders in Cloud, Packt Publishing.		
Online Learning Resources: 1. AWS Cloud Practitioner & Machine Learning Path – AWS Training 2. Google Cloud AI and ML Specialization – Coursera 3. Microsoft Azure AI Engineer Associate – Learn Portal 4. IBM Cloud and AI Learning – Cognitive Class 5. Cloud Computing and Distributed Systems (CLOUDS) Lab – University of Melbourne		



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Big Data Analytics & AI Applications (Professional Core)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3312T	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PCC
Course Objectives:					
This course will enable students:					
- To introduce the fundamentals of big data and its role in AI-driven applications. - To explore big data tools and technologies such as Hadoop, Spark, and NoSQL databases. - To enable students to build scalable AI pipelines for data analytics. - To apply AI/ML algorithms for real-time and batch processing environments. - To demonstrate use cases of big data in domains like healthcare, finance, and IoT using AI.					
Course Outcomes (CO):					
On completion of this course, student will be able to:					
- Understand the architecture and ecosystem of big data processing. - Analyze and manage large-scale datasets using Hadoop and Spark. - Apply AI/ML techniques to extract insights from big data. - Design and implement scalable data pipelines using distributed frameworks. - Solve real-world domain problems with AI-powered big data solutions.					
Syllabus				Total Hours:	
Module-I	Introduction to Big Data and Analytics Ecosystem			10 Hrs	
Definition and Characteristics of Big Data – Volume, Velocity, Variety, Veracity, Value, Types of Analytics: Descriptive, Diagnostic, Predictive, Prescriptive, Big Data Challenges and Opportunities, Hadoop Ecosystem Overview: HDFS, MapReduce, YARN, NoSQL Databases: Key-Value, Columnar, Document, Graph Models, Data Lake vs. Data Warehouse.					
Module-II	Big Data Tools and Frameworks			9 Hrs	
Apache Spark Architecture and RDDs, Spark SQL, DataFrames, and Datasets, Spark Streaming for Real-Time Analytics, Kafka for Data Ingestion and Message Queues, Hive, Pig, and Impala for Big Data Querying, Comparative Analysis of Hadoop vs. Spark.					
Module-III	Machine Learning on Big Data			9 Hrs	
Introduction to MLlib and Scikit-learn, Data Preprocessing for Big Data ML Pipelines, Supervised Learning: Classification and Regression on Large Datasets, Unsupervised Learning: Clustering and Dimensionality Reduction, Model Evaluation and Validation Techniques, Distributed Training and Optimization Techniques, Distributed Computing Frameworks					
Module-IV	AI Applications on Big Data			10 Hrs	
Predictive Maintenance using Big Data & AI, Fraud Detection in Banking with Machine Learning, AI in Healthcare: Diagnosis, Genomics, Patient Monitoring, Retail and E-commerce Analytics, AI for Smart Cities and IoT Sensor Data Analysis, Evaluation of Real-Time AI Applications on Streaming Data.					
Module-V	Advanced Topics and Case Studies			10 Hrs	
Deep Learning on Big Data using TensorFlow on Spark, Explainable AI (XAI) in Big Data Environments, Ethical Issues and Data Governance in Big Data AI, Edge Computing and AI for Low Latency Applications, Case Study 1: AI-Powered Big Data in Healthcare, Case Study 2: Big Data AI Solution in Smart Manufacturing.					
Text Books:					

1. Big Data: Principles and Paradigms by Rajkumar Buyya, Rodrigo N. Calheiros, Amir Vahid Dastjerdi – Wiley
2. Learning Spark: Lightning-Fast Big Data Analysis by Jules S. Damji et al. – O'Reilly
3. Data Science and Big Data Analytics by EMC Education Services – Wiley

Reference Books:

1. Designing Data-Intensive Applications by Martin Kleppmann – O'Reilly
2. Machine Learning with Spark by Rajdeep Dua, Tathagata Das – Packt Publishing
3. Streaming Systems by Tyler Akidau – O'Reilly Media
4. Artificial Intelligence for Big Data by Anand Deshpande – Packt

Web References:

1. <https://www.coursera.org/specializations/big-data>
2. <https://spark.apache.org/docs/latest/>
3. <https://www.edx.org/course/big-data-analysis-with-python>
4. <https://www.udacity.com/course/ai-for-business-leaders--nd088>
5. <https://www.kaggle.com/learn/intro-to-machine-learning>
6. <https://data-flair.training/blogs/apache-spark-tutorial/>



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Full Stack AI Development (Professional Core)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3313T	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PCC
Course Objectives:					
This course will enable students:					
- To equip students with knowledge and skills for building end-to-end AI-powered web applications. - To provide hands-on experience in integrating machine learning models with frontend and backend technologies. - To teach model deployment, version control, and MLOps best practices. - To expose students to full stack frameworks and cloud-based deployment platforms. - To prepare students for real-world AI applications in production settings.					
Course Outcomes (CO):					
On completion of this course, student will be able to:					
- Understand and apply full stack development principles in the context of AI solutions. - Build and serve machine learning models via RESTful APIs. - Design frontend interfaces for interaction with AI models. - Deploy AI applications using modern DevOps tools and cloud platforms. - Manage datasets, model versioning, and workflows in production-grade systems.					
Syllabus				Total Hours:48	
Module-I	Introduction to Full Stack AI Development			10 Hrs	
Overview of Full Stack Development in AI Context, Layers: Frontend, Backend, ML Layer, and Deployment Layer, Tools and Technology Stack (React, Node.js, Flask, Django, FastAPI, TensorFlow, PyTorch, MongoDB, PostgreSQL), Understanding Model Lifecycle and ML Ops, MLOps tools					
Module-II	Backend Development and API Integration			9 Hrs	
Introduction to Flask / FastAPI for model serving, RESTful API design and documentation (Swagger/OpenAPI), Connecting AI/ML models to APIs, Authentication, request handling, and session management, Error handling and response structuring					
Module-III	Frontend Development for AI Interfaces			10 Hrs	
Overview of frontend frameworks (React/Angular/Vue), Creating dynamic forms and dashboards for AI input/output, Data visualization using Chart.js, D3.js, Connecting frontend to API endpoints, Responsive design for AI application UX.					
Module-IV	Model Deployment and MLOps			9 Hrs	
Basics of CI/CD pipelines for AI models, Using Docker for containerization, Deployment on cloud platforms (Heroku, AWS, GCP), Introduction to MLflow, DVC, and model versioning, Logging, monitoring, and performance metrics.					
Module-V	Capstone Project and Case Studies			10 Hrs	
Full Stack AI Project Planning & Implementation, Use cases: Chatbot, Recommendation System,					

Image Classification App, NLP Web App, Industry-oriented workflows and best practices, Ethical considerations and data governance in AI applications

Textbooks:

1. "Full Stack Deep Learning" by Hamel Husain et al. (online version available at fullstackdeeplearning.com)
2. "Building Machine Learning Powered Applications" by Emmanuel Ameisen, O'Reilly.
3. "Flask Web Development" by Miguel Grinberg.

Reference Books:

1. "Machine Learning Engineering" by Andriy Burkov.
2. "Designing Data-Intensive Applications" by Martin Kleppmann.
3. "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron.

Online Resources:

1. Full Stack Deep Learning Course
2. FastAPI Documentation
3. Flask Mega-Tutorial

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Graph Neural Networks (Professional Elective-II)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3314Ta	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students: - To introduce the fundamentals of graph theory and graph-structured data. - To explore the concepts of neural networks extended to non-Euclidean domains. - To understand architectures and algorithms behind various types of GNNs. - To apply GNN models in real-world applications such as recommendation, social networks, and bioinformatics. - To enable students to build and evaluate GNN models using frameworks like PyTorch Geometric and DGL.					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Understand the basics of graph structures and their significance in machine learning. - Learn and implement different types of GNN architectures. - Apply GNNs to real-world structured data problems. - Use modern libraries and tools to train and evaluate GNNs. - Analyze the effectiveness and limitations of GNNs in different domains.					
Syllabus					Total Hours:48
Module-I	Fundamentals of Graph Theory and Machine Learning on Graphs				10 Hrs
Introduction to Graphs: Nodes, Edges, Adjacency Matrix, Types of Graphs: Directed, Undirected, Weighted, Bipartite, Graph Traversal Algorithms (BFS, DFS), Graph Representations for ML (Adjacency List, Matrix, Laplacian), Node, Edge, and Graph-level Prediction Problems, Motivation and Challenges for Learning on Graphs.					
Module-II	Spectral and Spatial Methods for Graph Learning				9 Hrs
Spectral Graph Theory Basics, Graph Convolution via Spectral Methods, Chebyshev and First-order Approximations, Spatial Graph Convolutions, Comparison of Spectral vs Spatial GNNs, Graph Laplacian and Eigenvalue Properties.					
Module-III	Graph Neural Network Architectures				9 Hrs
Graph Convolutional Networks (GCNs), Graph Attention Networks (GATs), GraphSAGE: Sampling and Aggregation, Graph Isomorphism Networks (GIN), Message Passing Neural Networks (MPNNs), Inductive vs Transductive GNN Learning.					
Module-IV	Applications of GNNs				10 Hrs
Node Classification (e.g., Cora, Citeseer), Link Prediction (e.g., Recommender Systems), Graph Classification (e.g., Molecule Property Prediction), Traffic Forecasting and Social Network					

Modeling, GNNs in Healthcare and Bioinformatics, Explainability and Interpretability in GNNs.		
Module-V	Implementation, Optimization, and Recent Advances	10 Hrs
Overview of PyTorch Geometric and DGL, GraphNet, Data Loading and Preprocessing for Graph Datasets, Model Training, Loss Functions, and Evaluation Metrics, Hyper parameter Tuning in GNNs, Recent Research Trends and Architectures (e.g., Heterogeneous GNNs, Graph Transformers), Challenges and Future Directions in GNNs.		
Text Books: 1. Zonghan Wu, Shirui Pan, Fengwen Chen, Guodong Long, Chengqi Zhang, Philip S. Yu, A Comprehensive Survey on Graph Neural Networks, IEEE Transactions on Neural Networks and Learning Systems, 2021. 2. Yao Ma, Jiliang Tang, Deep Learning on Graphs, Cambridge University Press, 2021. 3. William L. Hamilton, Graph Representation Learning, Morgan & Claypool Publishers, 2020.		
Reference Books: 1. Barrett, Jure Leskovec, Mining of Massive Datasets, Cambridge University Press. 2. Thomas Kipf, GCN and related papers and tutorials (arXiv). 3. Petar Veličković, Graph Attention Networks (original paper and slides). 4. Michael Bronstein et al., Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges (arXiv preprint).		
Online Learning Resources: 1. https://pytorch-geometric.readthedocs.io/ – PyTorch Geometric Docs 2. https://cs.stanford.edu/people/jure/ – Stanford GNN Projects 3. https://www.coursera.org/learn/graph-neural-networks – Coursera GNN Course by Stanford		



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Recommender Systems (Professional Elective-II)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3314Tb	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students: - To understand the theoretical foundations and practical techniques behind recommender systems. - To explore collaborative, content-based, and hybrid recommendation methods. - To apply matrix factorization and deep learning for building intelligent recommenders. - To analyze system performance using standard evaluation metrics. - To design and implement recommender systems for real-world applications.					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Explain the core concepts and types of recommender systems. - Implement basic collaborative and content-based filtering techniques. - Apply matrix factorization and deep learning models to recommendation problems. - Evaluate and optimize recommender systems using appropriate metrics. - Design scalable and context-aware recommender systems for diverse applications.					
Syllabus				Total Hours:48	
Module-I	Introduction to Recommender Systems			10 Hrs	
Introduction to Recommender Systems: Introduction to Information Filtering Systems, Types of Recommender Systems: Content-based, Collaborative, Hybrid, Data Sources: Explicit vs Implicit Feedback, Applications and Challenges in Recommendation, User and Item Profiling, Popularity, Personalization, and Serendipity Trade-offs.					
Module-II	Collaborative Filtering Techniques			9 Hrs	
Collaborative Filtering Techniques: User-based Collaborative Filtering, Item-based Collaborative Filtering, Similarity Measures: Cosine, Pearson, Jaccard, Neighborhood Selection and k-NN, Cold-start and Data Sparsity Issues, Memory-based vs Model-based Collaborative Filtering, Hybrid Models.					
Module-III	Content-based and Hybrid Systems			10 Hrs	
Content-based and Hybrid Systems: Item Feature Extraction and Vector Representation, TF-IDF and Cosine Similarity in Recommendations, User Profile Learning, Limitations of Content-based Filtering, Hybrid Recommender Architectures, Case Study: Netflix, Amazon Hybrid Systems.					
Module-IV	Matrix Factorization and Deep Learning Approaches			9 Hrs	
Latent Factor Models and SVD, ALS and SGD for Matrix Factorization, Non-negative Matrix Factorization (NMF), Neural Collaborative Filtering (NCF), Deep Learning Models: Auto encoders, CNNs, RNNs for Recommendations, Graph-based and Knowledge Graph Recommenders.					

Module-V	Evaluation, Ethics, and Industrial Applications	10 Hrs
Evaluation Metrics: Precision, Recall, F1, NDCG, MAP, A/B Testing in Recommender Systems, Explainability in Recommendations, Fairness, Bias, and Privacy in Recommenders, Scalability and Real-time Recommendations, Deploying Recommender Systems at Scale (e.g., Spotify, YouTube).		
Text Books: 1. Charu C. Aggarwal, Recommender Systems: The Textbook, Springer, 2016. 2. Francesco Ricci, Lior Rokach, and Bracha Shapira, Recommender Systems Handbook, Springer, 2nd Ed., 2015.		
Reference Books: 1. Jannach, Dietmar et al., Recommender Systems: An Introduction, Cambridge University Press, 2010. 2. Michael Ekstrand, Joseph A. Konstan, Collaborative Filtering Recommender Systems, Now Publishers, 2011. 3. Research papers from ACM RecSys Conference proceedings.		
Web Reference: 1. https://www.coursera.org/learn/recommender-systems–Coursera:UniversityofMinnesota 2. https://www.kaggle.com/learn/recommendation-systems–KaggleCourse 3. https://developers.google.com/machine-learning/recommendation–GoogleDevelopers		



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Predictive Analytics (Professional Elective-II)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3314Tc	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students: - To introduce the fundamental concepts and techniques of predictive analytics. - To apply statistical models and machine learning algorithms for prediction. - To interpret model performance using evaluation metrics. - To explore feature engineering, model tuning, and cross-validation. - To implement predictive solutions for real-world business and research problems.					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Understand the principles and importance of predictive analytics. - Apply regression and classification models for predictive tasks. - Perform data preprocessing, feature selection, and transformation. - Evaluate and validate models using standard metrics. - Design predictive solutions to solve domain-specific challenges.					
Syllabus				Total Hours:48	
Module-I	Introduction to Predictive Analytics			9 Hrs	
Introduction to Predictive Analytics and Business Intelligence, Types of Predictive Models: Classification, Regression, Time Series, Supervised vs Unsupervised Learning, Predictive Modeling Workflow, Applications in Marketing, Finance, Healthcare, Challenges in Predictive Analytics.					
Module-II	Data Preparation and Feature Engineering			10 Hrs	
Data Cleaning: Handling Missing, Noisy, and Inconsistent Data, Feature Selection and Dimensionality Reduction (PCA, LDA), Feature Scaling: Normalization, Standardization, Encoding Categorical Variables, Feature Extraction and Construction, Dealing with Imbalanced Datasets.					
Module-III	Predictive Modeling with Regression and Classification			10 Hrs	
Linear Regression and Polynomial Regression, Logistic Regression for Binary Classification, Decision Trees and Random Forest, k-Nearest Neighbors (k-NN) and Naïve Bayes, Support Vector Machines (SVM), Model Selection and Comparison.					
Module-IV	Model Evaluation and Validation			10 Hrs	
Training, Testing, and Validation Sets, Cross-Validation Techniques (k-Fold, Stratified, LOOCV), Evaluation Metrics: Accuracy, Precision, Recall, F1 Score, ROC-AUC, Confusion Matrix and Classification Report, Bias-Variance Trade-off and Overfitting, Hyperparameter Tuning: Grid Search, Random Search, Tools for Hyper parameter Tuning.					
Module-V	Advanced Topics and Applications			9 Hrs	
Ensemble Learning: Bagging, Boosting (AdaBoost, XGBoost), Predictive Analytics with Time Series (ARIMA, Prophet), Deep Learning for Predictive Modeling (ANNs, LSTM), Use of Predictive Analytics in IoT, Retail, and Healthcare, Ethics and Privacy in Predictive Analytics, Building and Deploying End-to-End Predictive Systems.					

Text Books:

1. Dean Abbott, Applied Predictive Analytics: Principles and Techniques for the Professional Data Analyst, Wiley, 2014.
2. John D. Kelleher, Brendan Tierney, Data Science: Predictive Analytics and Data Mining, MIT Press, 2018.

Reference Books:

1. Galit Shmueli et al., Data Mining for Business Analytics: Concepts, Techniques, and Applications in R, Wiley, 2017.
2. Eric Siegel, Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die, Wiley, 2016.
3. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, Springer, 2009.

Web References:

1. <https://www.coursera.org/specializations/predictive-analytics-CourseraSpecialization>
2. <https://www.edx.org/course/data-science-and-machine-learning-capstone> – edX Predictive Analytics Courses
3. <https://www.kaggle.com/learn/intro-to-machine-learning> – Kaggle Tutorials



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Block chain for AI (Professional Elective-II)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3314Td	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students: - To understand the foundational concepts of block chain technology and its architecture. - To explore smart contracts, consensus algorithms, and distributed ledger technology. - To investigate the integration of AI with block chain for secure, decentralized applications. - To develop block chain-enabled AI solutions for real-world use cases. - To understand the ethical, security, and scalability challenges in Blockchain-AI ecosystems.					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Explain the fundamentals of block chain and its components. - Analyze the role of consensus mechanisms in maintaining trust and decentralization. - Apply block chain for secure data sharing in AI systems. - Develop and deploy smart contracts using Ethereum/Solidity. - Evaluate block chain-based AI applications in healthcare, finance, and supply chains.					
Syllabus					TotalHours:48
Module-I	Block chain Fundamentals and Architecture				10 Hrs
Introduction to Block chain Technology, Components: Blocks, Hashing, Merkle Trees, Types of Block chains: Public, Private, Hybrid and Consortium, Distributed Ledger Technology (DLT) and P2P Networks, Block chain Structure and Mining, Use Cases and Evolution of Block chain.					
Module-II	Smart Contracts and Consensus Mechanisms				9 Hrs
Smart Contracts: Definition, Features, Use Cases, Ethereum and Solidity Basics, Consensus Algorithms: PoW, PoS, DPoS, PBFT, Gas, Transactions, and Events in Ethereum, Hyperledger Fabric: Architecture and Chaincode, Deployment and Testing of Smart Contracts.					
Module-III	Integration of Block chain and AI				10 Hrs
Motivation for Integrating Block chain with AI, Decentralized AI Models and Federated Learning, Secure Model Sharing and Provenance, Block chain for Data Integrity in AI Systems, AI for Block chain (e.g., optimizing consensus), Case Study: Decentralized AI Marketplace.					
Module-IV	Applications of Block chain in AI Systems				9 Hrs
Block chain for Explainable and Trusted AI, Applications in Healthcare and Genomics, Blockchain for Autonomous Vehicles and IoT, Financial AI Systems with Smart Contracts, Supply Chain and Logistics Intelligence, NFT-based AI Applications (Digital Identity, IP).					
Module-V	Security, Privacy and Challenges in Block chain-AI				10 Hrs
Security Challenges: Sybil Attacks, 51% Attacks, Privacy Preservation and Zero Knowledge Proofs, Scalability and Energy Concerns in Block chain-AI, Ethical and Legal Concerns in AI with Block chain, Interoperability of Block chain Platforms, Future Trends: Quantum-Resistant Block chain AI.					

Text Books:

1. Imran Bashir, Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications, Packt, 2020.
2. Melanie Swan, Blockchain: Blueprint for a New Economy, O'Reilly Media, 2015.
3. Joseph Holbrook, Architecting AI Solutions on Blockchain, Packt Publishing, 2020.

Reference Books:

1. ArshdeepBahga, Vijay Madisetti, Blockchain Applications: A Hands-On Approach, VPT, 2017.
2. Karamjit Singh, Blockchain for AI: Use Cases and Implementation, Springer, 2023.
3. Roger Wattenhofer, The Science of the Blockchain, 2016.

Web References:

1. <https://www.coursera.org/specializations/blockchain>
2. edX: Blockchain Fundamentals – UC Berkeley
3. <https://www.coursera.org/learn/ai-blockchain-ibm>



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Introduction to Quantum Computing (Professional Elective-III)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3315Ta	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students:					
• To introduce the principles and mathematical foundations of quantum computation. • To understand quantum gates, circuits, and computation models. • To explore quantum algorithms and their advantages over classical ones. • To develop the ability to simulate and write basic quantum programs. • To understand real-world applications and the future of quantum computing in AI, cryptography, and optimization.					
Course Outcomes:					
On completion of this course, student will be able to:					
• Explain the fundamental concepts of quantum mechanics used in computing. • Construct and analyze quantum circuits using standard gates. • Apply quantum algorithms like Deutsch-Jozsa, Grover's, and Shor's. • Develop simple quantum programs using Qiskit or similar platforms. • Analyze applications and challenges of quantum computing in real-world domains.					
Syllabus					Total Hours:48
Module-I	Fundamentals of Quantum Mechanics and Linear Algebra				10 Hrs
Classical vs Quantum Computation, Complex Numbers, Vectors and Matrices, Eigen values and Eigen Vectors, Hilbert Spaces and Dirac Notation, Quantum States and Qubits, Superposition and Measurement, Tensor Products and Multi-Qubit Systems.					
Module-II	Quantum Gates and Circuits				9 Hrs
Quantum Logic Gates: Pauli, Hadamard, Phase, Controlled Gates and CNOT, Unitary Operations and Reversibility, Quantum Circuit Representation, Quantum Teleportation, Simulation of Quantum Circuits.					
Module-III	Quantum Algorithms and Complexity				9 Hrs
Quantum Parallelism and Interference, Deutsch and Deutsch-Jozsa Algorithms, Grover's Search Algorithm, Shor's Factoring Algorithm, Quantum Fourier Transform, Complexity Classes: BQP, P, NP, and QMA.					
Module-IV	Quantum Programming and Simulation Platforms				10 Hrs
The Concept of Violence- Types of Gender-based Violence-Gender-based Violence from a Human Rights Perspective-Sexual Harassment - Domestic Violence - Different forms of violence against women - Causes of violence, impact of violence against women - Consequences of gender-based violence.					
Module-V	Applications and Future of Quantum Computing				10 Hrs
Quantum Machine Learning: Basics and Models, Quantum Cryptography and Quantum Key Distribution, Quantum Algorithms in AI and Optimization, Quantum Advantage and Supremacy, Ethical and Societal Impact of Quantum Technologies, Future Trends and Research Directions.					

Text Books:

1. Michael A. Nielsen, Isaac L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press, 10th Anniversary Edition, 2010.
2. Eleanor Rieffel and Wolfgang Polak, Quantum Computing: A Gentle Introduction, MIT Press, 2011.
3. Chris Bernhardt, Quantum Computing for Everyone, MIT Press, 2019.

Reference Books:

1. David McMahon, Quantum Computing Explained, Wiley, 2008.
2. Phillip Kaye, Raymond Laflamme, Michele Mosca, An Introduction to Quantum Computing, Oxford University Press, 2007.
3. Scott Aaronson, Quantum Computing Since Democritus, Cambridge University Press, 2013.

Web References:

1. IBM Quantum Experience and Qiskit Tutorials
2. Coursera – Quantum Mechanics and Quantum Computation by UC Berkeley
3. edX – The Quantum Internet and Quantum Computers
4. YouTube – Quantum Computing for the Determined by Michael Nielsen
5. Qiskit Textbook – IBM Quantum.



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AI for Finance (Professional Elective-III)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3315Tb	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students:					
- To introduce the role of Artificial Intelligence (AI) in financial applications and decision making. - To understand financial data types, sources, and processing methods. - To apply machine learning and deep learning models in various finance sectors. - To analyze risk, fraud detection, credit scoring, and portfolio management using AI. - To evaluate ethical and regulatory challenges in AI-enabled finance.					
Course Outcomes (CO):					
On completion of this course, student will be able to:					
- Describe the fundamentals of AI techniques applicable to finance. - Analyze financial time series data using AI-based models. - Apply machine learning for fraud detection and credit risk analysis. - Build predictive models for stock prices, trading, and customer segmentation. - Evaluate the limitations and ethical implications of AI in financial systems.					
Syllabus				Total Hours:48	
Module-I	Introduction to Finance and AI Applications			10 Hrs	
Introduction to Financial Markets and Instruments, Overview of AI Techniques in Finance, Types of Financial Data: Market, Transactional, Customer, Financial Statements and Key Indicators, AI Use Cases in Banking, Insurance, and Investment, FinTech and the Rise of Robo-Advisors.					
Module-II	Machine Learning in Finance			10 Hrs	
Supervised Learning for Credit Scoring, Unsupervised Learning for Customer Segmentation, Feature Engineering for Financial Data, Handling Imbalanced Datasets in Fraud Detection, Time Series Forecasting with Regression and ARIMA Model and its Variations, Model Validation and Backtesting in Finance.					
Module-III	Deep Learning and NLP in Finance			9 Hrs	
Introduction to Deep Learning for Finance, Stock Price Prediction using LSTM and RNNs, Sentiment Analysis from Financial News and Tweets, NLP for Document Classification: Earnings Reports, Chatbots and Virtual Assistants in Banking, Reinforcement Learning for Portfolio Optimization.					
Module-IV	AI-Driven Financial Applications			9 Hrs	

Fraud Detection Systems using ML and DL, Credit Risk and Loan Default Prediction, AI in Algorithmic and High-Frequency Trading, Robo-Advisors: Architecture and Optimization, Blockchain and AI Integration for Financial Security, Case Studies: AI in Wealth Management & Insurance.		
Module-V	Ethics, Regulation, and Future of AI in Finance	10 Hrs
Regulatory Frameworks in AI-based Finance, Explainability and Interpretability of Financial Models, Ethical Issues: Bias, Fairness, Transparency, Data Privacy and GDPR in Financial AI, Responsible AI Practices in Finance, Emerging Trends: Quantum AI, Decentralized Finance (DeFi).		
Text Books: 1. Yves Hilpisch, Artificial Intelligence in Finance: A Python-Based Guide, O'Reilly, 2020. 2. Yves Hilpisch, Python for Finance: Mastering Data-Driven Finance, O'Reilly, 2018. 3. Markus Loecher, Machine Learning for Finance, Packt Publishing, 2021.		
Reference Books: 1. A. W. Lo, The Evolution of Technical Analysis, Wiley Finance, 2010. 2. Tony Guida, Big Data and Machine Learning in Quantitative Investment, Wiley, 2019. 3. Tucker Balch, AI for Trading – Georgia Tech Specialization, Coursera.		
Web References: 1. Coursera: AI for Trading – by NYIF and Google Cloud 2. edX: Artificial Intelligence in Finance – NYIF 3. Udemy: Machine Learning and AI in Finance 4. DataCamp: Financial Trading with Python 5. YouTube: AI for Finance by Sentdex, Two Minute Papers, and DataProfessor		



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Social Network Analysis (Professional Elective-II)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3315Tc	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students:					
- To introduce the fundamentals and key concepts of social network theory and graph theory. - To analyze the structure and properties of large-scale social networks. - To apply centrality, influence, and community detection measures. - To model information diffusion and network dynamics. - To implement real-world social network analysis using tools and datasets.					
Course Outcomes (CO):					
On completion of this course, student will be able to:					
- Understand basic network models and social network structures. - Analyze key properties like centrality, clustering, and small-world effect. - Apply community detection algorithms and influence maximization. - Interpret diffusion models for viral marketing and information spread. - Use tools such as Gephi, NetworkX, or SNAP for real-world SNA.					
Syllabus				Total Hours:48	
UNIT-I	Introduction to Social Networks and Graph Theory			10 Hrs	
Basic Concepts: Graphs, Nodes, Edges, Directed/Undirected Graphs, Real-world Examples: Facebook, Twitter, LinkedIn, Adjacency Matrix and Graph Representation, Types of Social Networks: Ego, Bipartite, Multilayer, Degree Distribution, Path Length, and Connectivity, Random Graph Models: Erdős–Rényi and Watts-Strogatz.					
UNIT-II	Structural Properties of Networks			10 Hrs	
Network Centrality Measures: Degree, Closeness, Betweenness, Eigenvector Centrality and PageRank, Network Clustering and Community Detection Basics, Triadic Closure and Clustering Coefficient, Small-world Phenomenon and Milgram’s Experiment, Homophily, Influence, and Structural Balance.					
UNIT-III	Community Detection and Subgroup Analysis			10 Hrs	
Girvan–Newman Algorithm and Modularity, Label Propagation and Louvain Method, Clique Detection and k-Core Decomposition, Overlapping Communities and Fuzzy Clustering, Cohesive Subgroups and Structural Equivalence, Evaluation Metrics: NMI, Modularity Score, Conductance.					
UNIT-IV	Information Diffusion and Influence in Networks			9 Hrs	
Models of Diffusion: Linear Threshold and Independent Cascade, Influence Maximization and Viral Marketing, Contagion Models and Epidemic Spreading, Rumor Propagation and Cascade Models, Information Bottlenecks and Bridges, Measuring Influence and Reach.					
Module-V	Tools, Applications, and Ethics in SNA			9 Hrs	
SNA Tools: Gephi, Pajek, NetworkX, SNAP, Case Study: Twitter and Hashtag Analysis, LinkedIn Network Mining and Graph Features, Applications in Marketing, Security, and Epidemiology, Ethical Issues in Social Network Data Mining, Building and Visualizing Your Own Social Graph.					

Text Books:

1. Wasserman, S., & Faust, K., Social Network Analysis: Methods and Applications, Cambridge University Press, 1994.
2. Easley, D., & Kleinberg, J., Networks, Crowds, and Markets: Reasoning About a Highly
3. Connected World, Cambridge University Press, 2010.
4. Newman, M., Networks: An Introduction, Oxford University Press, 2010.

Reference Books:

1. Borgatti, S. P., Everett, M. G., & Johnson, J. C., Analyzing Social Networks, SAGE Publications, 2018.
2. Barabási, A.-L., Linked: How Everything Is Connected to Everything Else, Basic Books, 2014.
3. Hansen, D., Shneiderman, B., & Smith, M. A., Analyzing Social Media Networks with NodeXL, Elsevier, 2020.

Online Learning Resources:

1. Coursera – Social Network Analysis (University of Michigan)
2. [YouTube – NetworkX and Gephi Tutorials (freeCodeCamp, TheNetNinja)]
3. edX – Networks: Friends, Money, and Bytes (University of California, Berkeley)
4. Khan Academy – Graph Theory



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Cyber security & AI-Driven Threat Detection (Professional Elective-III)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3315Td	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students: - To provide a foundational understanding of cyber security principles and threat landscapes. - To explore the application of AI and machine learning techniques in detecting cyber threats. - To analyze malware behavior, intrusion patterns, and anomaly detection using intelligent systems. - To evaluate and build automated systems for real-time security analytics. - To understand the ethical, legal, and societal implications of AI-driven security system					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Understand cyber security frameworks, threat types, and vulnerabilities. - Apply AI/ML techniques for cyber threat identification and classification. - Analyze patterns in malware, network traffic, and security logs. - Design and evaluate intelligent intrusion detection and prevention systems. - Explore ethical hacking practices and policy aspects in AI-based security.					
Syllabus				Total Hours:48	
Module-I	Fundamentals of Cyber security			10 Hrs	
Introduction to Cyber security: CIA Triad, Threats & Vulnerabilities, Types of Attacks: Malware, Phishing, DDoS, Insider Threats, Security Policies and Access Controls, Risk Assessment and Vulnerability Management, Cryptography Basics: Symmetric, Asymmetric, Hash Functions, Cyber security Frameworks: NIST, ISO 27001, OWASP, COBIT..					
Module-II	Machine Learning for Cyber Threat Detection			9 Hrs	
Supervised and Unsupervised Learning in Security Contexts, Feature Engineering for Security Data, Classification Models for Intrusion Detection (SVM, RF, KNN), Clustering Techniques for Anomaly Detection, Evaluation Metrics: Accuracy, Precision, ROC, F1 Score, Case Study: AI for Email Phishing Detection.					
Module-III	Deep Learning in Threat Intelligence			9 Hrs	
Deep Neural Networks for Cyber security, RNNs and LSTMs for Log and Sequence Data,Auto encoders for Anomaly Detection, CNNs for Malware Classification using Binary Analysis, Adversarial Attacks on AI-based Security Systems, Case Study: Threat Detection using Deep Learning.					

Module-IV	Real-Time Threat Detection and SIEM Systems	10 Hrs
Security Information and Event Management (SIEM), Log Analysis and Real-Time Alerting, Threat Intelligence Platforms (TIPs), Integration of AI in SIEM Tools (Splunk, ELK Stack), Network Traffic and Packet Inspection using ML, SOC Operations and Automation using AI.		
Module-V	Ethical Hacking, Privacy, and Legal Aspects	10 Hrs
Penetration Testing & Ethical Hacking with AI Tools, Red Team vs. Blue Team Simulation, Data Privacy Regulations: GDPR, HIPAA, Cyber Laws, AI Bias and Fairness in Security Decision-Making, Case Study: Ethical Dilemmas in AI Security Systems, Future Trends: Zero Trust, AI SOC, Federated Threat Detection.		
Text Books: 1. Stallings, W., Network Security Essentials: Applications and Standards, Pearson Education. 2. Shon Harris & Fernando Maymi, CISSP All-in-One Exam Guide, McGraw Hill. 3. Emmanuel Tsukerman, Machine Learning for Cybersecurity Cookbook, Packt Publishing. 4. Clarence Chio & David Freeman, Machine Learning and Security, O'Reilly Media.		
Reference Books: 1. John Paul Mueller, Luca Massaron, Machine Learning for Dummies, Wiley. 2. Mark Stamp, Information Security: Principles and Practice, Wiley. 3. Bruce Schneier, Secrets and Lies: Digital Security in a Networked World, Wiley. 4. ShaiShalev-Shwartz and Shai Ben-David, Understanding Machine Learning, Cambridge 5. University Press.		
Web References 1. Online Learning Resources: 2. Coursera – AI for Cybersecurity (IBM) 3. edX – Cybersecurity Fundamentals by Rochester Institute of Technology 4. MIT OpenCourseWare – Computer and Network Security 5. [YouTube – Cybersecurity& AI Tutorials by Simplilearn, Great Learning] 6. Udemy – Machine Learning for Cybersecurity 7. Splunk Documentation – AI & Threat Detection		



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Big Data & Cloud Computing Lab (Professional Core)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3316P	0:0:3:1.5	3	CIE:30 SEE:70	3 Hours	PCC

Course Objectives:

This course will enable students:

- To provide hands-on experience in working with big data tools and cloud computing environments.
- To equip students with practical skills in data ingestion, transformation, analysis, and visualization using Hadoop and Spark ecosystems.
- To enable deployment and management of cloud services using AWS, Azure, or GCP.
- To expose students to cloud-native storage, computing, and container orchestration techniques.
- To integrate big data workflows with cloud infrastructure for scalable, distributed data processing.

Course Outcomes (CO):

On completion of this course, student will be able to:

- Students will be able to implement big data pipelines and cloud-based solutions using tools like Hadoop, Spark, and cloud platforms such as AWS, Azure, or GCP.
- Students gain proficiency in managing distributed data processing, scalable storage, cloud service provisioning, and deploying applications using containers and orchestration platforms.
- Students will understand the synergy between big data technologies and cloud computing to solve real-world problems efficiently.

List of lab Experiments:

Week:1

Installation and Configuration of Hadoop Cluster (Single Node & Multi-node) Hadoop HDFS setup, NameNode & DataNode configuration.

Week:2

Working with HDFS: File Operations Upload, read, delete, and replicate files in HDFS.

Week:3

MapReduce Programming Basics Word count, sorting, and filtering examples in Java/Python.

Week:4

Apache Hive & Pig for Querying Large Datasets Creation of tables, data loading, and running queries.

Week:5

Apache Spark Basics: RDDs and DataFrames Implement Spark transformations and actions.

Week:6

Data Preprocessing and Machine Learning using PySpark MLlib Classification or regression using MLlib pipelines (Cognitive Level: Apply & Evaluate)

Week:7

Introduction to Cloud Computing and AWS/Azure/GCP Console Creating virtual machines, basic

compute and storage services.

Week:8

Cloud Storage and Database Services Using S3 (AWS), Blob (Azure), or GCP buckets and Cloud SQL/NoSQL.

Week:9

Deploying Big Data Workloads on Cloud (EMR, HDInsight, Dataproc) Running Hadoop/Spark jobs in cloud-managed services.

Week:10

Cloud Function/Serverless Deployment.

Week:11

Building and deploying a serverless function (e.g., AWS Lambda) Containerization with Docker.

Week:12

Building, running, and managing Docker containers Orchestration with Kubernetes in the Cloud Deploy and manage a containerized application using GKE/EKS/AKS.

Text Books:

1. Tom White, Hadoop: The Definitive Guide, O'Reilly Media.
2. Rajkumar Buyya et al., Mastering Cloud Computing, McGraw-Hill Education.
3. Holden Karau et al., Learning Spark: Lightning-Fast Big Data Analysis, O'Reilly Media.

Reference Books:

1. Vignesh Prajapati, Big Data Analytics with R and Hadoop, Packt Publishing.
2. Benjamin Bengfort, Data Analytics with Hadoop, O'Reilly.
3. Srinivasan & J.Shrinivasan, Cloud Computing – A Hands-on Approach, Wiley India.

Online Learning Resources:

1. Big Data Specialization – Coursera (University of California San Diego)
2. Cloud Computing Basics – edX (LearnQuest)
3. Data Engineering with Google Cloud – Coursera (Google)



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FULL STACK AI LAB (Professional Core)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3316P	0:0:3:1.5	3	CIE:30 SEE:70	3 Hours	PCC
Course Objectives:					
This course will enable students: • Enable students to build end-to-end AI-powered web applications. • Integrate frontend, backend, database, and AI models in real-time. • Provide hands-on experience with Flask, Express, MongoDB, React, and ML models. • Develop and deploy AI applications using industry-standard practices.					
Course Outcomes (CO):					
On completion of this course, student will be able to: • Design frontend interfaces using React/HTML/CSS. • Build backend logic using Flask or Node.js APIs. • Integrate and deploy ML models with web services. • Store and retrieve data using MongoDB/MySQL. • Test, debug, and deploy AI-based web applications.					
List of Lab Experiments:					
Lab Experiments (12 Total)					
1. Setup Flask or Node.js server with React/HTML frontend. 2. Create login/signup system with Express/Flask and MongoDB. 3. Train and save ML model (e.g., Naive Bayes, Logistic Regression). 4. Build API to serve ML model predictions via Flask. 5. Integrate ML predictions in frontend using fetch/AJAX. 6. Create dynamic dashboard using Chart.js/Plotly. 7. Implement JWT tokens or sessions for authentication. 8. Add file upload functionality (image/text for prediction). 9. Store interactions/predictions in database and visualize history. 10. Create CI/CD pipeline using GitHub Actions/Heroku. 11. Build mini-project: News Classifier / Spam Detector / Fake News Detector. 12. Final Demo & Deployment on Render/Heroku/Vercel/localhost.					

Text Books:

1. “Full Stack Deep Learning” by Emmanuel Ameisen, O’Reilly, 2020
2. “Flask Web Development” by Miguel Grinberg, O’Reilly, 2018
3. “Python Machine Learning” by Sebastian Raschka, Packt Publishing

Reference Books:

1. “Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow” by Aurélien Géron
2. “MongoDB: The Definitive Guide” by Kristina Chodorow
3. “Node.js Design Patterns” by Mario Casciaro

Online Learning Resources:

Full Stack Web Development with Flask and Python- Udemy



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Soft Skills (Skill Enhancement Course) (Common to All Branches of Engineering)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0026T	0:1:2:2	2	CIE:30 SEE:70	3 Hours	SEC
Course Objectives:					
This course will enable students: - To encourage all round development of the students by focusing on soft skills - To make the students aware of critical thinking and problem-solving skills. - To enhance healthy relationship and understanding within and outside an organization. - To function effectively with heterogeneous teams					
Course Outcomes (CO):					
On completion of this course, student will be able to: - List out various elements of soft skills. - Describe methods for building professional image. - Apply critical thinking skills in problem solving. - Analyze the needs of an individual and team for well-being. - Assess the situation and take necessary decisions. - Create a productive work place atmosphere using social and work-life skills ensuring personal and emotional well-being					
Syllabus				Total Hours:48	
Module-I	Soft Skills & Communication Skills			9 Hrs	
Soft Skills - Introduction, Need - Mastering Techniques of Soft Skills – Communication Skills Significance, process, types - Barriers of communication - Improving techniques.					
Activities:					
Intrapersonal Skills- Narration about self- strengths and weaknesses- clarity of thought – self-expression – articulating with felicity (The facilitator can guide the participants before the activity citing examples from the lives of the great, anecdotes and literary sources)					
Interpersonal Skills- Group Discussion – Debate – Team Tasks - Book and film Reviews by groups - Group leader presenting views (non- controversial and secular) on contemporary issues or on a given topic.					
Verbal Communication- Oral Presentations- Extempore- brief addresses and speeches- convincing- negotiating- agreeing and disagreeing with professional grace.					
Non-verbal communication – Public speaking – Mock interviews – presentations with an objective to identify non- verbal clues and remedy the lapses on observation.					
Module-II	Critical Thinking			10 Hrs	

Active Listening – Observation – Curiosity – Introspection – Analytical Thinking – Open-mindedness – Creative Thinking - Positive thinking - Reflection

Activities:

Gathering information and statistics on a topic - sequencing – assorting – reasoning – critiquing issues placing the problem – finding the root cause - seeking viable solution – judging with rationale – evaluating the views of others - Case Study, Story Analysis

Module-III

Problem Solving & Decision Making

10 Hrs

Meaning & features of Problem Solving – Managing Conflict – Conflict resolution – Team building - Effective decision making in teams – Methods & Styles

Activities:

Placing a problem which involves conflict of interests, choice and views – formulating the problem – exploring solutions by proper reasoning – Discussion on important professional, career and organizational decisions and initiate debate on the appropriateness of the decision. Case Study & Group Discussion

Module-IV

Emotional Intelligence & Stress Management

10 Hrs

Managing Emotions – Thinking before Reacting – Empathy for Others – Self-awareness – Self Regulation – Stress factors – Controlling Stress – Tips

Activities:

Providing situations for the participants to express emotions such as happiness, enthusiasm, gratitude, sympathy, and confidence, compassion in the form of written or oral presentations.

Providing opportunities for the participants to narrate certain crisis and stress –ridden situations caused by failure, anger, jealousy, resentment and frustration in the form of written and oral presentation, Organizing Debates

Module-V

Corporate Etiquette

9 Hrs

Etiquette- Introduction, concept, significance - Corporate etiquette - meaning, modern etiquette, benefits - Global and local culture sensitivity - Gender Sensitivity - Etiquette in interaction- Cell phone etiquette - Dining etiquette - Netiquette - Job interview etiquette -Corporate grooming tips Overcoming challenges

Activities :

Providing situations to take part in the Role Plays where the students will learn about bad and good manners and etiquette - Group Activities to showcase gender sensitivity, dining etiquette etc. - Conducting mock job interviews - Case Study - Business Etiquette Games

NOTE-:

1. The facilitator can guide the participants before the activity citing examples from the lives of the great, anecdotes, epics, scriptures, autobiographies and literary sources which bear true relevance to the prescribed skill.

2. Case studies may be given wherever feasible for example for Decision Making- The decision of

King Lear.

Text Books:

1. James Momoh, "SMART GRID : Fundamentals of Design and Analysis", John Wiley and Sons, New York, 2012.
2. Janaka Ekanayake, Nick Jenkins, Kithsiri Liyanage, Jianzhong Wu, Akihiko Yokoyama, "Smart Grid: Technology and Applications", John Wiley & Sons, New Jersey, 2012.

Reference Books:

1. Power Grid Corporation of India Limited, "Smart Grid Primer", 1st Edition, Power Grid Corporation of India Limited, Bangalore, India, 2013.
2. Fereidoon.P.Sioshansi, "Smart Grid – Integrating Renewable, Distributed and Efficient Energy", 1st Edition, Academic Press, USA, 2011.
3. Stuart Borlase, "Smart Grids: Infrastructure, Technology and Solutions", 1st Edition, CRC Press Publication, England, 2013.
4. Phadke A G, Thorp J S, "Synchronized Phasor Measurements and Their Applications", 1st Edition, Springer, Newyork, 2012.



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Technical Report Writing & IPR (Audit Course)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0053T	2:0:0:0	0	CIE:30 SEE:70	3 Hours	AC
Course Objectives:					
This course will enable students: - To enable the students to practice the basic skills of research paper writing - To make the students understand the importance of IP and to educate them on the basic concepts of Intellectual Property Rights. - To practice the basic skills of performing quality literature review - To help them in knowing the significance of real life practice and procedure of Patents. - To enable them learn the procedure of obtaining Patents, Copyrights, & Trade Marks					
Course Outcomes:					
On completion of this course, student will be able to: - Identify key secondary literature related to their proposed technical paper writing - Explain various principles and styles in technical writing - Use the acquired knowledge in writing a research/technical paper - Analyse rights and responsibilities of holder of Patent, Copyright, Trademark, International Trademark etc. - Evaluate different forms of IPR available at national & international level - Develop skill of making search of various forms of IPR by using modern tools and techniques.					
Syllabus					Total Hours:48
Module-I					10 Hrs
Principles of Technical Writing: styles in technical writing; clarity, precision, coherence and logical sequence in writing-avoiding ambiguity- repetition, and vague language -highlighting your findings discussing your limitations -hedging and criticizing -plagiarism and paraphrasing.					
Module-II					10 Hrs
Technical Research Paper Writing: Abstract- Objectives-Limitations-Review of Literature-Problems and Framing Research Questions- Synopsis.					
Module-III					9 Hrs
Process of Research: publication mechanism: types of journals- indexing-seminars- conferences- proof reading –plagiarism style; seminar & conference paper writing; Methodology-discussion- results- citation rules.					
Module-IV					9 Hrs
Introduction to Intellectual property: Introduction, types of intellectual property, International organizations, agencies and treaties, importance of intellectual property rights Trade Marks: Purpose and function of trademarks, acquisition of trade mark rights, protectable matter, selecting and evaluating trade mark, trade mark registration processes.					
Module-V					10 Hrs
Law of copy rights: Fundamentals of copy right law, originality of material, rights of reproduction, rights to perform the work publicly, copy right ownership issues, copy right registration, notice of copy right, international copy right law Law of patents: Foundation of patent law, patent searching process, ownership rights and transfer. Patent law, intellectual property audits.					

Text Books:

1. Deborah. E. Bouchoux, Intellectual Property Rights, Cengage Learning India, 2013
2. Meenakshi Raman, Sangeeta Sharma. Technical Communication: Principles and practices. Oxford.

Reference Books:

1. R.Myneni, Law of Intellectual Property, 9th Ed, Asia law House, 2019.
2. Prabuddha Ganguli, Intellectual Property Rights Tata Mcgraw Hill, 2001
3. P.Naryan, Intellectual Property Law, 3rd Ed ,Eastern Law House, 2007.
4. Adrian Wallwork. English for Writing Research Papers Second Edition. Springer Cham Heidelberg New York ,2016
5. Dan Jones, Sam Dragga, Technical Writing Style

Web References:

1. <https://theconceptwriters.com.pk/principles-of-technical-writing/>
2. <https://www.ewh.ieee.org/soc/emcs/acstrial/newsletters/summer10/TechPaperWriting.html>
3. <https://www.ewh.ieee.org/soc/emcs/acstrial/newsletters/summer10/TechPaperWriting.html>
4. <https://www.manuscriptedit.com/scholar-hangout/process-publishing-research-paper-journal/>
5. <https://www.icsi.edu/media/website/IntellectualPropertyRightLaws&Practice.pdf>
6. <https://lawbhoomi.com/intellectual-property-rights-notes/>
7. <https://www.extension.purdue.edu/extmedia/ec/ec-723.pd>



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Semester-7 (Theory-6,SEC-1, AC-1)							
Sl. No.	Category	Course Code	Course Title	Hours per week			Credits
				L	T	P	
1	PCC	23A3319T	Generative AI	3	0	0	3
2	MEC	23A0049T 23A0050T 23A0048T	Management Course-II 1. Business Ethics and Corporate Governance 2. E-Business 3. Management Science	2	0	0	2
3	PEC	23A3320Ta 23A3320Tb 23A3320Tc 23A3320Td	Professional Elective-IV 1. Explainable AI & Model Interpretability 2. AI in Cyber Security 3. AI-driven Software Engineering & Dev Ops 4. AI for Robotics	3	0	0	3
4	PEC	23A3321Ta 23A3321Tb 23A3321Tc 23A3321Td	Professional Elective-V 1. ML Ops & AI Model Deployment 2. Data Wrangling 3. Healthcare AI 4. AI for Smart Cities & IoT Systems	3	0	0	3
5	OEC		Open Elective-III	3	0	0	3
6	OEC		Open Elective-IV	3	0	0	3
7	SEC	23A0540P	Skill Enhancement Course Prompt Engineering	0	1	2	2
8	AC	23A0054T	Audit Course Gender Sensitization	2	0	0	-
9	PCC	23A3322	Evaluation of Industry Internship	-	-	-	2
Total				19	1	02	21

Category	Credits
Professional Core Courses(PCC)	5
Professional Elective Courses(PEC)	6
Management Elective Courses(MEC)	2
Open Elective Courses(OEC)	6
Skill Enhancement Course(SEC)	2
Audit Course(AC)	-
Total	21



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Generative AI (Professional Core)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3319T	3:0:0:3	3	CIE:30 SEE:70	3Hours	PCC
Course Objectives:					
This course will enable students: - Introduce the fundamentals of Generative AI, including its principles, architecture, and evolution. - Provide a deep understanding of Large Language Models (LLMs) and their application in natural language generation tasks - Develop practical knowledge of Prompt Engineering, including prompt tuning, prompt design, and performance evaluation - Explore applications of Generative AI across various domains including code generation, art synthesis, content generation, and interactive systems. - Equip students with ethical, social, and safety considerations when designing and deploying generative AI applications.					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Demonstrate a strong understanding of the architecture and functioning of Generative AI models, including transformers and LLMs. - Be capable of applying prompt engineering techniques to steer model behavior for desired outputs across various tasks. - Design and fine-tune generative models for applications such as text generation, image creation, music synthesis, and conversational AI. - Analyze and evaluate the effectiveness of prompts and generated content, using relevant metrics and methodologies. - Apply ethical principles to ensure responsible development and deployment of generative AI systems.					
Syllabus				Total Hours:48	
Module-I	Introduction to Generative AI			9 Hrs	
Overview of AI and types of AI, What is Generative AI? Definitions and Concepts, Historical evolution of generative models, Types of generative models – GANs, VAEs, Autoregressive Models, Introduction to Transformers and LLMs, Applications and use cases of Generative AI, Challenges in Generative AI development, Introduction to text-to-image and image-to-text models.					
Module-II	Fundamentals of Prompt Engineering			10 Hrs	
Definition and significance of Prompt Engineering, Types of prompts: Zero-shot, One-shot, Few-shot, Techniques for effective prompt design, Prompt templates and chaining, Prompt tuning and parameter-efficient tuning, Evaluating prompt performance, Use of APIs for testing prompts (OpenAI, Cohere, Anthropic), Best practices and prompt libraries.					
Module-III	Working with Large Language Models (LLMs)			10 Hrs	
Overview of pre-trained LLMs: GPT, BERT, LLaMA, Claude, PaLM, Architectures and tokenization strategies, Fine-tuning vs. in-context learning, LLM-powered tools (ChatGPT, GitHub Copilot, Bard, Claude), Role of attention mechanism and transformer layers, Tools for model experimentation. (Hugging Face, LangChain), Performance metrics for LLMs, Case studies of model adaptation and deployment.					

Module-IV	Applications of Generative AI	10 Hrs
Text generation and summarization, Image and art generation (DALL·E, Midjourney, Stable Diffusion), Code generation and completion tools (Codex, Copilot), Music and video generation, Generative chatbots and customer service, Story generation and dialogue systems, Domain-specific applications (Legal, Healthcare, Education), Comparative study of generative models by task.		
Module-V	Ethics, Security & Responsible AI	9 Hrs
Bias and fairness in LLMs and generative systems, Explainability and transparency in generative AI, Copyright and originality issues, Adversarial use of generative models – deepfakes, misinformation, AI safety protocols and red-teaming, Regulatory and policy frameworks for generative AI, Responsible prompt crafting and moderation.		
Text Books: 1. Deep Learning with Python by François Chollet, Manning Publications 2. Transformers for Natural Language Processing by Denis Rothman, Packt 3. Practical Generative AI by AmitShukla, BPB Publications		
Reference Books: 1. Generative Deep Learning by David Foster, O'Reilly 2. Artificial Intelligence: A Guide for Thinking Humans by Melanie Mitchell 3. Machine Learning B.Techning by Andrew Ng (available online)		
Online Courses 1. Generative AI with Large Language Models – Coursera (AWS & DeepLearning.AI) 2. Prompt Engineering for ChatGPT – Coursera 3. Generative AI Fundamentals – Google Cloud Training 4. Hugging Face – Transformers Course 5. DeepLearning.AI Short Courses		



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Business Ethics and Corporate Governance Management Course - II (Common to CSE,AI&ML,CS,DS)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0049T	2:0:0:2	2	CIE:30 SEE:70	3 Hours	MEC
Course Objectives:					
This course will enable students: - To make the student understand the principles of business ethics - To enable them in knowing about the ethics in management - To facilitate the student‘ role in corporate culture - To impart knowledge about the fair-trade practices - To encourage the student in knowing about the corporate governance					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Understand the Ethics and different types of Ethics. - Understand business ethics and ethical practices in management - Understand the role of ethics in management - Apply the knowledge of professional ethics & technical ethics - Analyze corporate law, ethics, codes & principles - Evaluate corporate governance & corporate scams					
Syllabus				Total Hours:48	
Module-I	Ethics			10 Hrs	
Introduction – Meaning – Nature, Scope, significance, Loyalty, and ethical behavior. Value systems - Business Ethics - Types, Characteristics, Factors, Contradictions and Ethical Practices in Management -Corporate Social Responsibility – Issues of Management – Crisis Management.					
LEARNING OUTCOMES:-					
After completion of this unit student will - Understand the meaning of loyalty and ethical Behavior - Explain various types of Ethics - Analyze issues & crisis of management					
Module-II	Ethics in Management			9 Hrs	
Introduction- Ethics in production, finance, Human resource management and Marketing Management - The Ethical Value System – Universalism, Utilitarianism, Distributive Justice, Social Contracts, Individual Freedom of Choice, Professional Codes; Culture and Ethics –Ethical Values in different Cultures - Culture and Individual Ethics – professional ethics and technical ethics.					
LEARNING OUTCOMES:-					
After completion of this unit student will - Understand the meaning of Ethics in various areas of management - Compare and contrast professional ethics and technical ethics					

Develop ethical values in self and organization		
Module-III	Corporate Culture	10 Hrs
Introduction - Meaning, definition, Nature, and significance – Key elements of corporate culture, shared values, beliefs and norms, rituals, symbols and language - Types of corporate culture, hierarchical culture, market driven culture – Organization leadership and corporate culture, leadership styles and their impact on culture, transformational leadership and culture change. LEARNING OUTCOMES:- After completion of this unit student will - Define corporate culture - Understand the key elements of corporate culture - Analyze organization leadership and corporate culture		
Module-IV	Legal Frame Work	9 Hrs
Law and Ethics -Agencies enforcing Ethical Business Behavior - Legal Impact – Environmental Protection, Fair Trade Practices, legal Compliances, Safeguarding Health and wellbeing of Customers – Corporate law, Securities and financial regulations, corporate governance codes and principles. LEARNING OUTCOMES:- After completion of this unit student will - Understand Law and Ethics - Analyze Different fair trade practices - Make use of Environmental Protection and Fair Trade Practices		
Module-V	Corporate Governance	10 Hrs
Introduction - Meaning – Corporate governance code, transparency & disclosure -Role of auditors, board of directors and shareholders. Global issues, accounting and regulatory frame work - Corporate scams - Committees in India and abroad, corporate social responsibility. BoDs composition, Cadbury Committee - Various committees - Reports - Benefits and Limitations. LEARNING OUTCOMES:- After completion of this unit student will - Understand corporate governance code - Analyze role of auditors, board of directors and shareholders in corporate governance - Implementing corporate social responsibility in India.		
Text Books: - Murthy CSV: Business Ethics and Corporate Governance, HPH July 2017 - Bholananth Dutta, S.K. Podder – Corporation Governance, VBH. June 2010		
Reference Books: - Dr. K. Nirmala, KarunakaraReaddy. Business Ethics and Corporate Governance, HPH - H.R.Machiraju: Corporate Governance, HPH, 2013 K. Venkataramana, Corporate Governance, SHBP. 4. N.M.Khandelwal. Indian Ethos and Values for Managers		

Web References:

1. https://onlinecourses.nptel.ac.in/noc21_mg46/
2. <https://archive.nptel.ac.in/courses/110/105/110105138/>
3. https://onlinecourses.nptel.ac.in/noc21_mg54/
4. https://onlinecourses.nptel.ac.in/noc22_mg54/
5. <https://archive.nptel.ac.in/courses/109/106/109106117/>



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E-Business (Management Course – II) (Common to CSE,AI&ML,CS,DS)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0050T	2:0:0:2	2	CIE:30 SEE:70	3 Hours	MEC
Course Objectives:					
This course will enable students: - To provide knowledge on emerging concept on E-Business related aspect. - To understand various electronic markets & business models. - To impart the information about electronic payment systems & banking. - To create awareness on security risks and challenges in E-commerce. - To the students aware on different e-marketing channels & strategies.					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Remember E-Business & its nature, scope and functions. - Understand E-market-Models which are practicing by the organizations - Apply the concepts of E-Commerce in the present globalized world. - Analyze the various E-payment systems & importance of net banking. - Evaluate market research strategies & E-advertisements. - Understand importance of E-security & control					
Syllabus				TotalHours:48	
Module-I	Electronic Business			10 Hrs	
Introduction – Nature, meaning, significance, functions and advantages - Definition of Electronic Business - Functions of Electronic Commerce (EC)-Advantages & Disadvantages of E-Commerce – E-Commerce and E-Business, Internet Services, Online Shopping- E-Commerce Opportunities for Industries..					
Learning Outcomes: -After completion of this unit student - Understand the concept of E-Business - Contrast and compare E-Commerce & E-Business - Evaluate opportunities of E-commerce for industry					
Module-II	Electronic Markets and Business Models			9 Hrs	
Introduction –E-Shops-E-Malls E-Groceries - Portals - Vertical Portals-Horizontal Portals - Advantages of Portals -Business Models- Business to Business (B2B)-Business to Customers(B2C) - Business to Government(B2G)-Auctions-B2B Portals in India					
Learning Outcomes: -After completion of this unit student will - Understand the concept of business models - Contrast and compare Vertical portal and Horizontal portals - Analyze the B2B,B2C and B2G model					
Module-III	Electronic Payment Systems			10 Hrs	

Introduction to electronic payment systems (EPS) -Types of electronic payments - Credit/debit cards, e-wallets, UPI, and crypto currencies -Smart cards and digital wallets: Features and usage -Electronic Fund Transfer (EFT): Role in business transactions -Infrastructure requirements and regulatory aspects of e-payments. Learning Outcomes: -After completion of this unit student will • Understand the Electronic payment system • Contrast and compare EFT and smart cards • Analyze debit card and credit cards		
Module-IV	E-Security	9 Hrs
Security risks and challenges in electronic commerce - Cyber threats - Phishing, hacking, identity theft, and malware - Digital Signatures & Certificates - Security protocols over public networks (HTTP, SSL, TLS) -Firewalls in securing e-business platforms. Learning Outcomes: -After completion of this unit student will • Understand E-Security • Contrast and compare security protocols and public network • Evaluate on Digital signature		
Module-V	E-Marketing	10 Hrs
Introduction – Online Marketing – Advantages of Online Marketing – Internet Advertisement – Advertisement Methods – Conducting Online Market Research– – E-marketing planning: Online branding, social media marketing, and email marketing - E-business strategies: Digital advertising, content marketing, and analytics – E-Customer Relationship Management (eCRM) E-supply chain management (e-SCM) Learning Outcomes: -After completion of this unit student will • Understand the concept of online marketing • Apply the knowledge of online marketing • Compare e-CRM and e-SCM		
Text Books: 1. AratiOturkar & SunilKhilari. E-Business. Everest Publishing House, 2022 2. P.T.S Joseph. E-Commerce, Fourth Edition, Prentice Hall of India, 2011.		
Reference Books: 1. Debjani, Kamalesh K Bajaj. E-Commerce, Second Edition Tata McGraw-Hill's, 2005 2. Dave Chaffey.E-Commerce E-Management, Second Edition, Pearson, 2012. 3. Henry Chan. E-Commerce Fundamentals and Application, RaymondLeathamWiley India 2007 4. S. Jaiswal. E-Commerce GalgotiaPublicationPvt Ltd., 2003		
Web References: 1. https://www.slideshare.net/fatimahAlkreem/e-businessppt-67935771 2. https://www.slideshare.net/VikramNani/e-commerce-business-models 3. https://www.slideshare.net/RiteshGoyal/electronic-payment-system 4. https://www.slideshare.net/WelingkarDLP/electronic-security 5. https://www.slideshare.net/Ankitha2404/emarketing-ppt		



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Management Science (Management Course – II) (Common to CSE,AI&ML,CS,DS)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0048T	2:0:0:2	2	CIE:30 SEE:70	3Hours	MEC
Course Objectives:					
The aim of this course is to: - To provide fundamental knowledge on Management, Administration, Organization & its concepts. - To make the students understand the role of management in Production - To impart the concept of HRM in order to have an idea on Recruitment, Selection, Training & Development, job evaluation and Merit rating concepts - To create awareness on identify Strategic Management areas & the PERT/CPM for better Project Management - To make the students aware of the contemporary issues in modern management					
Course Outcomes (CO):					
On completion of this course, student will be able to - Remember the concepts & principles of management and designs of organization in a practical world - Understand the knowledge of Work-study principles & Quality Control techniques in industry - Apply the process of Recruitment & Selection in organization. - Analyze the concepts of HRM & different training methods. - Evaluate PERT/CPM Techniques for projects of an enterprise and estimate time & cost of project & to analyze the business through SWOT. - Create awareness on contemporary issues in modern management &technology.					
Syllabus					Total Hours:48
Module-I	Introduction to Management				9 Hrs
Management - Concept and meaning - Nature-Functions - Management as a Science and Art and both. Schools of Management Thought - Taylor’s Scientific Theory-Henry Fayol’s principles - Elton Mayo’s Human relations - Organizational Designs - Line organization - Line & Staff Organization - Functional Organization - Matrix Organization - Project Organization - Committee form of Organization - Social responsibilities of Management.					
Module-II	Operations Management				10 Hrs
Principles and Types of Plant Layout - Methods of Production (Job, batch and Mass Production), Work Study - Statistical Quality Control- Material Management - Objectives - Inventory-Functions - Types, Inventory Techniques - EOQ-ABC Analysis - Marketing Management - Concept -Meaning - Nature-Functions of Marketing - Marketing Mix - Channels of Distribution -Advertisement and Sales Promotion - Marketing Strategies based on Product Life Cycle.					
Module-III	Human Resources Management (HRM)				10 Hrs
HRM - Definition and Meaning – Nature - Managerial and Operative functions - Job Analysis - Human Resource Planning(HRP) - Employee Recruitment-Sources of Recruitment - Employee Selection - Process - Employee Training and Development - methods - Performance Appraisal Concept - Methods of Performance Appraisal – Placement - Employee Induction - Wage and Salary Administration.					

Module-IV	Strategic & Project Management	10 Hrs
Definition& Meaning - Setting of Vision - Mission - Goals - Corporate Planning Process - Environmental Scanning - Steps in Strategy Formulation and Implementation - SWOT Analysis - Project Management - Network Analysis - Programme Evaluation and Review Technique (PERT) - Critical Path Method (CPM) Identifying Critical Path - Probability of Completing the project within given time - Project Cost- Analysis - Project Crashing (Simple problems).		
Module-V	Contemporary Issues in Management	9 Hrs
Customer Relations Management(CRM) - Total Quality Management (TQM) - Six Sigma Concept - Supply Chain Management(SCM) - Enterprise Resource Planning (ERP) - Performance Management – employee engagement and retention - Business Process Re-engineering and Bench Marking - Knowledge Management – change management –sustainability and corporate social responsibility.		
Text Books: 1. Frederick S. Hillier, Mark S. Hillier. Introduction to Management Science, October 26, 2023. 2. A.R Aryasri, Management Science, TMH, 2019.		
Reference Books: 1. Stoner, Freeman, Gilbert. Management, Pearson Education, New Delhi, 2019. 2. Koontz & Weihrich, Essentials of Management, 6/e, TMH, 2005. 3. Thomas N. Duening & John M. Ivancevich, Management Principles and Guidelines, Biztantra. 4. Kanishka Bedi, Production and Operations Management, Oxford University Press, 2004. 5. Samuel C. Certo, Modern Management, 9/e, PHI, 2005.		
Web References: 1. https://www.slideshare.net/slideshow/introduction-to-management-and-organization-231308043/231308043 2. https://nptel.ac.in/courses/112107238 3. https://archive.nptel.ac.in/courses/110/104/110104068/ 4. https://archive.nptel.ac.in/courses/110/105/110105069/ 5. https://onlinecourses.nptel.ac.in/noc24_mg112/		



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Explainable AI & Model Interpretability (Professional Elective – IV)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3320Ta	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students: - To introduce the principles of interpretability and explainability in AI/ML models. - To analyze the trade-offs between model accuracy and interpretability. - To explore popular post-hoc and intrinsic explainability techniques. - To examine fairness, accountability, and transparency in AI systems. - To develop hands-on skills with interpretability tools and libraries.					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Understand the need for explainability in modern AI systems. - Differentiate between black-box and white-box models. - Apply interpretability techniques such as SHAP, LIME, and PDPs. - Evaluate the fairness and transparency of AI systems. - Use explainability tools for model auditing and deployment in high-stakes domains.					
Syllabus				TotalHours:48	
Module-I	Foundations of Explainable AI			10 Hrs	
Introduction to Explainability and Interpretability, Importance of XAI in Healthcare, Finance, and Law , White-box vs Black-box Models, Desiderata: Fairness, Accountability, Transparency, Human Centered AI and Trust ,Taxonomy of XAI Techniques (Global vs Local, Post-hoc vs Intrinsic), Regulatory and Ethical Implications (GDPR, AI Bill of Rights), Model Simplicity vs Predictive Power.					
Module-II	Model-Specific Explainability Techniques			9 Hrs	
Decision Trees and Rule-based Models, Linear Models and Feature Importance, Generalized Additive Models (GAMs), Visualization of Weights and Coefficients, Logistic Regression Coefficient Interpretation, Case Study: Credit Scoring using Transparent Models,Comparison of Interpretable ML Models, Use Cases and Trade-offs.					
Module-III	Model-Agnostic Explainability Techniques			10 Hrs	
Local Interpretable Model-agnostic Explanations (LIME), SHAP Values (SHapley Additive explanations), Partial Dependence Plots (PDPs), Individual Conditional Expectation (ICE) Plots,Anchors and Counterfactual Explanations, Feature Interaction and Permutation Importance, Comparative Analysis of SHAP, LIME, PDP, Model Debugging with XAI.					
Module-IV	Deep Learning Explainability			9 Hrs	
Visualizing CNNs: Filters, Feature Maps, Saliency Maps and Grad-CAM, Integrated Gradients, Explaining RNNs and LSTM Outputs, Concept Activation Vectors (TCAV), Attention-based Interpretability in Transformers, Explaining Language Models (BERT, GPT) Evaluation of Deep Model Explanations.					

Module-V	Fairness, Bias & Tools for XAI	10 Hrs
Fairness Metrics: Demographic Parity, Equal Opportunity, Sources of Bias in Data and Models, Discrimination Detection and Mitigation Strategies, Introduction to AIF360, What-If Tool, Fairlearn, Case Study: Bias in Hiring Algorithms, Explainability in ML Pipelines (MLFlow, Skater), XAI in Federated and Privacy-Preserving AI, Designing Interpretable AI Systems from Scratch.		
Text Books: 1. Christoph Molnar, —Interpretable Machine Learning, Leanpub. 2. Sameer Singh et al., —Explainable AI: Interpreting, Explaining and Visualizing Deep Learning, Springer. 3. Dan Roth, Zachary Lipton, and Been Kim, —Explainable AI: Foundations, Developments, Prospects, MIT Press (Online forthcoming).		
Reference Books: 1. Marco Tulio Ribeiro et al., —Why Should I Trust You? (LIME) – Research Paper 2. Scott Lundberg et al., —A Unified Approach to Interpreting Model Predictions (SHAP) – NIPS 3. A. Barredo Arrieta et al., —Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges, Information Fusion Journal. 4. Zachary C. Lipton, —The Mythos of Model Interpretability – Communications of the ACM		
Online Courses 1. Coursera – Explainable AI with Google Cloud 2. Udacity – AI for Everyone by Andrew Ng 3. HarvardX – Data Science: Machine Learning Interpretability 4. fast.ai – Practical Deep Learning Courses 5. Interpretable ML Book (https://christophm.github.io/interpretable-ml-book/) 6. Google PAIR: People + AI Research 7. Microsoft Fairlearn Documentation		



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AI in Cyber security (Professional Elective-IV)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3320Tb	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students: - To introduce the fundamental concepts of AI and their applications in cyber security. - To understand AI-driven techniques for threat detection, classification, and mitigation. - To explore machine learning and deep learning methods used for malware and intrusion detection. - To equip students with skills in building intelligent security systems. - To examine ethical, legal, and privacy aspects in AI-driven cyber security.					
Course Outcomes (CO):					
On completion of this course, students will be able to: - Understand AI principles and their relevance in cyber security. - Apply machine learning techniques to detect and respond to threats. - Analyze security incidents using intelligent tools and models. - Evaluate and implement AI models for malware detection and anomaly analysis. - Design AI-based cyber security frameworks for real-world scenarios.					
Syllabus				TotalHours:48	
Module-I	Introduction to AI in Cyber security			10 Hrs	
Role of AI in Modern Cyber security, Overview of Cyber Threats and Attack Vectors, Fundamentals of Machine Learning for Security, AI vs Traditional Security Techniques, AI-Based Cyber Defense Lifecycle, Threat Intelligence with AI, Cyber security Data Types and Challenges, Case Studies of AI- Driven Attacks and Defenses.					
Module-II	Machine Learning for Cyber Threat Detection			9 Hrs	
Supervised Learning for Intrusion Detection, Unsupervised Learning for Anomaly Detection, Feature Engineering from Network Traffic, Classification Algorithms: SVM, Decision Trees, Random Forests, Clustering Techniques: K-Means, DBSCAN, Ensemble Models and Model Evaluation Metrics, Real-Time Threat Detection Pipelines, Data Imbalance and Adversarial Sampling.					
Module-III	Deep Learning in Cyber security			10 Hrs	
Neural Networks for Threat Classification, CNNs for Malware Detection from Binary Files, RNNs/LSTMs for Sequential Log Analysis, Auto encoders for Anomaly Detection, GANs in Malware Evasion and Defense, Transfer Learning for Threat Signature Extraction, Deep Learning vs Traditional Models: A Comparative Study, Real-World Use Cases and Limitations.					

Module-IV	AI for Specific Security Domains	9 Hrs
AI for Phishing and Spam Detection, AI in Cloud Security and Edge Devices, Botnet and DDoS Attack Detection, AI-Driven Endpoint Security, Natural Language Processing for Threat Intelligence, Behavioral Biometrics and Fraud Detection, AI in Social Engineering Attack Prevention, Security Information and Event Management (SIEM) with AI.		
Module-V	Challenges, Ethics & Future of AI in Cyber security	10 Hrs
Explainable AI (XAI) in Cyber security, Adversarial Attacks and Defenses in AI Systems, Data Privacy and Federated Learning, Legal and Ethical Issues in AI Security Solutions, AI Model Bias and Fairness in Security Decisions, Securing AI Models Against Manipulation, Building Scalable AI- Powered SOC's, Future Trends: Autonomous Security, AI-Augmented Threat Hunting.		
Text Books: 1. Clarence Chio & David Freeman, —Machine Learning and Security, O'Reilly Media. 2. Xiaofeng Chen et al., —Artificial Intelligence and Big Data Analytics for Cybersecurity, Springer. 3. Mark Stamp, —Information Security: Principles and Practice, Wiley.		
Reference Books: 1. Sumeet Dua & Xian Du, —Data Mining and Machine Learning in Cybersecurity, CRC Press. 2. Shai Shalev-Shwartz & Shai Ben-David, —Understanding Machine Learning, Cambridge University Press. 3. Zhiwei Lin & Yang Xiang, —Cyber Security Intelligence and Analytics, Springer. 4. Bhavani Thuraisingham, —Data Mining for Malware Detection, CRC Press.		
Online Courses: 1. Coursera – —AI for Cybersecurity by University of Colorado 2. Udemy – —Machine Learning for Cybersecurity		



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AI-Driven Software Engineering & DevOps (Professional Elective-IV)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3320Tc	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students to: • To introduce the principles and practices of AI-driven software engineering and DevOps. • To explore how AI techniques can automate and optimize software development workflows. • To study intelligent tools for code generation, testing, monitoring, and deployment. • To equip students with skills in AI-powered CI/CD pipelines and operations. • To foster an understanding of ethical implications and reliability in intelligent software systems.					
Course Outcomes (CO):					
After completion of the course, students will be able to: • Understand AI's role in modern software development and operations. • Apply machine learning techniques to automate software engineering tasks. • Design and manage intelligent CI/CD and DevOps workflows. • Evaluate AI tools in software testing, refactoring, and monitoring. • Implement AI-based solutions for predictive maintenance and decision support in DevOps					
Syllabus					Total Hours:48
Module-I	Foundations of AI in Software Engineering				10 Hrs
Overview of Traditional vs AI-driven Software Development, AI Opportunities in Software Lifecycle Phases, Introduction to ML/DL Models in Engineering Tasks, Code Representation and Learning from Code, NLP for Source Code Understanding, Software Knowledge Graphs and Reasoning, Datasets and Benchmarks for Software Engineering AI, Case Studies of AI-Enhanced Development Tools.					
Module-II	AI in Code Generation and Refactoring				9 Hrs
Program Synthesis and Code Completion Models, Large Language Models (e.g., Codex, Code BERT) in IDEs, Code Clone Detection and Automated Refactoring, Learning-Based Bug Detection and Code Smell Identification, AI in Software Architecture Recommendations, Embedding Techniques for Source Code, Prompt Engineering for Software Tasks, Reliability and Safety in Generated Code.					
Module-III	Intelligent Testing, QA, and Debugging				9 Hrs
Test Case Generation Using AI, Automated Unit Testing, Regression Testing with ML, Learning Bug Patterns from Repositories, AI-Based Static and Dynamic Code Analysis, Fault Localization and Automated Debugging, Quality Assurance Metrics Enhanced by AI, Reinforcement Learning for Test Prioritization, Ethics and Bias in AI-Driven Testing – (E).					
Module-IV	AI in DevOps Automation and CI/CD				10 Hrs
DevOps Fundamentals and Integration with AI, Intelligent CI/CD Pipeline Design, Predictive Build Failure and Log Analysis, AI in Infrastructure-as-Code and Deployment Orchestration, Self-Healing Systems and AIOps Concepts, Log Analytics and Anomaly Detection in Production, AI in Monitoring, Tracing, and Feedback Loops, DevSecOps and AI for Security Automation.					

Module-V	Advanced Topics and Ethical Considerations	10 Hrs
Explain ability and Transparency in AI-Driven Tools, Ethical and Legal Aspects in Automated Engineering, Human-AI Collaboration in Software Teams, Risk Management in Autonomous Code Deployment, AI for Technical Debt Prediction and Management, AI for Developer Productivity Analytics, Research Trends and Challenges in AI for SE, Capstone: Building a Smart DevOps Workflow		
Textbooks: 1. Tim Menzies, Diomidis Spinellis, and Thomas Zimmermann, —Perspectives on Data Science for Software Engineering, Morgan Kaufmann. 2. André van der Hoek, Reid Holmes, —Software Engineering for Machine Learning, Springer. 3. Len Bass, Ingo Weber, Liming Zhu, —DevOps: A Software Architect's Perspective, Addison-Wesley.		
Reference Books: 1. Carlos Eduardo Parnin et al., —AI for Software Engineering: Foundations, Advances, and Trends, Springer. 2. Luciano Baresi et al., —Machine Learning Techniques for Software Quality Evaluation, Springer. 3. Gene Kim, Jez Humble, and Nicole Forsgren, —Accelerate: The Science of Lean Software and DevOps, IT Revolution.		
Online Courses: 1. Coursera – —AI for Software Engineering by DeepLearning.AI 2. edX – —DevOps for Developers by Microsoft 3. GitHub Copilot and OpenAI Codex documentation 4. PapersWithCode – AI for Software Engineering benchmarks 5. MIT OCW – —Software Systems and —DevOps and CI/CD 6. Udemy – —AI-Powered DevOps Pipelines and Automation 7. Google Cloud – AIOps and MLOps tutorials		



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AI for Robotics (Professional Elective-IV)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3320Td	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students to:					
- To provide foundational knowledge in Generative AI and its core architectures. - To explore applications of generative models in text, image, and audio domains. - To equip students with hands-on experience in training and fine-tuning generative models. - To teach techniques for deploying AI models efficiently and securely in real-world settings. - To analyze the ethical, interpretability, and security concerns surrounding generative models.					
Course Outcomes (CO):					
After completion of the course, students will be able to:					
- Understand and compare different generative architectures like GANs, VAEs, Transformers. - Build, fine-tune, and evaluate generative models for text, vision, and multimodal data. - Design secure and scalable workflows for model deployment (on cloud and edge). - Integrate monitoring, optimization, and feedback for deployed AI services. - Address ethical, interpretability, and regulatory aspects of model deployment.					
Syllabus				Total Hours:48	
Module-I	Foundations of Generative AI			10 Hrs	
Introduction to Generative AI: Concepts, Use Cases, Types of Generative Models: GANs, VAEs, Diffusion, Autoregressive, Mathematical Foundations: Probabilistic Modeling & Latent Spaces, Generative Loss Functions: Adversarial Loss, KL Divergence, ELBO, Model Training Challenges: Mode Collapse, Posterior Collapse, Tools & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, Evaluation Metrics: FID, BLEU, Perplexity, Case Studies of Generative AI Systems					
Module-II	Generative Modeling for Text, Images, and Audio			9 Hrs	
Transformer Models: GPT, T5, BERT for Text Generation, Vision Models: StyleGAN, DALL·E, Stable Diffusion, Audio Generation: WaveNet, Jukebox, Voice Cloning, Multi-modal Generation: CLIP, Flamingo, BLIP, Prompt Engineering & Controlled Generation, Fine-Tuning &LoRA Techniques, Evaluation Techniques for Generated Media, Comparison of Open vs Proprietary Foundation Models					
Module-III	Model Serving and Deployment Essentials			10 Hrs	
Overview of AI Deployment Pipelines, Model Packaging: ONNX, TorchScript, SavedModel, REST &gRPC API Serving using FastAPI, Flask, Triton Inference, Batch vs Real-time Inference, Asynchronous Processing, Containerization & Orchestration with Docker and Kubernetes, Cloud Deployment: AWS SageMaker, GCP Vertex AI, Azure ML, Cost Optimization and Resource Management, Edge AI: TinyML, Mobile Deployment using TensorFlow Lite, CoreML					
Module-IV	MLOps and Model Lifecycle Management			9 Hrs	
MLOps Lifecycle: Versioning, Experiment Tracking, CI/CD for ML: GitHub Actions, Jenkins, MLflow, Model Monitoring & Drift Detection, Logging and Metrics using Prometheus, Grafana,Auto-scaling and Load Balancing of Inference Services, Continuous Training and Feedback Loops, Model Governance and Audit Trails, Role of Explainability in Deployment					
Module-V	Challenges and Responsible Deployment			10 Hrs	

Interpretability of Generative Models, Security and Adversarial Attacks on Generative Models, Bias, Fairness, and Harmful Content Generation, Privacy-Preserving Techniques: Differential Privacy, Federated Learning, Ethical and Legal Frameworks for Generative AI, Open Source vs Proprietary Models: Deployment Implications, Responsible AI Guidelines & Compliance (GDPR, EU AI Act), Capstone: End-to-End Generative Model Deployment Project.

Textbooks:

1. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, —Deep Learning, MIT Press.
2. Aurélien Géron, —Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, O'Reilly Media.
3. O'Reilly, —Generative Deep Learning by David Foster.

Reference Books:

1. Packt Publishing, "Practical Deep Learning" by Ronald T. Kneusel.
2. "Transformers for Natural Language Processing" by Denis Rothman.
3. "Machine Learning Engineering" by Andriy Burkov.

Online Courses:

1. Coursera – "Generative AI with Large Language Models" by DeepLearning.AI
2. Hugging Face Course – <https://huggingface.co/learn>
3. Fast.ai – "Practical Deep Learning for Coders"
4. YouTube – Full Stack Deep Learning by Berkeley AI
5. AWS Machine Learning University – Model Deployment
6. Towards Data Science and Papers With Code – Latest research & code repositories



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MLOps & AI Model Deployment (Professional Elective-V)

CourseCode	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3321Ta	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC

Course Objectives:

This course will enable students to:

- To understand the principles and best practices of operationalizing machine learning models in production environments.
- To explore the life cycle of AI model development, deployment, monitoring, and maintenance using modern MLOps frameworks.
- To develop skills in CI/CD for ML, reproducibility, model versioning, and containerization using Docker and Kubernetes.
- To deploy machine learning models using cloud-native services and track their performance using real-time metrics.
- To address scalability, reliability, and ethical considerations in ML model deployment.

Course Outcomes (CO):

After completion of the course, students will be able to:

- Illustrate the lifecycle and pipeline components of MLOps and implement basic version control and orchestration for ML workflows.
- Package ML models using appropriate tools and deploy them using Docker and Kubernetes environments with effective resource management.
- Develop and deploy machine learning models as APIs using FastAPI/Flask and configure for real-time or batch inference scenarios.
- Monitor and log ML systems using modern tools and detect data/model drift with strategies for continuous evaluation and feedback.
- Implement end-to-end MLOps solutions using cloud platforms and CI/CD tools, and analyze deployment challenges in real-world use cases.

Syllabus		TotalHours:48
Module-I	Introduction to MLOps and Deployment Pipelines	10 Hrs
Definition and need of MLOps, ML system lifecycle and pipeline components, DevOps vs. MLOps: key differences, CI/CD for ML projects, Data versioning and model lineage, Introduction to DVC, Git, and MLFlow, Workflow orchestration using Apache Airflow, Automated testing in ML pipelines		
Module-II	Model Packaging and Environment Management	9 Hrs
Packaging ML models using Pickle, Joblib, ONNX, Python virtual environments, Conda, Pipenv, Introduction to Docker for ML workloads, Building Dockerfiles for ML apps, Using Kubernetes for orchestration, Security, logging, and resource management, Docker Compose and Helm charts for deployment, Hands-on: Containerize and deploy a scikit-learn model		
Module-III	Model Serving and APIs	10 Hrs

RESTful API design for ML models, Model deployment using FastAPI and Flask, TensorFlow Serving, TorchServe basics, Introduction to gRPC for ML deployment, Asynchronous inference and batch vs real-time serving, Load testing and benchmarking, Authentication and authorization in model APIs, Deploying models on edge devices.		
Module-IV	Monitoring, Logging, and Continuous Evaluation	9 Hrs
Importance of monitoring and alerting in MLOps, Data drift and model drift detection, Logging prediction results and metadata, Prometheus, Grafana, and ELK Stack, A/B testing and canary deployments, Shadow deployments and rollback strategies, Feedback loops for continuous learning, Integration with external monitoring tools.		
Module-V	Cloud-native MLOps and Case Studies	10 Hrs
ML deployment on AWS SageMaker, Azure ML, Google AI Platform, CI/CD using GitHub Actions, Jenkins, and GitLab CI, AutoML and model registry, Real-world case study: End-to-end MLOps pipeline, Challenges and limitations in enterprise ML deployment, Responsible AI in production systems, Future trends in MLOps, Capstone Project Planning.		
TextBooks: 1. Introducing MLOps: How to Scale Machine Learning Projects with DevOps Tools – Mark Treveil, Alok Shukla, and O’Reilly Media. 2. Machine Learning Engineering – Andriy Burkov, TrueShelf Publishing. 3. Designing Machine Learning Systems – Chip Huyen, O’Reilly Media		
ReferenceBooks: 1. Practical MLOps – Noah Gift, O’Reilly Media 2. Kubeflow for Machine Learning – Trevor Grant et al., O'Reilly 3. Hands-On MLOps: Implement Machine Learning in Production – Munn, Meza, Vohra, Packt Publishing 4. Research papers from arXiv, MLSys Conference, and ICML Industry Track		
Online Courses: 1. Coursera – MLOps Specialization by DeepLearning.AI 2. Google Cloud – MLOps: Continuous Delivery and Automation Pipelines 3. Udemy – MLOps: ML Pipelines, CI/CD, and Model Deployment 4. AWS – Machine Learning Engineering for Production (MLOps) 5. Microsoft Learn – MLOps with Azure ML		



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Data Wrangling (Professional Elective-V)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3321Tb	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students to:					
- To introduce the fundamental techniques for acquiring, cleaning, transforming, and manipulating data. - To enable students to handle real-world messy data for analysis and machine learning. - To teach efficient use of libraries like Pandas, NumPy, and SQL for data wrangling. - To promote understanding of handling missing values, outliers, and inconsistent formats. - To expose students to automation, reproducibility, and workflow design in data preprocessing.					
Course Outcomes (CO):					
On completion of this course, student will be able to:					
- Understand and apply core data wrangling techniques. - Clean, transform, and reshape data using Python and SQL. - Handle missing values, data inconsistencies, and outliers. - Merge and join multiple datasets from different sources. - Automate data pipelines and preprocessing workflows for analytics and ML.					
Syllabus				Total Hours:48	
Module-I	Introduction to Data Wrangling and Data Acquisition			10 Hrs	
Introduction to Data Wrangling: Importance and Use Cases , Types of Data: Structured, Semi Structured, Unstructured, Data Acquisition Techniques: APIs, Web Scraping, Reading Data from CSV, Excel, JSON, XML, Using Python libraries: pandas, requests, BeautifulSoup, Working with Databases using SQLAlchemy and pandas, Loading Large Datasets and Chunking, Exploratory Analysis Before Cleaning.					
Module-II	Handling Missing, Noisy, and Inconsistent Data			9 Hrs	
Identifying and Understanding Missing Data, Techniques for Imputing Missing Values, Handling Inconsistent Data: Dates, Texts, Units, Removing Duplicates and Irrelevant Data, Detecting and Treating Outliers, Normalization and Standardization Techniques, Regular Expressions for Text Cleaning, Visualizing Missing/Outlier Data.					
Module-III	Data Transformation and Feature Engineering			10 Hrs	
Data Type Conversion and Parsing, Feature Extraction from Text, Dates, and Strings, One-Hot Encoding, Label Encoding, Binning and Discretization, Data Aggregation and Grouping, Pivoting, Melting, and Reshaping Data, Handling Imbalanced Data, Creating Derived Features and Feature Selection.					
Module-IV	Data Integration, Joining, and Workflows			9 Hrs	

Merging and Joining Datasets (Inner, Outer, Left, Right), Concatenation and Appending DataFrames, Data Consistency and Referential Integrity, Resolving Schema Mismatches, Designing Reusable Data Wrangling Functions, Automating Workflows with Functions and Pipelines, Data Lineage and Documentation, Case Study: End-to-End Data Wrangling Pipeline.		
Module-V	Tools, Libraries, and Case Studies in Data Wrangling	10 Hrs
Pandas and NumPy Advanced Techniques, Pyjanitor, Dask, and Polars for Efficient Wrangling, Using OpenRefine for Data Cleaning, SQL vs NoSQL in Data Wrangling, Real-world Wrangling Case Studies (Finance, Healthcare, Retail), Best Practices and Common Pitfalls in Data Wrangling, Reproducibility and Versioning in Data Pipelines, Final Capstone: Build and Evaluate a Clean Dataset for ML.		
Text Books: 1. M. Heydt – Data Wrangling with pandas, O’Reilly Media. 2. Hadley Wickham – R for Data Science (Data Wrangling Chapters), O’Reilly. 3. J. VanderPlas – Python Data Science Handbook, O’Reilly Media.		
Reference Books: 1. Wes McKinney – Python for Data Analysis, O’Reilly. 2. Cathy O’Neil and Rachel Schutt – Doing Data Science, O’Reilly. 3. David Mertz – Cleaning Data for Effective Data Science, Packt.		
Web References: 1. Data Wrangling with pandas (Datacamp):https://www.datacamp.com/courses/datamanipulation-with-pandas 2. Coursera: Data Wrangling, Analysis and AB Testing with SQL: https://www.coursera.org/learn/data-wrangling-analysis-abtesting 3. edX: Data Wrangling with R: https://online.rice.edu/courses/data-wrangling-r 4. Real Python Tutorials on pandas: https://realpython.com/learning-paths/pandas/ 5. Kaggle Notebooks (Data Cleaning & Wrangling): https://www.kaggle.com/learn/pandas		



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Healthcare AI (Professional Elective-V)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3321Tc	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students to:					
• Introduce fundamental concepts and scalable algorithms used in mining massive datasets. • Enable understanding of key techniques like clustering, classification, frequent itemset mining, and graph analysis on large-scale data. • Familiarize students with distributed computing frameworks such as Hadoop and Spark. • Provide practical insights into web and social network mining. • Equip students with the ability to analyze massive datasets using real-world tools and platforms.					
Course Outcomes (CO):					
On completion of this course, the student will be able to:					
• Understand and explain the challenges involved in mining large-scale datasets. • Apply efficient algorithms for clustering, classification, and association rule mining in big data environments. • Analyze and implement scalable solutions using frameworks such as Map Reduce, Hadoop, and Spark. • Solve real-world problems involving link analysis, recommendation systems, and mining of web/social data. • Critically evaluate algorithms based on scalability, efficiency, and effectiveness in large datasets					
Syllabus					Total Hours:48
Module-I	Introduction to Massive Data and Map Reduce Model				10 Hrs
Types of Massive Data – Structured, Unstructured, and Semi-Structured, Challenges of Mining Massive Data Sets, Storage Systems – Distributed File Systems, HDFS, Introduction to Map Reduce Programming Model, Designing Map Reduce Algorithms, Matrix-Vector Multiplication by Map Reduce, Workflow Management in Hadoop, Limitations of Map Reduce.					
Module-II	Frequent Itemset and Association Rule Mining				9 Hrs
Market Basket Model, A-Priori Algorithm – Scalable Variants, Handling Large Datasets in Frequent Pattern Mining, Park-Chen-Yu Algorithm, SON Algorithm, Multistage and Multihash Algorithms, PCY Algorithm and its Enhancements, Association Rules – Concepts and Evaluation, Finding Frequent Itemsets in Streaming Data					
Module-III	Clustering and Classification Techniques				10 Hrs
Hierarchical and Partitional Clustering, K-Means Clustering and its Scalability, BFR and CURE Clustering Algorithms, Decision Trees and Rule-Based Classification, Naïve Bayes Classifier for Large Datasets, Logistic Regression and SVM for Massive Data, Parallel Clustering Techniques, Evaluation of Clustering Results					
Module-IV	Link Analysis and Mining of Web/Social Networks				9 Hrs
Web Graph Structure and Crawling, PageRank and its Variants, Hubs and Authorities (HITS Algorithm), Link Spam Detection, Community Detection in Large Graphs, Mining Social Network Graphs, Recommendation Systems – User-Based and Item-Based Collaborative Filtering, Content-Based Filtering.					

Module-V	Frameworks and Real-World Applications	10 Hrs
Introduction to Apache Spark and RDDs, Spark MLlib for Data Mining, Streaming and Real-Time Data Analysis, Mining on Cloud Platforms (AWS, GCP, Azure), Case Study: E-commerce, Finance, and Healthcare, Scaling Algorithms to Petabyte-Level Data, Big Data Ethics and Governance, Research Trends in Mining Massive Data Sets		
Textbooks: 1. "Mining of Massive Datasets" by Jure Leskovec, Anand Rajaraman, and Jeffrey Ullman 2. "Data Mining: Concepts and Techniques" by Jiawei Han, Micheline Kamber, and Jian Pei 3. "Big Data: Principles and Best Practices of Scalable Real-Time Data Systems" by Nathan Marz and James Warren		
Reference Books: 1. "Big Data Analytics with Spark" by Mohammed Guller 2. "Hadoop: The Definitive Guide" by Tom White 3. "Practical Machine Learning with Spark" by Ajay Ohri 4. IEEE/ACM Journals and Conference Proceedings on Data Mining and Big Data		
Online Courses: 1. Mining Massive Datasets – Stanford University (Coursera) 2. Big Data Analysis with Apache Spark – edX (BerkeleyX) 3. Data Mining Specialization – University of Illinois (Coursera)		



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AI for Smart Cities & IoT Systems (Professional Elective-V)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A3321Td	3:0:0:3	3	CIE:30 SEE:70	3 Hours	PEC
Course Objectives:					
This course will enable students to:					
- To introduce students to the integration of Artificial Intelligence and Internet of Things (IoT) technologies for developing smart city solutions. - To understand the design, development, and deployment of intelligent systems to enhance urban infrastructure, transport, healthcare, energy, and governance. - To explore edge and cloud computing techniques to optimize real-time AI-based decisions for IoT applications. - To enable students to apply data analytics, computer vision, NLP, and automation to solve real-world urban challenges. - To foster innovation using ethical AI frameworks in the context of sustainability, privacy, and smart governance.					
Course Outcomes (CO):					
On completion of this course, student will be able to					
- Understand the architecture and components of smart cities powered by AI and IoT. - Analyze and design AI-driven solutions for transportation, energy, healthcare, waste management, and smart governance. - Deploy IoT systems that integrate sensors, edge devices, and AI models. - Utilize AI algorithms (machine learning, NLP, and computer vision) for real-time smart city use cases. - Evaluate and implement data-driven smart systems ensuring privacy, efficiency, and sustainability. - Leverage cloud platforms and edge computing for scalable AIoT applications in urban environments.					
Syllabus				Total Hours:48	
Module-I	Introduction to AI in Smart Cities and IoT Systems			10 Hrs	
Smart City Concepts: Components, Infrastructure, and Urban Needs, Overview of IoT and AI Integration, Smart City Frameworks (India, Singapore, EU, etc.), IoT Architecture: Sensing, Network, Processing, and Application Layers, Role of AI in Urban Planning and Resource Optimization, Case Studies on AI in Smart Cities, Edge, Fog, and Cloud Computing Concepts for Smart Systems.					
Module-II	AI Applications in Smart Transportation and Mobility			9 Hrs	
Traffic Monitoring and Congestion Prediction using AI, Intelligent Traffic Signal Control using Reinforcement Learning, Autonomous Vehicles and AI Algorithms, Vehicle Detection and License Plate Recognition using CV, Public Transport Optimization using Predictive Analytics, Smart					

Parking and Navigation Systems, Use of Drones and AI for Traffic Surveillance		
Module-III	AI and IoT for Smart Energy, Waste, and Water Management	10 Hrs
AI for Smart Grids and Energy Consumption Prediction, Load Balancing and Demand Forecasting using ML, Waste Segregation and Collection Automation using CV, Water Quality Monitoring Systems using IoT Sensors, Leak Detection and Anomaly Detection Models, Smart Metering and Energy Theft Detection, Sustainability and Carbon Monitoring AI Models		
Module-IV	Smart Healthcare, Surveillance, and Public Safety	9 Hrs
IoT-based Health Monitoring and Alert Systems, Predictive Healthcare and Disease Outbreak Detection, AI for CCTV Surveillance, Crowd Monitoring, and Violence Detection, NLP for Emergency Response and Chatbot Assistance, Smart Ambulance Routing and Response Optimization, COVID-19 Contact Tracing and Monitoring via AI & IoT, Data Privacy, Security & Ethical Issues in Surveillance Systems		
Module-V	AIoT System Design, Deployment, and Governance	10 Hrs
AI Model Deployment on Edge Devices (Raspberry Pi, Jetson Nano), Smart City Dashboards and Data Visualization, Real-time Streaming and Analytics Platforms (Apache Kafka, Spark), Cloud Integration: AWS IoT, Google Cloud AI, Azure IoT Suite, Governance Frameworks, Data Privacy, and Policy Standards, Evaluation Metrics for Smart City Projects, Future Trends in AIoT and Smart Urban Living		
Textbooks: 1. Pethuru Raj & Anupama C. Raman, The Internet of Things: Enabling Technologies, Platforms, and Use Cases, CRC Press. 2. Janaka Ekanayake, Smart Grid: Technology and Applications, Wiley. 3. Rajkumar Buyya, Fog and Edge Computing: Principles and Paradigms, Wiley. 4. Adrian McEwen, Hakim Cassimally, Designing the Internet of Things, Wiley.		
Reference Books: 1. Mahalik N. P., Sensor Networks and Applications, McGraw Hill. 2. Kim F. Taylor, Urban Artificial Intelligence and Governance, Springer. 3. Dastbaz, J. & Pattinson, C., Smart Cities: Innovation and Sustainability, Springer. 4. Research papers from IEEE Smart Cities, AIoT Journal, and Springer Urban Tech.		
Online Courses: 1. Coursera – Smart Cities: Management of Smart Urban Infrastructures (EPFL) 2. edX – Internet of Things (IoT) Program – Curtin University		



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Prompt Engineering (Skill Enhancement Course)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0540P	0:1:2:2	2	CIE:30 SEE:70	3 Hours	SEC
Course Objectives:					
This course will enable students to: ● Prompt engineering principles, strategies, and best practices, a crucial aspect in shaping AI models' behaviour and performance. ● Understanding Prompt Engineering is a comprehensive course designed to equip learners with the knowledge and skills to effectively generate and utilize prompts in natural language processing (NLP) and machine learning (ML) applications. ● Prompt engineering principles, strategies, and best practices, a crucial aspect in shaping AI models' behaviour and performance.					
Course Outcomes (CO):					
On completion of this course, student will be able to ● Understanding the fundamentals and evolution of prompt engineering. ● Gaining the ability to craft effective closed-ended, open-ended, and role-based prompts. ● Learning to probe and stress-test AI models for bias and robustness. ● Applying prompt optimization techniques and performance evaluation methods. ● Mitigating bias and promoting ethical prompting practices in NLP/ML systems.					
Syllabus				Total Hours:48	
Module-I	Introduction to Prompt Engineering			8 Hrs	
Lesson 1: ○ Foundations of Prompt Engineering ○ Overview of prompt engineering and its significance in NLP and ML. ○ Historical context and evolution of prompt-based approaches.					
Module-II	Types of Prompts and Their Applications			8 Hrs	
Lesson 2: ○ Closed-Ended Prompts ○ Understanding and creating prompts for specific answers. ○ Applications in question- answering systems.					
Lesson 3: ○ Open-Ended Prompts ○ Crafting prompts for creative responses. ○ Applications in language generation models.					
Module-III	Strategies for Effective Prompting			8 Hrs	

Lesson 4: ○ Probing Prompts ○ Designing prompts to reveal model biases. ○ Ethical considerations in using probing prompts.		
Lesson 5: ○ Adversarial Prompts ○ Creating prompts to stress-test models. ○ Enhancing robustness through adversarial prompting.		
Module-IV	Fine-Tuning and Optimizing with Prompts	8 Hrs
Lesson 6: ○ Fine-Tuning Models with Prompts ○ Techniques for incorporating prompts during model training. ○ Balancing prompt influence and generalization.		
Lesson 7: ○ Optimizing Prompt Selection ○ Methods for selecting optimal prompts for specific tasks. ○ Customizing prompts based on model behavior.		
Module-V	Evaluation and Bias Mitigation	8 Hrs
Lesson 8: ○ Evaluating Prompt Performance ○ Metrics and methodologies for assessing model performance with prompts. ○ Interpreting and analyzing results.		
Lesson 9: ○ Bias Mitigation in Prompt Engineering ○ Strategies to identify and address biases introduced by prompts. ○ Ensuring fairness and inclusivity in prompt-based models.		
Module-VI	Real-World Applications and Case Studies	8 Hrs
Lesson 10: ○ Case Studies in Prompt Engineering ○ Exploration of successful implementations and challenges in real-world scenarios. ○ Guest lectures from industry experts sharing their experiences.		
Text Books: 1. Mathematics for Machine Learning by Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong, Cambridge University Press, 2020. 2. Pattern Recognition and Machine Learning by Christopher Bishop, Springer.		
Reference Books: 1. Gilbert Strang, Linear Algebra and Its Applications, Cengage Learning, 2016. 2. Jonathan Gross, Jay Yellen, Graph Theory and Its Applications, CRC Press, 2018.		
Web References: 1. MIT– Mathematics for Machine Learning https://ocw.mit.edu. 2. Stanford CS229 – Machine Learning Course https://cs229.stanford.edu/ 3. DeepAI – Mathematical Foundations for AI https://deepai.org		



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Gender Sensitization (Audit Course)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A0054T	2:0:0:0	0	CIE:30 SEE:70	3 Hours	AC
Course Objectives:					
This course will enable students to:					
- To enable students to understand the gender related issues, vulnerability of women and men - To familiarize them about constitutional safeguard for gender equality - To expose the students to debates on the politics and economics of work - To help students reflect critically on gender violence - To make them understand that gender identities and gender relations are part of culture as they shape the way daily life is lived in the family as well as wider community and the workplace.					
Course Outcomes:					
On completion of this course, student will be able to:					
- Understand the basic concepts of gender and its related terminology - Identify the biological, sociological, psychological and legal aspects of gender. - Use the knowledge in understanding how gender discrimination works in our Society and how to counter it - Analyze the gendered division of labour and its relation to politics and economics. - Appraise how gender-role beliefs and sharing behaviour are associated with - more well-being in all culture and gender groups - Develop students' sensibility with regard to issues of gender in contemporary India.					
Syllabus				Total Hours:48	
Module-I	Understanding Gender			10 Hrs	
Introduction: Definition of Gender-Basic Gender Concepts and Terminology-Exploring Attitudes towards Gender-Construction of Gender-Socialization: Making Women, Making Men - Preparing for Womanhood. Growing up Male. First lessons in Caste.					
Module-II	Gender Roles and Relations			9 Hrs	
Two or Many? -Struggles with Discrimination-Gender Roles and Relations-Types of Gender Roles-Gender Roles and Relationships Matrix-Missing Women-Sex Selection and its Consequences-Declining Sex Ratio- Demographic Consequences-Gender Spectrum -					
Module-III	Gender and Labour			10 Hrs	
Division and Valuation of Labour-Housework: The Invisible Labor- —My Mother doesn't Work. —Share the Load. —Work: Its Politics and Economics -Fact and Fiction- Unrecognizedand Unaccounted work -Gender Development Issues-Gender, Governance and Sustainable Development-Gender and Human Rights-Gender and Mainstreaming.					
Module-IV	Gender-Based Violence			9 Hrs	

The Concept of Violence- Types of Gender-based Violence-Gender-based Violence from a Human Rights Perspective-Sexual Harassment - Domestic Violence - Different forms of violence against women - Causes of violence, impact of violence against women - Consequences of gender-based violence.		
Module-V	Gender and Culture	10 Hrs
Gender and Film-Gender and Electronic Media-Gender and Advertisement-Gender and Popular Literature- Gender Development Issues-Gender Issues-Gender Sensitive Language- Just Relationships		
Text Books: 1. A.Suneetha, Uma Bhugubanda, et al. Towards a World of Equals: A Bilingual Textbook on Gender, Telugu Akademi, Telangana, 2015. 2. Butler, Judith. Gender Trouble: Feminism and the Subversion of Identity. UK Paperback Edn. March 1990		
Reference Books: 1. Wtatt, Robin and Massood, Nazia, Broken Mirrors: The dowry Problems in India, London : Sage Publications, 2011 2. Datt, R. and Kornberg, J.(eds), Women in Developing Countries, Assessing Strategies for Empowerment, London: Lynne Rienner Publishers, 2002 3. Brush, Lisa D., Gender and Governance, New Delhi, Rawat Publication, 2007 4. Singh, Direeti, Women and Politics World Wide, New Delhi, Axis Publications, 2010 5. Raj Pal Singh, Anupama Sihag, Gender Sensitization: Issues and Challenges (English, Hardcover), Raj Publications, 2019 6. A.Revathy& Murali, Nandini, A Life in Trans Activism(Lakshmi Narayan Tripathi). The University of Chicago Press, 2016		
Web References: 1. Understanding Gender chrome extension://kdpelmjpfafjppnhbloffcjpeomlnpah/https://www.arvindguptatoys.com/arvindgupta/kamla-gender1.pdf https://onlinecourses.swayam2.ac.in/nou24_hs53/preview 2. Gender Roles and Relations https://www.plannedparenthood.org/learn/gender-identity/sex-gender-identity/what-are gender-roles-and-stereotypes https://www.verywellmind.com/understanding-gender-roles-and-their-effect-on-our-relationships-7499408 https://onlinecourses.swayam2.ac.in/cec23_hs29/preview 3. Gender and Labour https://www.economicsobservatory.com/what-explains-the-gender-division-of-labour-and how-can-it-be-redressed https://onlinecourses.nptel.ac.in/noc23_mg67/preview 4. Gender-based violence https://eige.europa.eu/gender-based-violence/what-is-gender-based-violence?language_content_entity=en https://www.worldbank.org/en/topic/socialsustainability/brief/violence-against-women-and-girls https://onlinecourses.swayam2.ac.in/nou25_ge38/preview 5. Gender and culture		

<https://gender.study/psychology-of-gender/culture-impact-gender-roles-identities/>

<https://sociology.iresearchnet.com/sociology-of-culture/gender-and-culture/>

<https://archive.nptel.ac.in/courses/109/106/109106136/>

Abdulali Sohaila. —I Fought For My Life...and Won.¶Available online (at:
<http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulal/> 10



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Semester-8							
Sl. No.	Category	Course Code	CourseTitle	Hoursperweek			Credits
				L	T	P	C
1	PCC	23A3323	Internship	-	-	24	4
			Project	-	-	-	8
Total							12



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Open Electives

Open Elective – I

Sl.No.	Course Code	CourseTitle	Offered by the Dept
1	23A0148T	Green Buildings	CIVIL
2	23A0149T	Construction Technology and Management	CIVIL
3	23A0222T	Electrical Safety Practices and Standards	EEE
4	23A0319T	Sustainable Energy Technologies	ME
5	23A0442T	Electronic Circuits	ECE
6	23A0027T	Mathematics for Machine Learning and AI	Mathematics
7	23A0034T	Materials Characterization Techniques	Physics
8	23A0040T	Chemistry of Energy Systems	Chemistry
9	23A0044T	English for Competitive Examinations	Humanities
10	23A0051T	Entrepreneurship and New Venture Creation	Humanities
11	23A0547T	Quantum Technologies and Applications	CSE

Open Elective – II

Sl.No.	Course Code	CourseTitle	Offered by the Dept
1	23A0150T	Disaster Management	CIVIL
2	23A0151T	Sustainability in Engineering Practices	CIVIL
3	23A0232T	Renewable Energy Sources	EEE
4	23A0334Tb	Automation and Robotics	ME
5	23A0443T	Digital Electronics	ECE
6	23A0030T	Optimization Techniques	Mathematics
7	23A0035T	Physics of Electronic Materials and Devices	Physics
8	23A0041T	Chemistry Of Polymers And Applications	Chemistry
9	23A0045T	Academic Writing and Public Speaking	Humanities
10	23A0029T	Mathematical Foundations of quantum technologies	Mathematics



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Open Electives

Open Elective – III

Sl.No.	Course Code	CourseTitle	Offered by the Dept
1	23A0152T	Building Materials and Services	CIVIL
2	23A0121T	Environmental Impact Assessment	CIVIL
3	23A0241T	Smart Grid Technologies	EEE
4	23A0335T	3D Printing Technologies	ME
5	23A0416T	Microprocessors and Microcontrollers	ECE
6	23A0027T	Wavelet transforms and its Applications	Mathematics
7	23A0036T	Smart Materials And Devices	Physics
8	23A0042T	Green Chemistry And Catalysis For Sustainable Environment	Chemistry
9	23A0046T	Employability Skills	Humanities
10	23A0037T	Introduction to Quantum Mechanics	Physics

Open Elective – IV

Sl.No.	Course Code	CourseTitle	Offered by the Dept
1	23A0153T	Geo-Spatial Technologies	CIVIL
2	23A0154T	Solid Waste Management	CIVIL
3	23A0242T	Electric Vehicles	EEE
4	23A0334Td	Total Quality Management	ME
5	23A0444T	Transducers and Sensors	ECE
6	23A0032T	Financial Mathematics	Mathematics
7	23A0038T	Sensors And Actuators For Engineering Applications	Physics
8	23A0043T	Chemistry Of Nano materials and Applications	Chemistry
9	23A0047T	Literary Vibes	Humanities
10	23A3315Ta	Introduction to Quantum Computing	CSE(AI&ML)



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Open Electives offered by CSE to other departments

Sl.No.		Course Code	CourseTitle	L	T	P	Credits
1	Open Elective - I	23A0545T	Java Programming	3	0	0	3
2		23A0546T	Fundamentals of Artificial Intelligence	3	0	0	3
3	Open Elective - II	23A0511T	Operating Systems	3	0	0	3
4		23A0548T	Introduction to Machine Learning	3	0	0	3
5	Open Elective - III	23A0512T	Data Base Management Systems	3	0	0	3
6		23A0532Tv	Cyber Security	3	0	0	3
7	Open Elective - IV	23A0549T	Introduction to Computer Networks	3	0	0	3
8		23A0538Td	Internet of Things	3	0	0	3

COURSES OFFERED FOR HONOURS DEGREE IN CSE

Sl.No.	Course Code	Course Title	Hours per week			Credits
			L	T	P	C
1	23A33H01	Advanced Machine Learning & AI Systems	3	0	0	3
2	23A33H02	Deep Learning & Neural Networks	3	0	0	3
3	23A33H03	Reinforcement Learning & Decision Making	3	0	0	3
4	23A33H04	AI for Robotics & Automation	3	0	0	3
5	23A33H05	AI Ethics, Fairness & Explainability	3	0	0	3
6	23A33H06	AI & Machine Learning Lab			3	1.5
7	23A33H07	Robotics &Autonomous Systems Lab			3	1.5



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Advanced Machine Learning & AI Systems (Honours Course)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A33H01	3:0:0:3	3	CIE:30 SEE:70	3 Hours	
Course Objectives:					
This course will enable students: - To deepen understanding of algorithmic principles for designing scalable and efficient AI/ML solutions. - To explore advanced topics such as optimization algorithms, randomized and approximation algorithms, and online learning. - To analyze computational complexity, tractability, and convergence of AI models. - To apply graph-based, evolutionary, and heuristic approaches in solving real-world AI/ML problems. - To integrate algorithmic strategies for large-scale machine learning, reinforcement learning, and neural network training					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Analyze and apply classical algorithmic techniques including divide and conquer, dynamic programming, approximation, and randomized algorithms in the context of AI/ML. - Implement advanced graph algorithms for shortest paths, flows, and community detection, and apply them to AI problems like NLP and recommender systems. - Apply convex and non-convex optimization strategies, gradient-based learning, and regularization techniques to train and tune AI/ML models effectively. - Use evolutionary, swarm intelligence, and reinforcement learning-based metaheuristic methods for neural architecture search and complex optimization tasks in AI. - Evaluate and design scalable algorithmic solutions with fairness and interpretability for AI/ML applications, referencing case studies like AlphaGo, GPT, and AutoML systems					
Syllabus				Total Hours:48	
Module-I	Foundations of Advanced Algorithmic Techniques			10 Hrs	
Review of Time and Space Complexity, Divide and Conquer, Dynamic Programming, and Greedy Algorithms, Recurrence Relations and Master Theorem, Approximation Algorithms: Vertex Cover, TSP, Set Cover, Randomized Algorithms: Monte Carlo and Las Vegas Types, Probabilistic Analysis and Tail Bounds, Applications in ML Preprocessing and Feature Selection					
Module-II	Graph Algorithms and AI Applications			9 Hrs	
Graph Representations and Traversal Algorithms, Shortest Path: Dijkstra’s, Bellman-Ford, FloydWarshall, Minimum Spanning Trees: Kruskal and Prim, Network Flows and Max Flow-Min Cut Theorem, Graph-Based Semi-Supervised Learning, PageRank, Centrality, and Community Detection, Applications in NLP, Vision, and Recommender Systems					
Module-III	Optimization in AI/ML			9 Hrs	
Convex and Non-Convex Optimization, Gradient Descent Variants: SGD, Momentum, Adam, Convergence Analysis and Learning Rates, Duality and Lagrange Multipliers, Regularization: L1, L2, ElasticNet, Hyperparameter Optimization: Grid, Random, Bayesian, Constrained Optimization in SVMs and Deep Learning					

Module-IV	Evolutionary & Metaheuristic Algorithms	10 Hrs
Genetic Algorithms and Evolutionary Strategies, Swarm Intelligence: PSO, Ant Colony Optimization, Simulated Annealing and Tabu Search, Multi-objective Optimization, Reinforcement Learning and Policy Gradient Methods, Neuroevolution: Evolving Neural Networks, Use Cases in Feature Engineering and Neural Architecture Search (NAS)		
Module-V	Advanced Topics and Case Studies	10 Hrs
Online Learning and Regret Minimization, Bandit Algorithms: Multi-Armed Bandits, Thompson Sampling, Large-Scale Algorithms: MapReduce, Apache Spark MLlib, Algorithmic Fairness, Interpretability, and Ethics in AI, Case Studies: AlphaGo, GPT, BERT, Recommendation Engines, Research Trends in Algorithmic ML and AutoML, Capstone Problem Solving using Hybrid Algorithms		
Textbooks: 1. Introduction to Algorithms – Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein (MIT Press) 2. Algorithms for Machine Learning – Giuseppe Bonaccorso, Packt Publishing 3. Convex Optimization – Stephen Boyd and Lieven Vandenberghe, Cambridge University Press 4. Reinforcement Learning: An Introduction – Richard S. Sutton and Andrew G. Barto		
Reference Books: 1. Machine Learning: A Probabilistic Perspective – Kevin P. Murphy. The Elements of Statistical Learning – Trevor Hastie, Robert Tibshirani, Jerome Friedman. 2. Evolutionary Computation – Kenneth A. De Jong. 3. Handbook of Approximation Algorithms and Metaheuristics – Teofilo F. Gonzalez		
Recommended Online Courses & Resources: 1. Coursera – Advanced Algorithms and Complexity (UC San Diego) 2. edX – Algorithmic Design and Techniques (UC San Diego) 3. MIT OpenCourseWare – Advanced Algorithms 4. Udemy – Optimization Algorithms in Machine Learning 5. Stanford Online – Convex Optimization		



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Deep Learning & Neural Network Architectures (Honours Course)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A33H02	3:0:0:3	3	CIE:30 SEE:70	3 Hours	
Course Objectives:					
This course will enable students: - To introduce the fundamental concepts and mathematical foundations of deep learning. - To explore different neural network architectures including CNNs, RNNs, LSTMs, and Transformers. - To enable students to implement, train, and optimize deep neural networks. - To analyze the performance and limitations of various architectures in different AI tasks. - To develop the ability to apply deep learning models to real-world applications such as image recognition, language modeling, and autonomous systems.					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Understand the theoretical foundations of neural networks and deep learning. - Implement and train multilayer perceptrons, CNNs, RNNs, and other architectures. - Analyze and optimize deep learning models using advanced regularization and tuning techniques. - Evaluate the applicability of different neural network architectures for various AI problems. - Apply state-of-the-art models such as Transformers and GANs in real-world domains.					
Syllabus					Total Hours:48
Module-I	Foundations of Neural Networks				10 Hrs
Introduction to Artificial Neural Networks, Biological Neuron vs. Artificial Neuron, Perceptron, Multilayer Perceptron (MLP), Activation Functions: ReLU, Sigmoid, Tanh, Softmax, Backpropagation and Gradient Descent, Loss Functions: MSE, Cross Entropy, Overfitting, Regularization (L1/L2), Dropout					
Module-II	Convolutional Neural Networks (CNNs)				10 Hrs
Convolution Operation and Feature Maps, Pooling Layers: Max and Average Pooling, CNN Architectures: LeNet, AlexNet, VGG, ResNet, Transfer Learning and Fine-tuning, Image Classification, Object Detection Basics, Implementation with TensorFlow/PyTorch					
Module-III	Recurrent Neural Networks (RNNs) and Variants				9 Hrs
Sequential Data and Time Series, RNN Basics and Backpropagation Through Time (BPTT), Vanishing and Exploding Gradients, LSTM and GRU Architectures, Applications in Text, Speech, and Music, Sequence-to-Sequence Models					
Module-IV	Advanced Architectures & Optimization				9 Hrs
Autoencoders and Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), Deep Reinforcement Learning Overview, Batch Normalization, Early Stopping, Hyperparameter Tuning and Optimization, Performance Metrics and Evaluation					

Module-V	Transformer Models & Applications	10 Hrs
Attention Mechanism and Self-Attention, Transformers and BERT Architecture, Positional Encoding, Multi-head Attention, Pre-trained Language Models and Fine-Tuning, Applications in NLP: Text Classification, Translation, Large Language Models and Transfer Learning		
Text Books: 1. Deep Learning – Ian Goodfellow, YoshuaBengio, and Aaron Courville (MIT Press) 2. Neural Networks and Deep Learning – Michael Nielsen (Online Book) 3. Hands-On Machine Learning with Scikit-Learn, Keras& TensorFlow – AurélienGéron (O'Reilly)		
Reference Books: 1. Pattern Recognition and Machine Learning – Christopher M. Bishop 2. Deep Learning for Computer Vision – RajalingappaaShanmugamani 3. Natural Language Processing with Transformers – Lewis Tunstall, Leandro von Werra, Thomas Wolf 4. Reinforcement Learning: An Introduction – Richard S. Sutton and Andrew G. Barto		
Recommended Online Courses: 1. Deep Learning Specialization – Andrew Ng (Coursera) 2. CS231n: Convolutional Neural Networks for Visual Recognition (Stanford) 3. Fast.ai – Practical Deep Learning for Coders 4. Deep Learning with PyTorch (Udacity) 5. Transformers by Hugging Face (free course)		



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REINFORCEMENT LEARNING & AUTONOMOUS SYSTEMS (Honours Course)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A33H03	3:0:0:3	3	CIE:30 SEE:70	3 Hours	
Course Objectives:					
This course will enable students: - To understand the foundational concepts of reinforcement learning (RL) and its mathematical formulation. - To explore model-free and model-based approaches for learning optimal policies. - To study deep reinforcement learning and its application to autonomous systems. - To develop the ability to design and implement RL algorithms in simulated and real environments. - To apply RL techniques in robotics, autonomous vehicles, and decision-making systems.					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Explain the basic concepts of reinforcement learning, including rewards, value functions, and policies. - Apply dynamic programming, Monte Carlo, and temporal difference methods for policy evaluation and control. - Design and implement model-free learning algorithms such as Q-learning and SARSA. - Utilize deep reinforcement learning frameworks (e.g., DQN, A3C, PPO) to solve complex tasks. - Integrate reinforcement learning into autonomous systems like robots and self-driving cars.					
Syllabus				Total Hours:	
Module-I	Introduction to Reinforcement Learning			10 Hrs	
What is Reinforcement Learning (RL), Agent-Environment Interface, Markov Decision Processes (MDPs), Rewards, States, Actions, Policies, Value Functions: State and Action Values, Exploration vs. Exploitation, Bellman Equations					
Module-II	Dynamic Programming and Model-Free Methods			9 Hrs	
Dynamic Programming for Policy Evaluation and Improvement, Policy Iteration and Value Iteration, Monte Carlo Methods: Prediction and Control, Temporal Difference Learning: TD(0), TD(λ), SARSA and Q-Learning Algorithms, Eligibility Traces					
Module-III	Deep Reinforcement Learning			9 Hrs	
Function Approximation and Neural Networks, Deep Q-Networks (DQN), Experience Replay and Target Networks, Policy Gradient Methods, Actor-Critic Architecture, Advanced Algorithms: A2C, A3C, DDPG, PPO					
Module-IV	Applications in Autonomous Systems			10 Hrs	
Autonomous Decision-Making Frameworks, RL in Robotics: Path Planning and Motion Control, RL in Autonomous Vehicles, Multi-Agent Reinforcement Learning, Safety and Ethical Issues in Autonomous RL, Transfer and Meta-Reinforcement Learning					
Module-V	Simulation, Tools, and Real-World Deployment			10 Hrs	
Simulation Environments: OpenAI Gym, PyBullet, CARLA, Implementing RL with TensorFlow and PyTorch, Case Studies: Robotic Manipulation, Game Playing (Atari, Go), Performance Metrics and					

Benchmarking, Deployment in Edge and Real-Time Systems, Challenges in Scaling RL
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Textbooks:

- | |
|---|
| <ol style="list-style-type: none">1. Richard S. Sutton and Andrew G. Barto – Reinforcement Learning: An Introduction, 2nd Edition, MIT Press2. Prashant Reddy, Balaraman Ravindran – Reinforcement Learning with TensorFlow (Packt)3. Laura Graesser and Wah Loon Keng – Foundations of Deep Reinforcement Learning |
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Reference Books:

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| <ol style="list-style-type: none">1. Ian Goodfellow, YoshuaBengio, Aaron Courville – Deep Learning, MIT Press2. Alexander Zai, Brandon Brown – Deep Reinforcement Learning in Action3. Lucian Busoniu et al. – Reinforcement Learning and Dynamic Programming Using Function Approximators |
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Recommended Online Courses & Resources:
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| <ol style="list-style-type: none">1. CS50's Introduction to AI with Python (Harvard)2. Deep Reinforcement Learning Specialization – Coursera (by University of Alberta)3. Udacity – Deep Reinforcement Learning Nanodegree4. Spinning Up in Deep RL – OpenAI5. David Silver's RL Lectures – DeepMind/YouTube |
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AI for Robotics & Automation (Honours Course)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A33H04	3:0:0:3	3	CIE:30 SEE:70	3 Hours	
Course Objectives:					
This course will enable students:					
• To introduce the principles of software engineering augmented with artificial intelligence. • To explore AI-driven tools for software requirement analysis, design, testing, and maintenance. • To understand DevOps practices and integrate AI techniques for CI/CD, deployment, and monitoring. • To automate software lifecycle management using machine learning and data-driven insights. • To develop intelligent pipelines for software delivery with adaptive testing and feedback systems.					
Course Outcomes (CO):					
On completion of this course, student will be able to:					
• Describe the role of AI in modern software engineering processes and lifecycle stages. • Apply AI/ML models for requirements gathering, code generation, defect prediction, and testing. • Implement intelligent DevOps practices including CI/CD, release automation, and anomaly detection. • Analyze data from software pipelines to drive informed decisions and improve quality • Develop an end-to-end AI-enabled software delivery pipeline with automated learning-based optimizations.					
Syllabus					Total Hours:48
Module-I	Foundations of Neural Networks				10 Hrs
Introduction to Software Engineering Lifecycle, Traditional vs. AI-Driven Software Development, AI/ML in Software Engineering: Overview and Scope, Natural Language Processing (NLP) for Requirements Engineering, AI for Software Design Recommendation, Intelligent Code Completion.(e.g., GitHub Copilot)					
Module-II	AI in Testing and Defect Prediction				10 Hrs
Static and Dynamic Testing with AI, Automated Test Case Generation, Defect Detection and Prediction using ML Models, Sentiment and Bug Report Analysis, AI in Refactoring and Code Review, Tools: SonarQube, DeepCode.					
Module-III	DevOps Principles and Practices				9 Hrs
DevOps Overview: CI/CD Pipelines, Infrastructure as Code (IaC), Configuration Management Tools: Ansible, Puppet,Monitoring and Logging Tools: Prometheus, Grafana, Containerization and Orchestration: Docker, Kubernetes, Agile and Lean Practices in DevOps.					
Module-IV	AI for DevOps Automation and Intelligence				10 Hrs
.Predictive Analytics for Deployment Success, AI for Log Analytics and Root Cause Analysis, Self-Healing Systems and Auto-Scaling, Feedback Loops in DevOps using Reinforcement Learning, Data-Driven Decision Making in Release Management, ChatOps and AIOps Platforms.					

Module-V	Case Studies and Emerging Trends	9 Hrs
Case Study: AI-Augmented DevOps in Enterprises, ML-Ops vs. DevOps vs. DataOps, Security in DevOps (DevSecOps), Explainability and Ethics in AI-Driven Software Engineering, Generative AI in Software Development, Future Trends and Industry Standards.		
Text Books: 1. Tim Menzies, Diomidis Spinellis – Artificial Intelligence and Software Engineering: Status and Future Directions 2. Len Bass, Ingo Weber, Liming Zhu – DevOps: A Software Architect's Perspective, Addison-Wesley 3. Thomas Erl, Ricardo Puttini, Zaigham Mahmood – AI & Analytics for DevOps, Pearson		
Reference Books: 1. Carlos Nunes Silva – AI in Software Engineering 2. Gene Kim, Jez Humble, Patrick Debois, John Willis – The DevOps Handbook 3. Andrew Ng – Machine Learning B.Techning (AI Systems Engineering Perspective)		
Online Resources & Courses 1. Coursera – AI for Software Engineering (IBM) 2. DevOps with Microsoft Azure – edX 3. Udacity – AI for DevOps Engineers Nanodegree		



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AI Ethics, Fairness & Explain ability					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A33H05	3:0:0:3	3	CIE:30 SEE:70	3 Hours	
Course Objectives:					
This course will enable students:					
- To understand ethical concerns and responsibilities in the development and deployment of AI systems. - To study fairness, bias, and accountability issues in AI and machine learning models. - To explore techniques and frameworks for interpreting and explaining AI decisions. - To analyze societal impacts of AI and build trust through transparent systems. - To promote responsible and inclusive AI development aligned with human values.					
Course Outcomes (CO):					
On completion of this course, student will be able to:					
- Describe the ethical principles, historical context, and responsibilities associated with AI deployment across domains like healthcare and law enforcement. - Identify different forms of bias in datasets and algorithms, and apply fairness metrics and mitigation strategies to ensure equitable AI systems. - Demonstrate the need for explainability in AI models and utilize tools such as LIME, SHAP, and Grad-CAM to generate local and global model explanations. - Design AI systems with accountability by integrating human oversight, ethical documentation (e.g., Model Cards, Datasheets), and adherence to global guidelines. - Critically assess the broader societal and legal implications of AI in areas such as surveillance, misinformation, and inclusivity, and explore international policy frameworks.					
Syllabus					TotalHours:48
Module-I	Foundations of AI Ethics				10 Hrs
Historical background of AI ethics, Core principles: beneficence, non-maleficence, autonomy, justice, Moral and legal responsibilities in AI systems, Risk assessment and governance in AI, Ethical AI case studies from healthcare, policing, hiring					
Module-II	Fairness and Bias in AI				9 Hrs
Types of bias: dataset bias, label bias, historical bias, Fairness definitions: demographic parity, equal opportunity, individual fairness, Disparate impact and fairness metrics, Algorithmic audits and bias detection, Fairness-aware learning and mitigation strategies					
Module-III	Explainable Artificial Intelligence (XAI)				10 Hrs
Need for interpretability in AI models, Taxonomy of XAI methods: model-agnostic, model-specific, LIME, SHAP, Grad-CAM, Partial Dependence Plots, Local vs Global explanations, Trade-offs: accuracy vs interpretability					
Module-IV	Accountability and Responsible AI Design				9 Hrs
Transparent AI systems, Human-in-the-loop and AI-assisted decision-making, Accountability frameworks (e.g., IEEE, NIST, EU Guidelines), Documentation tools: Datasheets for datasets, Model Cards, Responsible AI lifecycle management					

Module-V	Societal Impacts and Policy Considerations	10 Hrs
AI in surveillance, misinformation, and social manipulation, Ethical implications in autonomous systems (vehicles, weapons), AI and inclusion: accessibility, gender, race, socioeconomic impacts, Public policy, legal frameworks, and global initiatives, Future challenges and global governance of AI		
Text Books: 4. Virginia Dignum – Responsible Artificial Intelligence: How to Develop and Use AI in a Responsible Way, Springer 5. Cathy O’Neil – Weapons of Math Destruction, Crown Publishing 6. Mark Coeckelbergh – AI Ethics, The MIT Press		
Reference Books: 1. Nick Bostrom & Eliezer Yudkowsky – The Ethics of Artificial Intelligence 2. Shalini Kantayya (Film) – Coded Bias (documentary, 2020) 3. Floridi, Luciano – Ethics of Information, Oxford University Press		
Online Courses & Resources: 1. Coursera – AI For Everyone (Andrew Ng) 2. edX – Ethics of AI and Big Data (Linux Foundation)		



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AI & Machine Learning Lab (Honours Course)

Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A33H06	0:0:3:1.5	1.5	CIE:30 SEE:70	3 Hours	

Course Objectives:

This course will enable students:

- To provide hands-on experience in implementing AI and machine learning algorithms.
- To develop and evaluate models using real-world datasets.
- To introduce optimization and hyper parameter tuning techniques.
- To build intelligent systems for classification, prediction, and clustering.

Course Outcomes (CO)

On completion of this course, student will be able to:

- Implement key machine learning algorithms from scratch and using libraries.
- Pre-process data and select suitable features for modelling.
- Train, test, and evaluate models for accuracy and performance.
- Apply AI techniques to solve classification, regression, and decision-making problems.
- Develop simple AI agents and use neural networks for predictive tasks.

Tools Required

1. Python (NumPy, Pandas, Scikit-learn, TensorFlow/Keras, OpenCV)
2. Jupyter Notebook / Google Colab
3. Datasets from UCI, Kaggle, Scikit-learn
4. Anaconda / VS Code

List of 12 Experiments

1. **Data Pre-processing** – Cleaning, normalization, encoding, and splitting data.
2. **Linear Regression** – Implement simple and multiple linear regression.
3. **Logistic Regression** – Binary classification on datasets like breast cancer or Titanic.
4. **K-Nearest Neighbors (KNN)** – Classification task with evaluation metrics.
5. **Decision Trees and Random Forests** – Tree-based classification and visualization.
6. **Support Vector Machines (SVM)** – Margin classification with kernel trick.
7. **Naive Bayes** – Text classification with spam dataset.
8. **K-Means Clustering** – Unsupervised clustering with customer segmentation.
9. **Principal Component Analysis (PCA)** – Dimensionality reduction and visualization.
10. **Artificial Neural Networks (ANNs)** – Implement basic neural network using Keras.

11. **Model Evaluation & Tuning** – Use cross-validation, GridSearchCV, and confusion matrices.

12. **AI Agent Search Algorithms** – Implement A*, DFS, BFS for path finding problems.



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Robotics & Autonomous Systems Lab (Honours Course)					
Course Code	L:T:P:C	Credits	Exam Marks	Exam Duration	Course Type
23A33H07	0:0:3:1.5	1.5	CIE:30 SEE:70	3 Hours	
Course Objectives:					
This course will enable students: - To provide hands-on experience in deploying machine learning models into production environments. - To introduce the tools and practices of MLOps for automating ML workflows. - To train students in containerization, orchestration, monitoring, and CI/CD pipelines. - To understand model versioning, reproducibility, and lifecycle management. - To develop skills in using cloud platforms and APIs for scalable AI applications					
Course Outcomes (CO):					
On completion of this course, student will be able to: - Package and deploy AI models using tools such as Flask, Fast API, and Docker. - Automate machine learning workflows using CI/CD pipelines and MLOps tools. - Monitor and manage deployed models in real-time environments. - Apply version control and model registry techniques effectively. - Deploy models on cloud platforms like AWS, Azure, or GCP and use ML flow, Kubeflow, etc.					
List of 12 Lab Experiments:					
- Experiment 1: Build a simple ML model and serve it via Flask or FastAPI. - Experiment 2: Containerize the model application using Docker. - Experiment 3: Deploy a Dockerized model on a local or cloud-based Kubernetes cluster. - Experiment 4: Implement CI/CD pipeline using GitHub Actions or GitLab CI. - Experiment 5: Track experiments and manage model versions using ML flow. - Experiment 6: Use DVC (Data Version Control) for tracking data and pipeline stages. - Experiment 7: Automate model retraining and deployment using Jenkins. - Experiment 8: Model monitoring using Prometheus and Grafana. - Experiment 9: Introduce model drift detection and retraining triggers. - Experiment 10: Deploy a model on a cloud platform (e.g., AWS Sage Maker, GCP AI Platform). - Experiment 11: Use Kubeflow pipelines for end-to-end ML workflow management. - Experiment 12: Capstone: Full-cycle ML project from training to monitoring using MLOps best practices.					
Text Books:					
- Mark Treveil and Alok Shukla, AI and Analytics in Production: How to Implement Successful AI and Analytics Applications, O'Reilly Media. - Emmanuel Ameisen, Building Machine Learning Powered Applications, O'Reilly Media.					
Reference Books:					
Chris Fregly and Antje Barth, Data Science on AWS: Building End-to-End Applications, O'Reilly.					

2. Alfredo Deza and Noah Gift, Practical MLOps, O'Reilly Media.
3. Soham Kamani, Learning MLOps, Packt Publishing